

The Response of Debtors to Rate Changes^{*}

Andreas Fuster[†] Virginia Gianinazzi[‡] Andreas Hackethal[§]
Philip Schnorpfel[¶] Michael Weber^{||}

October 5, 2026

How borrowers respond to future changes in the interest rate on their debt matters for the transmission of monetary policy and for household financial stability. Combining bank data, a letter RCT, and a survey, we study this question in the context of the German mortgage market, where since 2022 borrowers have faced high interest rates when their rate fixation period ends. We find that borrower actions substantially reduce the impact of higher rates on monthly payments. Survey responses corroborate high informedness and a strong propensity to prepare for rate changes. The letter intervention does not affect rate beliefs but increases awareness of available options and refinancing among borrowers close to expiration of their rate fixation.

^{*}We thank our partner bank for the collaboration. The bank data were obtained under a non-disclosure agreement. The bank did not review the results prior to public dissemination. We are also grateful for helpful comments from Manuel Adelino, Claes Bäckman, João Cocco, Zeno Enders, Mattia Girotti, Carolin Hoeltken, Ragnar Juelsrud, Ankit Kalda, Edward Kim, Lorenz Kueng, Lena Liebich, Xu Lu, Pierre Mabilie, Max Miller, José-Luis Peydró, Christophe Spaenjers, Fabrice Tourre, Erkki Vihriälä, as well as from seminar and conference participants at Banco de España, Bundesbank, BYU Marriott School of Business, CEPR Advanced Forum for Financial Economics, CUNEF, Leibniz Institute for Financial Research SAFE, Nanyang Technological University, National University of Singapore, Nova SBE, Rotterdam School of Management, Tinbergen Institute, Universitat Pompeu Fabra, University of Basel, WHU, the 2026 Aarhus Finance Forum, 2026 Advances with Field Experiments Conference, 2025 Boulder Summer Conference, 2025 CEPR–Tilburg Finance Summit, 2026 CEPR Workshop on Frontiers in Measurement and Survey Methods, 2026 European Economic Association, 2025 Frankfurt Macro Seminar, 2026 Frankfurt–Mannheim Macro Workshop, 2025 Future of Finance Fest, 4th Georgia Tech–Atlanta Fed Household Finance Conference, 2025 Ghent University Workshop on Empirical Macroeconomics, 2026 Helsinki Finance Summit, 2025 ifo & BERD Workshop on “Financial Data for Research,” 15th ifo Conference on Macroeconomics and Survey Data, 2026 ITAM Finance Conference, 11th IWH–FIN–FIRE Workshop on “Challenges to Financial Stability,” 2026 Midwest Finance Conference, 2026 Montana Financial Decision-Making Conference, 2026 Mortgage Market Research Conference, 2025 Paris December Finance Meeting, 2024 Research in Behavioral Finance Conference, 9th SAFE Household Finance Workshop, 2026 SFS Cavalcade, 2026 Swiss Society for Financial Market Research, 8th Workshop on Subjective Expectations, and 1st Zurich–Bern Real Estate Finance and Economics Conference. The letter experiment received IRB approval from Goethe University Frankfurt and is preregistered in the AEA RCT Registry (ID: 14796). We gratefully acknowledge research support from the Swiss National Science Foundation.

[†]EPFL, Swiss Finance Institute, and CEPR. Email: andreas.fuster@epfl.ch.

[‡]Nova SBE. Email: virginia.gianinazzi@novasbe.pt.

[§]Goethe University Frankfurt. Email: hackethal@finance.uni-frankfurt.de.

[¶]Goethe University Frankfurt. Email: schnorpfel@finance.uni-frankfurt.de.

^{||}ESMT Berlin, Purdue University, CEPR, and NBER. Email: weber366@purdue.edu.

1 Introduction

The refinancing of debt with an expiring fixed-rate period is a key moment when higher market interest rates—for instance, following tighter monetary policy—transmit to households and corporations. The extent and nature of this transmission, however, depend crucially on how borrowers prepare for refinancing and whether they adjust loan balances or other contract terms when entering the new contract. These borrower actions can have important consequences for monetary policy and financial stability, yet our current understanding of them remains limited.

We study how households think about, prepare for, and ultimately react to increases in the interest rate on their debt. Our setting is the German mortgage market, where loans typically have fixed rates for five, ten, or more years, but amortize over a longer horizon.¹ At the end of the rate-fixation period, borrowers must refinance the remaining balance at prevailing market rates unless they repay the loan.

We combine three sources of evidence. First, we use data from one of the largest German banks. These data allow us to observe internal refinancing choices, including changes in payments, balances, interest rates, and other contract terms, and span both the pre-2022 low-rate environment and the subsequent higher-rate period. Second, we field a survey among the bank’s mortgagors to measure rate beliefs, preparation, and planned actions, including open-text explanations and a randomized vignette experiment. Third, we conduct a large-scale randomized letter intervention among more than 35,000 borrowers. The letter illustrates the increase in mortgage rates, explains its potential implications for payments, and describes actions borrowers can take in response. Linking the intervention to survey responses and subsequent bank outcomes allows us to study both engagement with the communication and letter-induced behavior.

We report four main findings. First, borrowers are much more active than a passive-refinancing view would suggest. Among loans with fixation periods up to 10 years, less than 45% refinance internally with the current lender; the others switch lenders or pay off the mortgage. About half of the internal refinancers act more than six months before expiration, using *forward mortgages*—a distinctive feature of the German setting—to lock in a future rate up to four years ahead of the end of the existing contract. Only 15% of all borrowers refinance internally in the final six weeks or later, after the bank mailed its

¹In contrast to the US, where the interest rate is usually fixed for the entire mortgage term, in most other countries with fixed-rate mortgages the loan *term* is typically substantially longer than the *fixation period*.

standard prolongation offers, and only 3% wait until after expiration and move to a high variable rate. Borrowers with initial fixation periods above 10 years, whose loans can be prepaid without penalty after 10 years, also behave actively: when market rates are below their contract rates, most borrowers exercise this option promptly.

The survey and letter RCT corroborate the picture of aware and active borrowers. Most survey respondents have accurate perceptions of the current mortgage rate environment and report preparing for the expiration of their rate fixation, by gathering information, saving, repaying, or locking in a rate early. The randomized letter leaves beliefs about mortgage rates and knowledge of contractual options largely unchanged; such limited treatment-induced updating suggests respondents were well informed to begin with (e.g., [Haaland et al., 2023](#)). Active preparation is not universal, however: it is more common among respondents who are aware of past rate increases or expect higher rates, appear more financially literate, and have higher income. Respondents with larger outstanding balances are also better informed about recent rate increases, consistent with endogenous information acquisition (e.g., [Link et al., 2025](#); [Maćkowiak et al., 2023](#)).

Our second finding is that borrower actions substantially dampen the passthrough of higher market rates to monthly mortgage payments. We compare realized payment changes to a passive counterfactual in which borrowers simply roll over the remaining debt at expiration by accepting the bank's standard prolongation offer that keeps the loan balance and term unchanged. For loans expiring after mid-2022, the passive counterfactual implies an average payment increase of €1,044 per year. The realized increase is only €276 per year. Borrowers thus offset roughly three quarters of the potential increase in their payments.² Three margins contribute to the dampening: locking in a rate early, before the full increase in market rates; reducing the loan balance at refinancing; and slowing amortization relative to the pace borrowers chose before 2022, though still repaying faster than the bank's offer implies.³ Reducing the balance takes liquid assets and repaying faster takes income, which suggests balance-sheet room rather than strain.

Our third finding is that the prevailing interest-rate environment strongly affects behavior. Forward-mortgage uptake is higher when the available market rate is low relative

²These numbers apply to internal refinancers. For borrowers who refinance externally, we cannot observe the change in monthly payments, but to the extent that borrowers switch to obtain a better rate elsewhere, this would further dampen the passthrough. Finally, borrowers who pay off their loan completely of course see a reduction of their monthly payment to zero.

³The survey suggests that these actions reflect a payment-smoothing motive. For instance, most name avoiding a large increase in the monthly payment as an important reason for reducing the loan balance.

to a borrower’s existing rate, and spikes when market rates increase rapidly, as in 2022. Once rates settle at a higher level, forward demand falls sharply, and more borrowers either switch lenders, pay off their mortgage, or wait until closer to expiration.⁴ A vignette module embedded in our survey provides causal evidence that preparation intensifies as the anticipated rate increase grows larger, matching the patterns in the bank data.

These findings imply that the passthrough of market rates to payments is largely a choice, not a contract term. Mapping market (or policy) rates to household cash flows via contract features, as is common in existing work, can therefore be inaccurate.⁵ In particular, repricing outstanding balances at market rates overstates payment burdens, as many borrowers offset much of the contract-implied increase. Importantly, however, dampened payment passthrough does not mean dampened transmission. Of the three margins borrowers use, only early rate lock-in blunts the effect of higher market rates on cash flows. Reducing the balance at refinancing trades liquid assets for lower debt, possibly accompanied by higher saving to rebuild liquid buffers. Slower amortization smooths payments by keeping leverage higher for longer. Higher rates thus reach mortgagors’ finances through saving and leverage rather than through forced payment changes.

Fourth, lender communication works primarily as a reminder rather than as new information. Recipients’ beliefs about mortgage rates and familiarity with mortgage features as measured in the survey are mostly unchanged. Yet the letter prompts action where the reset is imminent: recipients whose fixation period ends within six months of the mailing are about 40% more likely than corresponding control borrowers to meet a bank advisor and refinance shortly thereafter. Recipients further from expiration do not respond, so average effects over the six months after the mailing are small.

Finally, we document selection into effective exposure to the communication. The randomized letter increases the share of survey respondents who remember reading a bank letter by 30–37 percentage points. Importantly, borrowers who remember reading the letter are disproportionately those with larger loans, past prepayment activity, higher self-reported financial literacy, higher income, and more accurate beliefs about mortgage rates. This selection has a direct policy implication. Lender or regulator communication may predominantly reach borrowers who are already attentive and informed, while borrowers with lower awareness are harder to reach. Information provision alone may thus

⁴Among borrowers with longer fixation periods, the propensity to exercise the prepayment option when it becomes available also strongly depends on how available rates compare to a borrower’s current rate.

⁵For example, [Auclert \(2019\)](#) takes the share of adjustable-rate mortgages as a key determinant of how strongly monetary policy transmits, and [Cloyne et al. \(2020\)](#) infer payment changes from contract terms.

have limited power if the main friction is attention rather than availability.

We focus on German mortgagors for three reasons. First, Germany is the largest of many economies in which mortgage rates are typically fixed for shorter periods than the loan term. Second, the German market makes anticipation observable. For example, forwards allow locking in a new rate several years ahead of expiration, and loans with long fixation periods become freely prepayable after 10 years. Such actions date preparation in the bank data long before payments change. Third, the setting allows us to combine loan-level bank data, a customized survey, and a large-scale field experiment for the same mortgagors. The bank data show what borrowers do, the survey why they act and what they know, and the RCT how lender communication can move beliefs and refinancing.

Nevertheless, our setting has limitations. First, the German mortgage market has distinctive features, such as forward mortgages. But most mortgage markets do, from teaser rates in the UK and offset accounts in Australia to mortgage buybacks in Denmark, and such features shape which actions borrowers take. Second, homeownership in Germany is less than 45% (e.g., [Happel et al., 2024](#); [Malmendier and Wellsjo, 2024](#)) and thus lower than in most other advanced economies, suggesting potentially different selection into taking out a mortgage. Third, our data come from a single bank. We hence cannot distinguish external refinancing from full repayment, and we observe payment changes only for internal refinancers. Incomplete debt records are less of a concern: a property is financed by one lender only, and few borrowers pledge several properties as collateral. Finally, much of the analysis relies on borrowers who refinance internally and on survey respondents, who differ from other mortgagors, for instance in their attachment to the bank. Yet key choices barely vary with borrowers' attachment to the bank, borrowers who refinance into our bank from another lender adjust their loans similarly as rates rise, and survey respondents' choices resemble those of the wider bank sample.

Related literature We add to four strands of the literature. First, we contribute to work on the transmission of monetary policy through housing and mortgage markets (e.g., [Cloyne et al., 2020](#); [Flodén et al., 2021](#); [Foulis et al., 2026](#); [Jappelli and Scognamiglio, 2018](#)). Much of this literature investigates realized reductions in monthly payments (e.g., [Di Maggio et al., 2017](#); [Fuster and Willen, 2017](#)). More recent work studies the post-pandemic tightening cycle ([Adelino et al., 2025](#); [Andersen et al., 2026](#); [Bracke et al., 2024](#); [Elias et al., 2025](#)).⁶ These papers document important margins that mitigate the effect of higher rates

⁶[Kartashova and Zhou \(2025\)](#) analyze asymmetric effects of earlier rate increases and decreases in Canada.

on payments or spending, with variation across institutional settings.⁷ Our contribution is to combine loan-level data, a survey, and a letter RCT to document the adjustments that dampen payment passthrough, measure borrowers' beliefs and preparation, identify the causal effects of lender communication, and study selection into effective exposure to it.

Second, related literature studies spending responses to predictable changes in debt service or disposable income, such as mortgage amortization starts, final debt repayments, tax changes, or unemployment-benefit exhaustion (e.g., [Andersen et al., 2024](#); [Baker et al., 2026](#); [Ganong and Noel, 2019](#); [Jørring, 2024](#); [Kueng, 2018](#); [Stephens, 2008](#); [Scholnick, 2013](#)). A challenge in this literature is that standard event-study approaches often difference out any anticipatory behavior that occurs before the event window. Our setting and data allow us to observe several actions borrowers take ahead of the reset, and to measure directly whether borrowers are aware of and prepare for the future payment change. The closest paper in this respect is [Druehl et al. \(2022\)](#), who study spending responses to expected future changes in adjustable-rate mortgage payments in Denmark using naturally occurring information shocks from bank letters. We take a broader view of the actions borrowers can take before a rate reset, in a setting with a rich menu of refinancing and repayment options, and we combine bank data with direct survey evidence and a randomized field experiment.

Third, we build on the mortgage-refinancing literature, which often emphasizes sluggish refinancing in response to rate decreases (e.g., [Campbell, 2006](#); [Fisher et al., 2024](#); [Gianinazzi, 2023](#); [Keys et al., 2016](#)). Several frictions contribute, including inattention ([Andersen et al., 2020](#)), low financial literacy ([Bajo and Barbi, 2018](#)), distrust of lender motives ([Johnson et al., 2019](#)), and unawareness of mortgage terms ([Bucks and Pence, 2008](#)). Recent work by [Byrne et al. \(2023\)](#) shows that letters and reminders can increase refinancing among Irish borrowers with an incentive to refinance. Our findings complement this work—e.g., by establishing an important role of selection in engagement with lender communication—but suggest that attention and awareness can vary across environments. In our German setting, many borrowers are aware of the rate environment, actively prepare for the rate change, and their preparation intensifies when rates are higher. These findings are consistent with the anticipatory spending responses ob-

⁷For instance, [Bracke et al. \(2024\)](#) show that many UK borrowers extract home equity at refinancing to mitigate spending effects of rate increases (see also [Foulis et al., 2026](#)). Equity extraction is not available to German borrowers, who can instead lock in rates years ahead of the reset. [Elias et al. \(2025\)](#) find that in Australia, where rates pass fully and immediately into required payments, liquid buffers embedded in the mortgage itself mute the tightening effect on spending.

served for Danish borrowers ([Andersen et al., 2026](#)) as well as with Portuguese borrowers becoming more active after a rate increase ([Adelino et al., 2025](#)).

Fourth, we contribute to the broader literature using randomized information provision to study how households perceive and respond to the economic environment and policy (e.g., [Chopra et al., 2025](#); [Coibion et al., 2022](#)).⁸ Much of this work relies on survey experiments and measures effects on beliefs or stated intentions. We provide evidence from a field experiment that is linked to both survey measures and realized bank-data outcomes. The design allows us to study an often implicit first stage of information interventions: effective exposure to the communication, measured by respondents' recollection of having read the letter. We find that exposure is selected toward borrowers who are already better informed and more financially sophisticated. This selection may limit the ability of disclosure-based interventions to reach the borrowers who stand to benefit most—a margin that is mechanically suppressed in typical information-provision experiments in which participants are required to view the information.

2 Institutional setting

In Germany, residential mortgages typically have initial rate fixation periods of 5–15 years, while the terms over which the loans amortize usually span 25–35 years. At fixation end, borrowers have to refinance the residual amount at prevailing market rates, unless they choose full repayment. Refinancing can occur internally with the original lender or externally with a different bank.⁹ When refinancing, borrowers can freely reduce the loan balance and accelerate amortization by shortening the term. Increasing the balance or extending the term beyond the original contractual maturity is difficult.

Importantly for our analysis, German mortgages generally do not allow full prepayment during the fixation period, unlike fixed-rate mortgages in the US and Denmark.¹⁰ This identifies when borrowers face interest-rate risk: regardless of movements in market

⁸For reviews, see [Coibion and Gorodnichenko \(2026a\)](#), [Fuster and Zafar \(2023\)](#), and [Haaland et al. \(2023\)](#).

⁹While refinancing internally is free, switching lenders incurs monetary costs (though these are typically less than 0.5% of the new loan amount; [Lukas and Nöth, 2019](#)) and nonmonetary search costs. Moreover, online brokers facilitating mortgage comparisons are common.

¹⁰Prepayment during the fixation period is generally subject to a penalty that compensates the lender for lost interest income (*Vorfälligkeitsentschädigung*) and thus eliminates the monetary incentive for the borrower to refinance into a lower-rate loan before expiration.

rates, they continue to pay the fixed rate until the end of the fixation period, providing a clean setting to study how borrowers prepare for upcoming changes in mortgage costs.

There are situations in which prepayments without penalties during the fixation period are possible. First, about half the mortgage contracts at our partner bank include an annual partial prepayment right (*Sondertilgung*), typically allowing 5% (or rarely 10%) of the original loan amount to be repaid early per year. Second, loans can be prepaid in case the property is sold. Third, loans with an initial fixed-rate period of more than 10 years can be prepaid without penalty after 10 years, with six months' notice.

Borrowers can use a *forward mortgage* to lock in the subsequent fixed rate well ahead of expiration. For example, a borrower with a 10-year fixed-rate mortgage set to expire in 2025 could agree in 2022 on the fixed rate for a new 10-year fixation beginning in 2025, thus avoiding uncertainty about future market rates. This instrument allows us to study borrowers' anticipatory actions that affect future mortgage costs. Forwards can be originated from 3–5 years up to six months before the end of the fixation period. After that, internal refinancing occurs via a regular prolongation loan. Early rate fixation is priced directly through the interest rate to reflect prevailing market forward rates. For instance, fixing a 10-year interest rate one year in advance yields the current 10-year mortgage rate that the bank would offer for such a loan plus the one-year forward premium. Other than early lock-in and pricing, a forward does not differ from a regular prolongation loan.

Another instrument for financing property purchases in Germany is the home-savings loan (*Bausparen*). Contracts with immediate borrowing eliminate rate risk from origination. During an initial savings phase, borrowers build up savings and only pay interest on the loan. At expiration of the loan, typically after 10 years, accumulated savings and a new fixed-rate loan combine for repayment. There is no rate risk because the interest rate on the new loan is specified at origination of the initial loan. We hence exclude these loans are from our loan-level analysis.¹¹

The bank contacts borrowers through standardized letters. Each January, borrowers receive a mandatory statement of the outstanding balance and the interest, principal, and prepayments of the past year. Borrowers who have not objected to non-mandatory mail also receive a one-page reminder three, two, and one year before expiration. The reminder

¹¹Home-savings contracts also exist without immediate borrowing (*Bausparen ohne Vorfinanzierung*). While they can be used to mitigate rate risk, these products often purely serve to accumulate savings. We omit this product from the analysis because only 9% of borrowers in our sample have had it at some point and less than one-third of those might have been used to manage rate risk.

states the expiration date, notes that a forward loan can lock in current rates, invites the borrower to contact a bank advisor, and quotes no rate. Three to four months before expiration, as required by law, the bank informs all borrowers of the upcoming expiration and quotes an indicative rate at which it would continue the loan. Six weeks before expiration, borrowers who have not yet refinanced receive two prolongation offers, valid for 10 days, which they accept by ticking one and returning the signed letter. Other terms, such as a balance reduction or a higher amortization rate, require a meeting with a bank advisor. Borrowers who neither refinance nor repay move to a high variable rate.¹²

3 Patterns from bank data

In this section, we study borrower activity over the period 2018–2025. [Section 3.1](#) describes the bank data. [Section 3.2](#) explores time variation in refinancing. [Section 3.3](#) studies how the change in monthly payments upon refinancing evolves over time, and which borrower actions explain this evolution.

3.1 Bank data

Our partner is among the largest banks in Germany, offering a full suite of banking services nationwide. We draw on the bank’s proprietary data that provides detailed information on its mortgage customers. Our sample includes all clients with a mortgage that is outstanding at some point between December 2017 and October 2024 and was originated through the bank’s main brand and distribution channel.¹³ The sample contains 240,000 mortgagors for which we get updates until May 2025, when the dataset ends. In total, the data cover 396,000 regular mortgage loans originated between 1997 and May 2025, with each internal refinancing recorded as a separate loan.

Our loan-level analysis covers regular annuity mortgages; we exclude home-savings loans (*Bausparen*), held by 21% of borrowers and typically paired with a regular mortgage, because these are not subject to interest-rate risk: the rate on the initial loan and the post-expiration loan are both set at contract origination. Home-savings loans thereby

¹²This rate was 5% before 2022 and 7–8% thereafter. Below we show that less than 3% of borrowers remain on variable rates for extended periods.

¹³This distribution channel includes some mortgage brokers in addition to the bank’s retail business; it excludes independent asset managers that distribute customized bank products.

shield households from unexpected rate increases—a payment-change-mitigation channel at the household level that our loan-level estimates hence do not capture. Households with home-savings loans remain in the sample through their standard mortgages, and we control for home-savings holdings throughout. The main findings are similar when we exclude these households (not reported).¹⁴

For each regular mortgage, we observe origination characteristics and annual updates on outstanding balances. In cases of internal refinancing, we link the original loan to the subsequent loan. This allows us to precisely study the timing of refinancing decisions, combining information on the expiration date of the fixed rate period, the origination date of the new loan, and the date at which the new interest rate becomes effective (the latter two dates differ in cases of early refinancing). Conditional on refinancing, we can also observe changes in interest rates, monthly payments, balances, and fixation periods. For loans that are repaid without internal refinancing, we cannot determine whether repayment reflects external refinancing or the use of own funds.

Demographic information includes borrower age, gender (male, female, or couple), and zip code, as well as the bank’s internal measure of borrower default risk and a proxy for neighborhood quality. We also observe annual financial balances of clients’ deposit accounts, securities accounts, and other debt holdings with the bank, as well as information on clients’ meetings with bank advisors, including the meeting date and the stated purpose(s) of each meeting (e.g., mortgage financing, savings, or investments).

Sample characteristics Table 1 presents summary statistics for the main variables in the bank dataset. Panel A is on mortgage characteristics. The average borrower in the sample has €168,630 in mortgage debt at loan origination, an interest rate of 2.8%, monthly payments of €895, a fixation period of 11.6 years, and a total maturity of 20 years.¹⁵ On average, the residual balance at the end of the fixation period is €105,770. About 56% of borrowers are allowed to make annual partial prepayments (*Sondertilgungen*) during the fixation period, typically capped at 5% of the original balance (rarely up to 10%). 32%

¹⁴We also exclude public-subsidy loans from Germany’s state-owned development bank (KfW), which finance specific purposes—predominantly energy-efficient construction and renovation—and typically constitute a complementary tranche within a larger financing. 12% of mortgagors in our sample hold a KfW loan and these loans account for 6% of outstanding mortgage balances. Their refinancing process differs from that of regular mortgages and our data do not fully cover loan terms. Appendix B provides institutional details. Households with KfW loans remain in the sample through their standard mortgages, we control for KfW holdings, and our results are similar when we drop these households (not reported).

¹⁵This maturity may be shorter than the initial maturity, as the borrower may already have refinanced (in which case the remaining maturity is recorded). The same applies to the loan amount.

of borrowers have a forward loan at some point during our observation period, and 30% have a loan on a (partially or fully) rented out property.

Panel B shows borrower characteristics. The average borrower is 51 years old and has been a client of the bank for 13 years. The bank's estimate of borrowers' probability of default (PD) is heavily right-skewed.¹⁶ 41% of borrowers have a deposit account at the partner bank. Among these, the average balance is €27,350, while the median is much lower at €6,540. 7% of borrowers have a brokerage account at the bank. Borrowers meet with a bank advisor 0.36 times per year on average.

Sample representativeness We compare our bank sample to mortgagors in the Bundesbank's representative Panel on Household Finances (PHF). The latest PHF data are from 2021, so we report 2021 mortgage characteristics for the subsample with a mortgage at that time in our bank data; we only observe current customer characteristics otherwise.

Table A.1 shows summary statistics on both samples. The statistics indicate that average borrower age, mortgage balance, fixation period, and interest rate are quite close. The property value is larger in the bank sample, but needs to be interpreted with caution.¹⁷ The monthly payment is larger in the PHF, which might reflect that some borrowers have mortgage debt with multiple lenders, as indicated by the larger fraction of borrowers with more than one source of collateral in the PHF. Overall, mortgages in our bank sample are broadly representative of German mortgages.

3.2 Refinancing activity

The refinancing options for German mortgage borrowers depend on the length of their fixed-rate period. Borrowers with fixation periods of up to 10 years can take out a forward loan up to four years before the end of their contract, allowing them to lock in market rates in advance. However, they cannot prepay their loan before expiration without facing a penalty. If they do not act before the fixation ends, they must refinance at the prevailing market rate at expiration, or repay in full then. Borrowers with longer fixations (typically

¹⁶The PD measure is as of December 2017, i.e. prior to our sample period, for most borrowers. We also have access to a PD measure as of October 2024, which we use when December 2017 is not available (18% of the borrowers). We prefer the earlier PD vintage when available to reduce potential endogeneity concerns. Given the skew in the measure, in our analysis we separate borrowers into PD deciles.

¹⁷We report the value of the property if bought after 2015, which makes it difficult to compare the number to other mortgage characteristics, reported independent of the year of purchase. In the PHF, respondents self-report the purchase price of the property. In our bank data, we report the initial value of the collateral.

15 or 20 years) have more flexibility: they can prepay without penalty after 10 years, giving them a broader window to refinance or take out a forward loan, especially if market rates fall. If they do nothing, they continue paying the original fixed rate until the end of their fixation. Given the differential incentives to act, we analyze the refinancing choices of the two groups separately. Since almost two-thirds of loans have fixation periods up to 10 years, we put more focus on this group.

3.2.1 Refinancing activity of borrowers with fixation up to 10 years

We begin by examining refinancing decisions for borrowers with fixed-rate periods of up to 10 years. To exclude loans that would naturally amortize by the end of the fixation period, we restrict the sample to loans with a (projected) remaining balance of at least € 10,000. We additionally require that loans survive until at least two years prior to expiration to (partially) address left truncation.¹⁸

Figure 2 displays the cumulative share of borrowers who refinance their loan internally (either via a forward mortgage, or a regular prolongation), by year of expiration of the rate fixation. Several features stand out. First, less than 45% of borrowers refinance internally; the others either pay down the loan at expiration, or refinance with another provider.¹⁹ Second, a large share of internal refinancing happens over the last months prior to the expiration of the old loan, but there are also many borrowers that refinance their mortgage one year or more prior to expiration, via a forward contract.²⁰ Third, looking at the expiration vintages 2023 to 2025, there is an unusual increase in refinancing activity around three years (2025), two years (2024) and one year (2023) prior to expiration. This increase can be ascribed to borrowers refinancing via forward mortgages once market rates started increasing rapidly in early 2022, as we explain below. Fourth, in more recent expiration vintages, fewer borrowers refinance internally—consistent with the higher market rates leading borrowers to seek alternatives.

Table 2 further quantifies the distribution of refinancing outcomes by expiration year of the old loan. The share of internal refinancing decreased by about 6–8 pp (or over 15%) from pre- to post-2022 (column 3). Among internal refinancings, half take place at least six months before expiration via a forward loan (column 4) and half within the last

¹⁸For example, of all loans originated by the bank with expiration in 2019, we observe only those still outstanding in December 2017, i.e., loans that had not exited early due to property sales.

¹⁹As explained earlier, the data do not allow us to reliably distinguish the two cases.

²⁰Figure A.1 shows the precise distribution of internal refinance timing in the pooled data across years.

six months via a regular prolongation (column 5), although the ratio between the two (column 9) fluctuates substantially over time (see below). Only 15% of borrowers overall refinance internally within the last six weeks of loan fixation (column 6), that is, after the bank has sent its prolongation offers; roughly half of these appear to accept one of the bank’s default offers (column 7).²¹ Only 3% refinance after the old loan has expired (column 8), meaning they pay the high variable rate for some time. Thus, the share of borrowers that appear “passive” is low. However, we also note that the share of late prolongations substantially increases in 2024 and especially 2025, potentially suggesting more passive behavior when market rates are high. Our survey and RCT results will allow us to assess this possibility more directly.

The final column of [Table 2](#) shows particularly high forward activity for the 2022 and 2023 expiration vintages, while forward originations are lower for later vintages. [Figure 3](#) illustrates this pattern more directly by plotting monthly forward originations. As market interest rates rise sharply in the first half of 2022, forward-mortgage originations spike; as rates increase further in the second half of 2022 and remain elevated from 2023 onward, forward originations decline substantially. The surge in originations in Q1 and Q2 of 2022 cannot be explained by a reduction in the cost of these contracts as measured by forward spreads, nor do we observe an increase in the cost that could account for the subsequent decline in forward originations ([Figure A.2](#)).

Importantly, we do not take a stance on whether taking out a forward mortgage is ex-ante optimal at a given point in time, as the optimal option exercise depends on borrower-specific parameters—such as risk aversion and subjective beliefs about future rates—that we cannot observe.²² Nor do we mean to suggest that borrowers in 2022 were able to predict that long-term rates would rise further. Rather, the evolution of forward activity suggests that the interest rate environment is an important determinant of borrowers’ decisions, as we study more formally next.

Econometric analysis We now turn to a more formal analysis of how strongly and why interest rates matter for internal refinancing activity. To do so, we estimate Cox proportional hazard models, which posit that the hazard rate of loan i refinancing τ months

²¹We classify a borrower as likely accepting one of the bank’s offers if they did not alter the loan amount or the remaining term relative to what the bank would offer them; see [Section 3.3](#) for further discussion.

²²Optimal decisions can easily be characterized in some special cases. A risk-neutral borrower who believes that mortgage rates follow a random walk should never take out a forward mortgage if there is a positive forward premium. In contrast, a risk-averse borrower with random-walk beliefs who faces no forward premium should take out a forward mortgage as early as possible.

prior to expiration is given by

$$h(\tau|R_t^m, R_i, \mathbf{X}_{it}) = h_0(\tau) \cdot \exp(\beta_1(R_t^m - R_i) + \beta_2\Delta R_t^m + \Gamma\mathbf{X}_{it}). \quad (1)$$

We estimate the models separately for the time window during which borrowers can enter a forward contract (from 48 months to 6 months prior to the expiration of their fixation) and for the window during which they can enter a regular prolongation (the last 6 months). Our main variables of interest are $(R_t^m - R_i)$, which is the gap in pp between the market rate on a 10-year mortgage originated by the partner bank in calendar month t and the rate a borrower pays on their old loan, and ΔR_t^m , which is the three-month change in the market rate. The vector $\mathbf{X}_{it}$ covers borrower and loan characteristics, including the loan size, that could also influence refinancing decisions. The baseline hazard $h_0(\tau)$ is unrestricted, and we allow it to vary by mortgage origination year in some specifications to control for potential unobservable effects that differ across origination vintages. Standard errors are two-way clustered by loan and by calendar month.

The coefficient β_1 indicates how sensitive borrowers' refinancing actions are to rate differences, and β_2 whether recent changes in market rates affect borrower behavior. In some specifications, we include year-month fixed effects that absorb any aggregate effects that could influence refinancing (including but not limited to market interest rates). β_1 is then identified from variation in R_i , meaning it reflects only variation in the rate that different borrowers pay on their original mortgage. A significant β_1 coefficient in these specifications suggests that the original rate matters for refinancing choices, perhaps by acting as a reference rate. Also note that β_2 is not identified in specifications with year-month fixed effects.

Panel A of [Table 3](#) shows results for the forward window. Column 1 indicates that borrowers who pay a rate below the current market rate are less likely to take out a forward with the bank—per one-percentage-point increase in the gap, the hazard of taking out a forward decreases by about 22%.²³ Column 2 shows that forward-taking hazards are higher when market interest rates have recently increased (which is the case in particular in 2022), consistent with borrowers rushing to lock in a rate at such times, perhaps fearing that market rates will increase further. The first two findings may appear to contradict each other, because increases in market rates, which we find to increase forward-taking,

²³The table reports hazard ratios minus one, meaning proportional changes in the refinancing hazard associated with a one-unit increase in a regressor.

lead to larger rate gaps, which in turn reduce it. Column 3 shows that when we control for both of them jointly, both coefficients increase in magnitude, corroborating these countervailing effects.²⁴ Column 4 shows that effects are stable when we add a rich set of control variables.

In column 5, we add the year-month fixed effects, so that all aggregate variation is absorbed. The coefficient on the rate gap of -0.2 therefore effectively reflects that borrowers with a 1 pp higher rate on their existing loan (R_i) have a 20% higher propensity to take out a forward (in relative terms). Column 6 additionally allows for origination-vintage-specific baseline hazards (akin to origination-year effects), which further absorbs some variation in R_i across origination times; the fact that the coefficient remains strongly significant supports the conclusion that borrowers are highly sensitive to their existing rate when deciding whether to take out a forward at a given point in time.²⁵

In panel B of [Table 3](#), we report results of the same specifications estimated over the prolongation window only. There, the directional effect of the rate gap on internal refinancing propensities is the same—meaning borrowers are less likely to refinance internally when rates are higher compared to what they pay on their old loan—but the magnitude is smaller. Recent changes in market rates do not matter for prolongations.

Heterogeneity The results above indicate that refinancing decisions are highly sensitive to the evolution of market interest rates. It is interesting to ask whether there are specific borrower types for whom this sensitivity is stronger. In particular, one hypothesis may be that the spike in activity in 2022 is driven by the bank reaching out to its customers, rather than customers being informed and taking action. While we cannot directly observe bank outreach, we can distinguish between customers that have a more vs. less extensive relationship with the bank, based on whether they hold other products and services at the bank. Those customers that do hold other products have a higher overall propensity to refinance internally. However, they are not differentially sensitive to interest rates, as shown in [Figure A.3](#).

The figure also reports two other subsample analyses: first, based on whether the underlying home is owner-occupied or not. This split is interesting because one may sus-

²⁴The magnitude of the estimated β_2 may appear very large—a one-percentage-point increase in the market rate increases the hazard by 223% (Column 3) or 238% (Column 4). Note though that these are changes relative to a low baseline hazard—about 0.5% per month during the forward window.

²⁵In Appendix [Table A.2](#), we check whether the effect of R_i might reflect unobservable characteristics associated with borrowers obtaining different rates at origination, by replacing R_i with the market rate at the time of origination, R_{orig}^m . The coefficient remains strongly significant and of similar magnitude as in [Table 3](#).

pect that investment-property borrowers could be more sophisticated; on the other hand, their interest payments are tax-deductible while this is not the case for owner-occupied homes. Maybe due to these offsetting effects, patterns are very similar for the two groups of loans. Second, we cut the sample based on the remaining loan balance. There, we see graphically that loans with a higher balance have much higher internal-refinancing propensities and are more likely to take out forward loans in 2022, consistent with the borrowers having stronger incentives to pay attention.²⁶

Finally, we turn to a simpler approach to study heterogeneity in refinancing outcomes along various dimensions. Table A.3 reports regressions of the different outcomes seen in Table 2 on borrower and loan characteristics, controlling for expiration-quarter-by-location fixed effects. Results confirm that the existing loan’s interest rate and residual balance correlate positively with the probability of refinancing internally, especially via a forward. Borrowers with a higher probability of default are more likely to refinance internally, and tend to do so closer to expiration of the old loan, or even post-expiration. Thus, riskier borrowers appear more passive.²⁷ Similarly, single borrowers appear more passive than couples, which are less likely to refinance internally and, conditional on doing so, take action earlier. The depth of borrowers’ relationship with the bank matters—borrowers with a checking/savings account or other mortgage debt at the bank are more likely to refinance internally and tend to do so earlier, while having a brokerage account is associated with higher tendency to pay off or refinance externally. Finally, loans on rented-out properties are less likely to be refinanced internally, especially via forwards.

In sum, the refinancing patterns of borrowers with fixations up to 10 years suggest that most of them actively engage with future loan rate resets, as only a small share of loans are refinanced “passively” at the end of their original fixation. The rate environment strongly affects behavior—borrowers are less likely to refinance internally when market rates are higher, and forward activity shoots up when rates rise rapidly. At the same time, a minority of borrowers may be inattentive and passive, and this is correlated with some observable characteristics such as borrower risk.

²⁶In the regressions reported at the bottom of Figure A.3, the coefficient on ΔR_t^m is only slightly larger for large loans; this is because the baseline hazard for small loans is much lower, such that relative increases are of similar size.

²⁷This could partly reflect that it may be more difficult for risky borrowers to obtain a loan from another lender, though note that such borrowers also have a higher propensity to passively take the bank offer or even wait post-expiry conditional on refinancing internally.

3.2.2 Refinancing activity of borrowers with fixation above 10 years

While the results above suggest many borrowers are active leading up to loan expiration, quantifying awareness and activity prior to expiration is difficult because inaction can be optimal. For instance, a borrower might observe rising rates in 2022 but forgo a forward loan because its price reflects the expected path of rates. We next examine a situation in which the gain from acting can be unambiguous: refinancing without prepayment penalties. Borrowers with fixation periods longer than 10 years can freely refinance after a 10-year threshold. Thus, a borrower with a 15-year fixation period reaching that threshold in the low-rate period of 2019–21 could switch to a much lower mortgage rate five years ahead of loan expiration. After rates rise in 2022, the incentive to refinance early diminishes.²⁸ A further feature that makes this group informative is that the bank’s expiration-related communication, from the reminders three years ahead to the prolongation offers, starts only years after the threshold.²⁹

Panel A of Figure 4 shows the cumulative share of loans repaid, by year in which the loan can be prepaid penalty-free, for the 29,600 loans with fixation periods of 15 years or longer and a projected residual balance of at least €10,000. Repayment comprises internal refinancing, external refinancing, repayment from own funds, and property sales, which we cannot separate. Two results stand out. First, for the 2019–21 vintages that can refinance at much lower rates, around 70% repay in the year they reach the threshold. Second, the share repaid falls strongly for loans that arrive at the threshold in or after 2022, as the gap between the market rate and the existing rate closes and then reverses.

Panel B shows the cumulative share of loans refinanced internally, by month relative to the end of the prepayment-penalty period. For the 2019–21 vintages, about 30% of loans refinance internally. Strikingly, most of these refinancings, about 85%, are arranged prior to the threshold through forwards. The rate surge largely ends early lock-in for all vintages: the curves for loans reaching the threshold in 2023 and 2024 flatten in 2022. Almost all internal refinancings of these two vintages are hence forwards secured before mid-2022; they account for about one third of the loans repaid in 2023 and 2024 in Panel

²⁸Loans on long fixation periods becoming penalty-free in 2019–21 carry interest rates well above prevailing market rates of 1–2% at that time (the average rate is 5.0%, 4.5%, and 4.4% for loans reaching the penalty-free threshold in 2019, 2020, and 2021, respectively). In contrast, most loans reaching the threshold in 2022–24 are on rates below the then-prevailing market rate of about 4% (average rate is 3.6%, 3.3%, and 3.2% for loans reaching the penalty-free threshold in 2022, 2023, and 2024, respectively).

²⁹While we cannot rule out that bank advisors contact borrowers ahead of the threshold, the bank indicates that such contacts are rare.

A. Overall, the results corroborate that many borrowers are aware of the rate environment and their options. They act when the gain is large, even without expiration-related bank communication. Once the gain disappears, their option goes unexercised, so the rate environment determines behavior in both directions.

3.3 Borrower actions and impact on monthly payments

This section asks how much of the rise in market rates reaches monthly payments. Payments are the cash-flow channel through which higher rates reach spending, and the object that assessments of household financial stress track. Existing work maps market rates to payments through the contract: it reprices the outstanding balance at the new rate, holding balance and term fixed (e.g., [Auclert, 2019](#); [Cloyne et al., 2020](#)). In our setting, however, borrowers choose when to fix the new rate, how much to repay, and how fast to amortize. We compare the payment change the contract implies to the realized change among borrowers who refinance internally, for loans with fixation periods up to 10 years, and attribute the difference to these three choices.

Payment changes Panel A of [Figure 5](#) reports the distribution of *realized* monthly payment changes upon refinancing, calculated as the difference between the monthly payment on the new contract and that on the old contract, by quarter of expiration of the fixed rate period. Despite interest rates post-2021 being substantially above their level 10 years before ([Figure 1](#)), we observe no significant payment increase for most borrowers. Specifically, the median monthly payment change is €0 for loans expiring in 2022 and 2023, €11 for loans expiring in 2024, and €26 for loans expiring in 2025 (through March).

Actual vs. counterfactual payments We measure the contract-implied payment change with a passive counterfactual: the borrower makes no extra payments during the fixation period and, at expiration, rolls over the residual balance on the bank’s standard prolongation offer, keeping the residual term and the fixation length of the expiring loan. This corresponds to a borrower who waits for the two offers the bank mails six weeks before expiration and returns one signed.³⁰ The bank prices these offers off loan characteristics (amount, term, fixation length) and market conditions, not off borrower characteristics. We can therefore estimate the offered rate with a random forest model trained on observed prolongations. [Appendix C](#) details the estimation and shows that predicted rates

³⁰One offer fixes the rate for the residual term, the other for a shorter period. Since preferences for fixation lengths are plausibly stable, we assume borrowers select the fixation of the expiring loan when available.

closely match rates on realized passive rollovers.

Panel B of [Figure 5](#) plots the average payment change under the passive counterfactual (blue) and the average realized change (red) by expiration quarter. For loans expiring from mid-2022 onward, the counterfactual implies an increase of €87 per month, or €1,044 per year. The realized increase is €23 per month, or €276 per year. Borrowers who refinance internally thus offset about three quarters of the contract-implied increase. Payments also move little in the low-rate period 2018–21, when the counterfactual implies decreases of €70–100 per month, consistent with monthly payment targeting ([Argyle et al., 2020](#)).³¹ Borrowers thus smooth payments in both directions.

Decomposing the impact on payments Realized payments can deviate from the passive counterfactual through three choices. The rate differs when borrowers fix it ahead of expiration through a forward, choose a different fixation length, or negotiate terms in a meeting with an advisor. The term changes when borrowers adjust the maturity at refinancing. The balance decreases when borrowers make partial prepayments during the fixation period, where permitted, or repay part of the loan at refinancing.

To attribute the deviation to each choice, we vary one component at a time. Let $f(L', r', T')$ denote the passive counterfactual payment, where L' is the residual balance, r' the estimated passive rate, and T' the residual term, and let $f(L_1, r_1, T_1)$ denote the realized payment, where L_1 , r_1 , and T_1 are the chosen balance, rate, and term. We define

$$\begin{aligned} \text{Rate effect} &= f(L', r_1, T') - f(L', r', T') \\ \text{Term effect} &= f(L', r', T_1) - f(L', r', T') \\ \text{Amount effect} &= f(L_1, r', T') - f(L', r', T') \end{aligned} \tag{2}$$

[Figure 6](#) plots the average of each effect by expiration quarter, together with the full effect.³² The term effect is positive throughout: borrowers amortize faster than under the passive offer, adding €100–150 per month before 2022 and about €50 for loans expiring from 2023 onward. Borrowers thus scale back faster amortization when rates rise. The rate effect is close to zero before 2022, turns negative in 2022, and reaches about –€60

³¹This partly reflects an increase in loan amortization. We explore the drivers of this pattern in separate work.

³²The decomposition is not exact. Because the payment is nonlinear in its arguments, the three effects do not sum to the full effect, and each depends on the values at which the other two components are held. For instance, the rate effect could instead be computed as $f(L_1, r_1, T_1) - f(L_1, r', T_1)$. Interactions between choices, such as a shorter fixation lowering the rate, are assigned to the component that changes, here the rate. The alternatives differ quantitatively but yield the same qualitative conclusions.

per month for loans expiring in early 2023: these borrowers fixed their rate before the full rise in market rates, in line with the surge in forward originations in 2022 (Figure 3). The effect then fades to about $-\text{€}15$ for loans expiring in early 2025. The amount effect lowers payments by $\text{€}25\text{--}50$ per month throughout and by slightly more after 2022.

Figure 7 shows more directly how different loan characteristics at refinancing change with the level of rates. Panel A illustrates that once rates increase, borrowers become much less likely to shorten the term of their mortgage by accelerating amortization. Panel B shows the evolution of the share of borrowers that reduces the loan balance by 5% or more upon internal refinancing. This share is broadly stable over time, between 12% and 16%. Finally, Panel C shows that with the rate increase, the average fixation period of the refinancing loan drops substantially, from over 7.5 years to around 5.5 years. In Table A.4, we present complementary regression evidence that studies the dependence of these actions on the change in the market rate between the time a borrower takes out their original loan and the refinancing loan, with and without additional controls and month-of-refinancing fixed effects.³³

To summarize, borrowers who refinance internally offset about three quarters of the payment increase the contract implies, and the median borrower faces no increase. The three margins behind this offset differ in kind. The rate effect is hedging: borrowers who fixed the rate before mid-2022 are shielded from the increase, a stock of early lock-ins that runs off for later vintages. The amount effect is a balance-sheet reallocation: reducing the balance moves liquid assets into home equity, leaving net worth unchanged but liquidity lower. The term effect is a saving decision, determining the speed of repayment. Relative to the passive offer, borrowers carry a smaller balance into the new contract and repay it faster. Compared with low-rate refinancing, they reduce the balance slightly more at the reset but repay the remainder more slowly, which smooths the payment and leaves the higher rate to be paid on the remaining balance for longer.

Three conclusions follow. First, how much of a rate increase reaches the payment is a choice borrowers make, not a term of the contract. Second, the small passthrough to payments does not mean transmission is small. Only early lock-in blunts the cash-flow

³³For term adjustments, we robustly find that a more positive market rate change is associated with less term shortening. The association with balance reductions is zero without controls, but becomes significantly positive once controls are added. For rate fixation periods, the sign on the rate-change coefficient is unconditionally negative, in line with the graphical evidence, but turns positive once we include month-of-refinancing fixed effects, suggesting that the shorter fixation chosen after rates increased may reflect aggregate factors rather than individual circumstances.

channel, and it fades. Balance reductions draw down liquid assets, which borrowers build ahead of the reset or possibly rebuild after it, and slower amortization keeps leverage higher for longer. Higher rates thus reach these households through liquidity, saving, and leverage rather than through the payment. Third, repricing outstanding balances at market rates overstates payment burdens. The offset itself is evidence of balance-sheet room rather than strain: reducing the balance requires liquid assets, and repaying faster than the offer requires income to spare. Strain, where it exists, concentrates in the upper quartile, whose payments rise by €50 per month or more.

3.4 Completeness and selection in single-lender data

Data from a single lender come with three limitations. The first is that we do not observe what borrowers do when not refinancing internally at expiration. However, both alternatives, repayment in full and refinancing with a different lender, are active choices, so this gap does not affect our conclusion that borrowers actively manage the rate reset.

The second is possibly incomplete records of a household's mortgage debt. This concern is limited, as a regular mortgage on a property comes from a single lender. Mortgage debt spread across lenders hence requires several properties, and few borrowers pledge more than one: only 4% in the bank data and 9% in the representative PHF do ([Table A.1](#)).

The third is that selection into staying with the bank could distort the behavior we document. Borrowers who stay may be more attached to the bank. Indeed, those with a checking or savings account at the bank are 5.7 pp, or 13%, more likely to refinance internally ([Table A.3](#)). More attached borrowers might be less price-sensitive, or better informed because they meet advisors more often.

Our evidence suggests that selection into staying does not materially affect behavior. Recall from [Section 3.2.1](#) that borrowers who are likely more attached to the bank, as proxied by them holding products with the bank beyond mortgage-related ones (i.e. checking/savings account, insurance, or a securities account; 53% of sample) do not differ in terms of how sensitive their forward-origination activity is to interest rates. In addition, [Figure A.4](#) repeats the decomposition of payment changes following refinancing ([Figure 6](#)) for both groups. The patterns are similar: term reductions fade and rate effects turn negative for loans expiring after 2022 in both groups, and the overall mitigation of payment increases is of equal size.

It might still be the case that borrowers who stay with the bank are different from those who switch. While we cannot compare those who stay with those who leave, we can compare them to borrowers who refinance a mortgage held externally into our bank, which we observe for forward loans. [Figure A.5](#) compares these switchers to internal forward borrowers by origination quarter. The forward period, that is, how far ahead of expiration borrowers lock in the new rate, is similar in both groups throughout, including the surge in early lock-ins in 2022. External forward borrowers amortize more slowly in every quarter, but both groups slow amortization once rates rise.

4 Survey

The bank data show what borrowers do at the bank. They do not show what borrowers know, why they act, or what they do elsewhere. For instance, activity ahead of expiration is consistent with informed borrowers but also with borrowers who act once the bank prompts them, and saving, information acquisition, and search across lenders are mostly unobservable in the bank data. We therefore survey a subset of the bank's mortgagors. [Section 4.1](#) describes the design of the survey. [Section 4.2](#) discusses survey administration, sample selection and representativeness, and reliability of responses. [Section 4.3](#) focuses on respondents' beliefs about mortgage rates and preparation for rate changes. [Section 4.4](#) studies respondent behavior in a hypothetical refinancing situation.

4.1 Survey design

The survey consists of three sections: a pre-vignette section on beliefs about mortgage rates and preparing for the possibility of higher future rates; a vignette section in which respondents confront a hypothetical refinancing situation; and a post-vignette section on economic beliefs and preferences, household finances, and demographics. [Appendix D](#) reproduces the questions, translated from German.

Pre-vignette section The survey opens with beliefs about mortgage rates: the perceived change over the past three years and its effect on the household's finances, point estimates of the current rate and of the rate in two and five years, and the perceived likelihood of a rate 2 pp above the two-year estimate. Placing these questions first lets us measure letter effects on beliefs ([Section 5](#)) before the survey itself can prime respondents.

Respondents then report whether they have prepared for the end of current their fixation period, plan to, or do not prepare, and explain how and why in an open-text field. Open text captures what is on top of respondents' minds and avoids priming on a list of response options (e.g., [Haaland et al., 2025](#)). We also ask about the timing of preparation and expected monthly payment changes.

The section closes by asking whether respondents have received and read a letter from their bank about rising mortgage rates over the past two months (not received; received but not read; received and read; do not remember). The mention of rising rates identifies our letter without describing its content, which could shape the responses that follow. The question is the first stage of the letter RCT: it measures how many and which recipients read the letter, which separates non-exposure from ex-ante informedness when the letter leaves beliefs unchanged, and it scales intention-to-treat effects to the effect of reading.

Vignette section The vignette, or “strategic survey,” is a between-subject experiment; we follow [Andre et al. \(2022\)](#), [Fuster et al. \(2021\)](#), and [Schnorpfeil et al. \(2025\)](#), among others. In the baseline (vignette 1), respondents imagine a €100,000 mortgage balance whose fixation ends in one year, with the rate expected to rise from 1.5% to 3.5%. They rate how likely they would be to (i) meet an advisor at their current lender, (ii) compare offers from other lenders, (iii) reduce the balance through prepayments or a partial repayment at expiration, (iv) lock in the future rate with a forward loan, (v) reduce spending, and (vi) increase income, and they choose a fixation period for the new loan. Structured follow-ups elicit the reasons for (i) and (iii).

Two variants change one feature of the baseline. Vignette 2 moves the fixation end from one year to three years. Its action questions refer to the next 12 months—except for balance reductions, which can occur at any time until expiration—so that respondents in vignettes 1 and 2 choose over the same horizon. Vignette 3 keeps the one-year horizon but sets the current rate to 3%, so the increase is 0.5 pp instead of 2 pp. The post-reset rate is 3.5% in all variants, close to the rate at the time of the survey ([Figure 1](#)). The variants thus differ in the borrower's situation but not in the macroeconomic environment.

There are two main reasons for integrating the vignette into the survey. First, the hypothetical scenario applies to everyone, so all respondents make hypothetical choices and convey the reasoning behind them. In reality, instead, mortgage setups are very heterogeneous. This heterogeneity means many borrowers do not face each decision, such as those who are close to repayment. The second, related reason is that the hypothetical loan mitigates idiosyncratic factors that contribute to noise, such as the size of the loan. The

structured questions about the hypothetical refinancing situation also complement our open-ended question about the preparation for borrowers' real-world refinancing.

Post-vignette section The final section elicits additional respondent characteristics, beliefs, and preferences: risk tolerance, which governs exposure to interest-rate risk (e.g., [Campbell and Cocco, 2003](#)); debt aversion; trust in the lender (e.g., [Johnson et al., 2019](#)); whether higher rates have raised saving; self-assessed financial knowledge; income, discretionary income, real-estate and financial holdings, and education; and the perceived effect of an unexpected rate increase on inflation, growth, stock prices, and house prices, which serves as a further proxy for financial literacy.

4.2 Survey administration, sample selection, and representativeness

Survey administration We run the survey in partnership with the same German bank, allowing us to combine bank data, letter RCT, and survey for the same set of mortgage customers. In November 2024, the bank sent a short email to around 46,000 customers, inviting them to participate in a survey on mortgages run by Goethe University Frankfurt. The bank sent a reminder email 1.5 weeks after the initial invitation. The survey remained open for 2.5 weeks. Overall, 1,968 customers completed the survey. The response rate is 4.3%, which is slightly above other surveys of the bank. The median response time is 14.2 minutes. Participants received a €10 online-shopping voucher for survey completion.

Sample selection Among those who complete the survey, we omit respondents who take less than six minutes (1.6% of the sample) or more than 120 minutes (0.6%). This step leaves 1,924 respondents in the baseline survey sample. The estimation sample varies across analyses, mainly with whether we restrict it to respondents who hold an active regular mortgage in the bank data. 1,595 respondents do; the others hold only home-savings (*Bausparen*) or public-subsidy (KfW) loans, or their regular mortgage recently ended. We note this and further sample restrictions in the individual exhibits.

Sample representativeness Mortgages in our survey sample are broadly comparable to representative mortgages, but respondents are positively selected in terms of education and income and have a closer relation to the partner bank. [Table A.1](#) compares survey respondents to mortgagors in the Bundesbank's PHF and to the average mortgagor in the bank data ([Section 3.1](#)). Relative to PHF mortgagors, respondents are of roughly similar age, but better educated and with higher household net income. Mortgage characteristics are comparable; for example, the current amount is slightly higher and the payment

slightly smaller in our sample. Relative to all mortgagors of the bank, those invited to the survey and particularly those who complete it are younger and more likely to be male. Their mortgages are larger (current amount: €176k vs. €148k), though the collateral value is smaller. Most noticeably, 61% of respondents also have a checking account with the bank, whereas only 41% of all mortgagors do.

We explore how selection into the survey affects the response to changes in mortgage rates. [Figure A.6](#) repeats the decomposition of payment changes following refinancing for survey respondents. Because few refinance internally in a given quarter, we show the decomposition by year of expiration. Respondents slow down debt repayment and lock in rates early for loans expiring after 2022, so the overall mitigation of payment increases in the survey sample is comparable to the full sample ([Figure 6](#)).

Importantly, receiving the letter as part of our RCT does not affect the propensity to participate in the survey—see [Table A.5](#) for regressions of participation indicators on letter dummies with and without additional controls.

Reliability of survey responses Choices we observe in the bank data predict choices in the hypothetical refinancing, which supports the reliability of responses ([Dutz et al., 2026](#); [Schnorpfeil et al., 2023](#)). In [Table A.6](#), respondents who have made prepayments, that is, voluntary extra payments, are more likely to state they would likely prepay in the vignette; forward and home-savings (*Bausparen*) loans in the bank data predict a forward-loan preference in the vignette; and forward loans—which imply fixing rates early—and rate-fixation periods in the bank data explain the fixation choice in the vignette.

Regression controls The survey analyses share a common set of controls. Controls from the survey include the perceived past-three-year change in the mortgage rate (five-point scale); point estimates of the rate today and in two years (1% tails winsorized); uncertainty about the rate in two years (five-point scale); self-reported risk tolerance, debt aversion, trust in banks, and financial literacy (all on a five-point scale); buckets of household income net of regular expenses; and indicators for university education, non-owner-occupied property, traditional savings products (e.g., term deposit), life insurance, stocks, crypto, and completing the survey after the bank sent a reminder email. Bank controls include log total mortgage debt outstanding; the mortgage rate and fixation period, averaged across loans for borrowers with multiple loans; borrower age; the probability-of-default decile; and indicators for a prepayment right, multiple mortgage loans, a home-savings plan (*Bausparen*), a public-subsidy (KfW) loan, a checking or savings account, meeting an advisor (past 12 months), and an online-banking login (past

month). Further controls enter as fixed effects: mortgage-product status,³⁴ account type (female, male, couple), neighborhood quality (nine categories), federal state of residence, and letter group. We note any deviations from this specification in the individual exhibits.

4.3 Beliefs about interest rates and preparation for rate changes

Interest rate perceptions and expectations While the bank-data analyses suggest many borrowers are aware of the rate environment and active, the survey allows us to test for rate awareness directly. [Figure 8](#) reports statistics for respondents’ beliefs about mortgage rates.³⁵ The first stacked bar from the left shows beliefs about how mortgage rates have changed relative to three years ago (in 2021), which we elicit on a five-point scale ranging from “today’s rates are much lower” to “much higher.” 83% of respondents believe mortgage rates are somewhat or much higher than they were three years ago. Less than 10% of respondents perceive rates to have decreased.

The second bar documents the distribution of respondents’ point estimates of the current rate on a new mortgage with a 10-year fixation period. Actual rates at the time of the survey in November/December 2024 are around 3.5%, and the median perceived rate in our survey is 3.5%. Dispersion is limited: 4% of respondents perceive rates to be below 2.5%, and 8% perceive rates of 5% or higher. The dispersion of mortgage rate “nowcasts” is considerably tighter than for mortgagors in the US, based on data from the NY Fed Survey of Consumer Expectations.³⁶

The third and fourth bars report mortgage-rate expectations. We ask for point predictions of rates in two and five years, which we contrast with perceived current rates, respectively. Expected rates are close to the perceived rate on average. 14% of respondents expect rates in two years to deviate more than 1 pp from their current perception. The five-year median expectation is similar to the median perceived current rate, but dispersion in the forecast increases. In sum, most respondents are aware of the recent increase

³⁴For the 17% of survey respondents without an active regular mortgage, we do not observe mortgage terms and set them to zero. The mortgage-status fixed effects (no active mortgage product, regular mortgage, public-subsidy loan only, home-savings loan only) absorb these groups, so coefficients on mortgage terms are identified from the 1,595 respondents with an active regular mortgage.

³⁵We study beliefs of all respondents, including letter recipients, because the letter treatment does not have significant effects on interest-rate beliefs on average, as discussed in [Section 5.2](#) below.

³⁶This point is illustrated in [Figure A.7](#), which shows that the gap between the 10th and 90th percentile of mortgage rate nowcasts among mortgagors in the US in 2024 is 2.8 pp; it averages 2.5 pp over 2014–24 and never falls below 2 pp. In contrast, in our survey data, the corresponding gap is 1.6 pp.

in rates, have accurate perceptions of current mortgages rates, and expect rates to remain at a similar level; however, some heterogeneity in awareness exists.

Preparation for rate reset We next study whether and how mortgagors prepare for the future interest-rate reset. Our objective is twofold. First, we speak to the question of what is behind the relatively large fraction of late/passive refinancing when rates are high (shown in [Section 3.2](#)). Second, we shed light on margins of adjustment beyond those observable in the bank data. To focus on respondents subject to rate risk, we require at least €10,000 expected mortgage debt at reset. We develop a coding scheme that classifies the open-text responses into nine categories and apply the scheme using both a human coder and a large language model (GPT-4o). 79% of responses can be classified into at least one category, in line with existing work (e.g., [Andre et al., 2026a,b](#)).

[Figure 9](#) plots the categorized open-text explanations of whether and how mortgagors prepare for the rate reset. 18% of borrowers select that they do not prepare (top-left panel). Among them, 23% explain they will actively pay off (prepay) their loan, which could also be interpreted as a preparation. 23% say they repay without mentioning active repayment.³⁷ That is, nearly half of respondents who do not prepare will repay their loan at the end of the fixation period. 7% do not prepare because they perceive the outstanding loan to be small. 32% state they take no action, at least until they are close to the reset.

44% of borrowers plan to prepare in the future (top right panel). Among them, the most common form of preparation is information acquisition (64% of responses), which for many borrowers entails searching for the best loan offers or consulting a mortgage advisor. Apart from that, some respondents consider fully or partially paying off the loan (10% respectively), locking in the future mortgage rate ahead of the rate reset using a forward loan or home-savings loan (*Bausparen*) (13%),³⁸ or increasing savings (20%).

38% of borrowers state they have prepared already (bottom left panel). There is more heterogeneity in how they prepare. A significant fraction has locked in their rate (38%), which is consistent with the surge in demand for forward loans once rates started to increase ([Figure 3](#)). Many borrowers fully (26%) or partially (12%) pay off their mortgage, which again aligns with actual behavior, such as the increased likelihood of loan-balance reductions ([Table A.4](#)). 17% mention they have increased savings.

³⁷Many of these answers are ambiguous (such as “loan will be fully repaid”), so they could indicate active or natural repayment. Natural repayment should not happen given the minimum-debt restriction we impose. Some responses indicate natural repayment nonetheless, but the overall fraction is small (<5%).

³⁸A reference to home-savings loans often entails rate-risk mitigation at loan origination, not a dynamic response to a change in the rate environment.

Finally, we plot the frequency of categorized responses for borrowers pooled across types of preparation (bottom right panel), which illustrates the heterogeneity in how borrowers prepare: they rely on (partial) repayments, locking in interest rates early, increased savings, and acquiring information. Overall, the open-text explanations on preparing for higher future interest rates indicate most borrowers actively prepare, often taking steps to mitigate the payment impact of higher rates. These data thus suggest a larger fraction of late refinancing in the high-rate environment observed in the bank data need not reflect greater inactivity. Instead, passive refinancing might rather be due to a lack of attractive options (e.g., locking in high rates using a forward). Moreover, acquiring information and saving are additional margins of preparation. Nonetheless, a non-negligible minority indicates either no preparation, or preparation through information acquisition, which while possibly meaningful does not constitute active financial preparation.

Heterogeneity What determines whether borrowers prepare for the rate reset and beliefs about mortgage rates? Column 1 of [Table 4](#) shows results of regressing an indicator for whether borrowers prepare—by selling the property, partially or fully repaying, locking in rates, or saving—on a large set of correlates. Most noteworthy, awareness of rate increases between 2021–24 and higher expected rates over the next two years predict preparing. While most borrowers are aware of the rate environment and prepare for the rate reset, some borrowers might not prepare because they are unaware of rate increases or forecast lower rates. We also find proxies for financial literacy—including whether respondents associate unexpected rate increases with decreases in (property) prices—and income predict preparation. When we omit respondents who state they fully prepay from the estimation sample—which is naturally less common for large mortgages—the outstanding balance has no predictive power (Column 2). Patterns are more pronounced for borrowers closer to the end of the fixation period (Column 3).

Columns 4 and 5 report determinants of rate beliefs. Importantly, respondents with a larger outstanding balance are more likely to perceive recent rate increases and forecast lower rates. Distance to reset does not predict beliefs. Loan size correlating with informedness about recent rate increases is consistent with the literature on endogenous information acquisition: borrowers more exposed to interest rates pay more attention (e.g., [Link et al., 2025](#); [Maćkowiak et al., 2023](#)). Moreover, financial literacy correlates with perceived past rate increases and expected rate decreases.³⁹

³⁹The Q4 2024 ECB Survey of Professional Forecasters expects two-year decreases in deposit rates. The easing cycle was priced into the long end already, so the two-year-ahead path for mortgage rates was flat.

4.4 Hypothetical refinancing situation

Refinancing choices In the baseline scenario (vignette 1), borrowers have a hypothetical €100,000 mortgage with an interest rate of 1.5% that resets in one year to a rate of 3.5%. In this scenario, 71% of respondents state they would “very likely” compare loan offers from multiple lenders, and 16% would “rather likely” compare offers. Relatedly, 78% would likely meet an advisor from their current lender. Prepayments of any form are common as well (71% would likely prepay), consistent with their prominence in the bank data. We find similar likelihoods of reducing expenses, raising income, and taking out a forward loan, of around 47%. Respondents would hence consider various actions to prepare for the higher-rate reset. See [Figure A.8](#) for details.

Varying the refinancing situation We study between-respondent effects of varying the situation. To ease the interpretation of the effects, the outcomes are indicators equal to one if an action is very or rather likely, and zero otherwise. Because we randomize vignette assignment, controls help with precision but have no material effect on the estimates. [Table 5](#) shows the results. First, we study how increasing the distance to reset from one year (vignette 1) to three years (vignette 2) affects the likelihood of action over the next 12 months. A reset in three years reduces the likelihood of consulting a bank advisor by 15 pp (20%) and searching for loan offers by 6 pp (7%). Respondents are also 5 pp (or 11%) less likely to reduce expenses and 7 pp (15%) less likely to try to raise income. However, a later rate reset does not affect the likelihood of prepayments, for which we do not restrict the choice horizon, and of taking out a forward. The results indicate respondents become more active as they approach the rate reset.

Second, varying the size of the hypothetical increase in interest rates at reset matters for choices as well. Specifically, vignette-3 respondents confront a smaller rate increase, from 3% to 3.5%. This smaller increase lowers the likelihood of consulting a bank advisor by 5 pp (6%) and of raising income by 6 pp (12%). Instead, they are 10 pp (21%) more likely to lock in the 3.5% using a forward loan. Preparation thus intensifies with the size of the anticipated rate increase, with one exception: respondents lock in the new rate more readily when it is close to their existing rate, in line with the hazard analysis ([Section 3.2](#)).

We also ask for how long respondents would want to fix the new 3.5% interest rate, with possible options of one, five, 10 years and “as long as possible.” We define a dummy equal to one for those that select 10 years or longer. The last column of [Table 5](#) shows that in vignette 3 (smaller rate increase), respondents are about 6 pp (11%) more likely to

choose such a long fixation, suggesting they are less willing to lock in a rate for a long time when that rate represents a larger increase to their old rate.

The results support two further interpretations. First, they help interpret the bank-data pattern that fewer borrowers lock in early and more wait until closer to expiration once rates are high. Since a larger rate increase in the vignette shifts preparation away from early lock-in and toward other margins, waiting when rates are high need not reflect inattention. Second, while not the focus of the paper, changes to consumption-savings decisions appear as another relevant margin of adjustment ahead of higher-rate refinancing. Respondents' open-text explanations of their own preparation point to saving as well. Anticipatory decreases in spending could speed up the transmission of monetary policy.

Reasoning behind choices We document how respondents in vignette 1 reason about their hypothetical refinancing choices.⁴⁰ [Figure A.9](#) is on meeting an advisor from the current lender. For the 78% of respondents who would likely arrange a meeting, the most important reasons are to negotiate a better mortgage deal and better understand mortgage options (Panel A). The remaining 22% who are unlikely to meet often state they are well-informed already or want to acquire information themselves (Panel B).

[Figure A.10](#) reports on the reasons for (not) making prepayments, defined as voluntary, extra reductions in the loan balance. 71% would likely prepay, and most name avoiding a large increase in the monthly payment and reducing the interest burden as important reasons (Panel A). This finding is consistent with the limited passthrough from higher rates to payments we observe in the bank data ([Figure 5](#)). The 29% unlikely to prepay mostly reason based on opportunity costs: they have other priorities for their savings (e.g., durables purchase) or prefer to invest their savings differently (Panel B).

5 Letter RCT

In this section, we present results from the letter experiment. [Section 5.1](#) describes the letter objective, sample, and content. [Section 5.2](#) shows effects of the letter intervention on survey outcomes. [Section 5.3](#) reports intervention effects on bank-data outcomes. [Section 5.4](#) explores selection into reading the letter.

⁴⁰Effects of varying the hypothetical refinancing situation on reasoning are insignificant. The single exception is that respondents who confront a smaller rate increase (vignette 3) are more likely to view "tolerating higher monthly payments due to higher rates" as an important reason for being unlikely to prepay.

5.1 Letter setup

In November 2024, the partner bank mails a letter to a random three quarters of about 48,000 mortgagors; the remaining quarter receives no letter. We design the letter, and the bank sends it under its letterhead, so it reaches borrowers as regular lender communication rather than as researcher-provided information. The letter documents the rise in mortgage rates since 2022, illustrates what a higher rate can imply for monthly payments after the rate reset, lists options to respond, and offers a meeting with a bank advisor. Any letter effect is on top of existing lender communication ([Section 2](#)).

The letter can change behavior through two channels. It can inform borrowers about mortgage rates, their payment impact, and the available options, or it can remind borrowers who are already informed to act. We distinguish the channels by measuring both beliefs and actions. The survey, fielded about two weeks after the mailing, measures whether the letter moves beliefs, such as on mortgage rates and features. If borrowers are well informed to begin with, beliefs should move little (e.g., [Coibion and Gorodnichenko, 2026a](#); [Weber et al., 2025](#)). The bank data record advisor meetings, refinancing, prepayments, and products taken out over the six months after the mailing. If the letter changes actions but not beliefs, it works as a reminder rather than as new information. The survey also measures whether respondents remember reading the letter. Reading is voluntary, unlike in survey experiments that require participants to view the information, so we can study who selects into effective exposure.

Sample The bank selects eligible borrowers in October 2024 using three criteria. First, the borrower has not objected to non-mandatory mail. Second, the borrower holds a regular mortgage originated before June 2022 and thus has not yet refinanced at higher rates. Third, the expected balance at the end of the fixation period is at least €10,000, so the borrower faces meaningful rate risk. Of 140,000 active mortgagors, about 48,000 meet all criteria; the first and third exclude the most borrowers.

We randomly assign borrowers to four groups of about 12,000 each. The control group receives no letter. Variant 1 contains all four sections described below, variant 2 omits the section on rate uncertainty, and variant 3 omits the payment example. The variants let us test whether a specific piece of content drives effects while still pooling all recipients. [Table A.7](#) shows that the groups are largely balanced on loan and borrower characteristics.

Content The two-page letter is titled “How do you deal with higher mortgage rates?” and has four sections ([Appendix E](#)). We keep it short to limit the cost of engaging with

it (Stantcheva, 2023). The first section states and illustrates (Panel A of Figure A.11) that market rates have risen a lot since early 2022.⁴¹ It reports an average rate on new mortgages of 4% and mentions the bank’s cooperation with Goethe University Frankfurt.⁴²

The second section notes that the rate at which the bank continues the loan can differ from 4% because future rates are uncertain. It cites a Bundesbank survey that shows heterogeneity in rate expectations among German households. This section is omitted for variant 2 to assess whether information about high rate uncertainty drives effects, in light of the importance of macro uncertainty for household behavior (Coibion et al., 2024).

The third section sketches how higher interest rates can affect monthly payments. An increase in rates by 2 pp—a plausible rate change—on a €100,000 loan initially raises interest expenses by €2,000 per year, or €167 per month. A small graphic makes these numbers salient (Panel B of Figure A.11). Variant 3 omits this section to gauge whether information about concrete payment changes matters.

The fourth section lists options to respond to higher interest rates. We mention (i) reducing the balance through annual partial prepayments (*Sondertilgung*) or repayment at expiration, (ii) locking in the future rate using a forward loan or a home-savings contract (*Bausparen*) to have certainty about future payments, and (iii) increasing savings to afford higher future payments. The letter closes by offering a meeting with a bank advisor to discuss which option fits best, and lists the advisor’s contact details.

5.2 Letter effects in survey data

Table 6 reports letter effects on survey outcomes. We regress the outcomes on indicators for each letter variant (Panel A) or for any letter (Panel B), for the 881 respondents who are part of the letter experiment. Controls are the pre-letter bank controls described in Section 4.2, as well as household income and university education. We omit the survey-based belief controls because the letter could affect them. The letter does not affect survey participation (Table A.5), mitigating concerns about differential selection into the survey.

Column 1 captures whether respondents report having read a bank letter on rising mortgage rates in the past two months. The letter raises this share by 33 pp, from 8% in

⁴¹The partner bank asks us not to show the original letter, so letter graphics are separate from the text.

⁴²The 4% refer to July 2024, the latest Bundesbank statistic when the letter went to print in September. Rates had fallen to around 3.5% by the time of the survey (Figure 1).

the control group.⁴³ The effect ranges from 30 to 37 pp across variants, and we cannot reject that it is identical. Two thirds of recipients thus do not remember reading the letter a few weeks after the mailing, and [Section 5.4](#) studies who does. If the letter affects only those who report reading it, its effect is about three times the coefficients we report.

Columns 2–3 show insignificant letter effects on beliefs about mortgage rates. We ask for the perceived rate change over the past three years on a five-point scale, from “much lower” to “much higher,” and an indicator for a perceived current rate of 3–4% on a new mortgage with a 10-year fixation period.⁴⁴ We treat this range as accurate because the letter states 4%, while rates had fallen to around 3.5% by the time of the survey. Limited updating in response to provided information indicates that respondents were informed to begin with (e.g., [Coibion and Gorodnichenko, 2026b](#); [Weber et al., 2025](#)). Indeed, as we show in [Figure 8](#), most respondents have well-calibrated rate beliefs. Any letter effects on actions are hence unlikely to reflect increased awareness of the rate environment.

Column 4 presents letter effects on expected changes in monthly payments upon refinancing. We ask for expected changes in euros (in buckets of 50); the change is zero if they expect payments to remain roughly the same (42% of respondents); and we exclude respondents who report they do not need to refinance. We scale the expected payment change by the current payment. The letter does not affect expectations about payment changes, despite explicitly outlining how higher rates can feed into payments.

Columns 5–8 document letter effects on familiarity with prepayments and forward loans and on the corresponding hypothetical choices. Respondents rate their familiarity on a four-point scale from “completely unknown” to “completely known.” The letter raises familiarity with prepayments by 0.16 standard deviations, significant at the 10% level, and familiarity with forwards by an insignificant 0.11 standard deviations. The likelihood of prepaying as part of the hypothetical refinancing situation ([Section 4.4](#)) is also positively affected, while the likelihood of taking out a forward is unchanged.

In sum, the letter reaches one third of its recipients and does not change what they believe about mortgage rates or future payments, consistent with borrowers who know the rate environment. Familiarity with contractual options rises modestly at most. Letter effects on actions, to which we turn next, are therefore unlikely to reflect new information about rates, and would instead point to the reminder channel ([Section 5.1](#)).

⁴³12% of control-group respondents report having received such a letter, as the bank regularly sends marketing and contract material. The effect on reporting receipt, read or not, is 33–38 pp.

⁴⁴The letter does not affect the first and second moment of expected mortgage rates either.

5.3 Letter effects in bank data

We next examine letter effects on borrower actions in the bank data over the six months after the mailing, until the data end in May 2025. Borrowers can respond by meeting a bank advisor (in the pursuit of information acquisition), refinancing, making extra payments where the contract permits, or building savings. Which actions are available depends on the distance to expiration. A regular prolongation is possible only within six months of expiration, and earlier refinancing requires a forward loan.

The outcomes are indicators for whether the borrower meets a bank advisor at least once; refinances with the bank; does so at least six weeks before expiration, that is, ahead of the bank's prolongation offers; meets an advisor and refinances within seven days; makes an extra payment; and opens a new savings product (savings account, securities account, or home-savings loan (*Bausparen*)). We regress each outcome on an indicator that pools the three letter variants and on the controls listed in the table notes.

Panel A of [Table 7](#) presents average letter effects. Column 1 shows no effect on meeting a bank advisor. Effects on refinancing internally and on refinancing ahead of the bank's prolongation offers are positive but insignificant (Columns 2–3). However, the letter raises the joint outcome of meeting an advisor and refinancing shortly thereafter by 0.23 pp, a 33% increase relative to the control-group mean (Column 4). We find no effects on extra payments and new savings products (Columns 5–6). Small average effects are unsurprising because most recipients cannot refinance during the six months.

We then examine heterogeneity in treatment effects by distance to expiration. While advisor meetings, extra payments (when permitted), and savings products are available to all borrowers, refinancing options vary with distance to expiration. We thus classify borrowers into three groups based on the actions they could take post-letter. The first group comprises the 2,200 borrowers (5%) whose fixation period ends by May 2025. They are within six months of expiration when the letter arrives, have not yet refinanced, and move to the high variable rate if they do not act. In the same months, they receive the bank's expiration notice and, if they wait, its prolongation offers ([Section 2](#)). The second group comprises the 15,400 borrowers (32%) whose fixation period ends between June 2025 and May 2029 and who can refinance only early, mostly through a forward loan. The third group comprises the 29,900 borrowers (63%) whose fixation period ends later and who cannot yet lock in a new rate.

Panel B of [Table 7](#) reports results. The letter prompts action only in the first group. Re-

cipients in this group are 4.4 pp (12%) more likely than corresponding control borrowers to meet an advisor and 3.5 pp (16%) more likely to refinance internally, both significant at the 10% level. They are 2 pp more likely to refinance ahead of the bank’s prolongation offers, which roughly doubles this probability, and 4.5 pp (40%) more likely to meet an advisor and refinance shortly thereafter. Effects on extra payments and savings products are insignificant. The other two groups do not respond on any margin.⁴⁵ If one third of recipients read the letter, effects on readers are plausibly about three times as large.

These results suggest that the letter works as a reminder rather than as new information. First, the letter does not change beliefs about mortgage rates or payments. Second, the letter modestly raises familiarity with prepayments in the survey, yet extra payments do not respond in the bank data. What responds are advisor meetings and refinancing among borrowers who face the decision anyway. Third, borrowers close to expiration are neither less informed about rates nor more likely to read the letter than other borrowers. They likely respond because their decision is imminent, not because they learn more. The reminder effect relates to [Byrne et al. \(2023\)](#), who find that a reminder letter raises refinancing among Irish borrowers more than improvements to the disclosure itself.

5.4 Selection into reading the letter

[Table 8](#) studies who reads the letter. The dependent variable is one if the respondent reports having read a bank letter on rising mortgage rates in the past two months, and zero otherwise (no letter received, letter received but not read, does not remember). We regress it on an indicator for receiving any of our letters, a loan trait (Columns 1–3) or borrower trait (Columns 4–7), and their interaction. Because the letter is randomized, the interaction measures how the letter’s effect on reading varies with the trait. The traits are not randomized, so the differences describe who selects into reading.⁴⁶

We observe systematic selection into reading the letter. Borrowers with one-standard-deviation higher outstanding debt are 7 pp (21%) more likely to read the letter, and those

⁴⁵Toward the end of our observation period, the first borrowers in group 2 enter the six-month window in which regular refinancing is possible, which likely explains why non-recipients in group 2 are more likely to act than non-recipients in group 3 (the omitted group). Nonetheless, at the time of letter reception, refinancing for group 2 would only be possible using a forward.

⁴⁶We cannot separate reading from remembering the letter. Both are required for the letter to inform, and we refer to the combination as effective exposure. Survey respondents are positively selected on education and attachment to the bank ([Section 4.2](#)), so the differences may not carry over one-for-one to all recipients.

who have made prepayments in the past are 11 pp (35%) more likely. In contrast, a rate reset within two years does not predict reading. These patterns align with our results on who is aware of the recent rate increases (Table 4): outstanding debt predicts informedness, but distance to expiration does not. Borrowers with above-median self-reported financial literacy and those with above-median household income net of expenses are 18 pp (76%) and 15 pp (60%) more likely to read the letter, respectively. Finally, recipients who correctly perceive mortgage rates to have risen since 2021 and the current rate to be 3–4% are 17 pp (88%) and 13 pp (57%) more likely to read it.⁴⁷ In a regression on all traits jointly among recipients, outstanding debt, financial literacy, and perceived rate increases remain significant predictors (Table A.8).

Selection into reading limits what the letter can achieve. The letter reaches the borrowers who have the least to learn from it, which helps explain why beliefs do not move, while borrowers with lower awareness are harder to reach. Information provision alone may thus have limited power if the main friction is attention rather than availability. The pattern is consistent with endogenous attention: reading rises with the stakes, measured by outstanding debt, and with the ability to process the content, measured by financial literacy (e.g., Maćkowiak et al., 2023). Selection into effective treatment exposure also speaks to evidence on state-dependent attention to the macroeconomy. Households pay more attention to inflation when it is high (Bracha and Tang, 2024; Link et al., 2025; Pfäuti, 2026; Weber et al., 2025). They are then easier to reach, but communication moves their expectations less (Coibion and Gorodnichenko, 2026b). We find the cross-sectional counterpart of this tradeoff between reach and impact: at a given point in time, lender communication reaches the borrowers whose beliefs it is least likely to change.

6 Conclusion

Based on a unique combination of loan-level data from a German bank, a survey of its borrowers, and a letter RCT, we study how borrowers prepare for and respond to changes in the interest rate on their mortgage. Our results indicate that borrowers are generally well-informed about the interest-rate environment, and often prepare for the expiration of their rate fixation. Their various actions mitigate the impact of rate changes on monthly

⁴⁷To be clear, this reflects selection, not a treatment effect from the letter: Table 6 shows that estimated treatment effects on rate awareness are directionally negative and far from statistically significant.

payments, resulting in much smaller variation in payments than in a counterfactual world with passive refinancing. Thus, the transmission of monetary policy changes to monthly payments is substantially dampened, and strains on household finances appear limited.

Borrower awareness of the many options the German institutional setting provides is not perfect, however, as indicated by the significant effects of the letter RCT intervention on some survey responses and on the refinancing behavior of the selected subsample of borrowers that had not taken action until shortly before expiration of their old loan. Nevertheless, we interpret our findings as contrasting to some extent with many results in the literature that emphasize limits to borrower awareness and understanding. It is an interesting open question to what extent these seemingly different results reflect cross-country variation in institutional settings, the time period studied, or the selection of households that own homes with a mortgage (since German homeownership is lower than in most other advanced economies).

References

- ADELINO, M., M. FERREIRA, AND S. ZHAO (2025): “Monetary Policy Pass-Through Under Adjustable Rate Mortgages,” Working paper.
- ANDERSEN, A. L., A. JAKOBSEN, M. R. JØRGENSEN, AND N. JOHANNESSEN (2026): “Consumption Responses to Rising Mortgage Rates: Unpacking the Cash-Flow Channel of Monetary Policy,” Working paper.
- ANDERSEN, H. Y., S. L. BECH, AND A. DE STEFANI (2024): “Spending Response to a Predictable Increase in Mortgage Repayments: Evidence from Expiring Interest-Only Loans,” *Review of Economics and Statistics*, 106, 277–285.
- ANDERSEN, S., J. Y. CAMPBELL, K. M. NIELSEN, AND T. RAMADORAI (2020): “Sources of inaction in household finance: Evidence from the Danish mortgage market,” *American Economic Review*, 110, 3184–3230.
- ANDRE, P., I. HAALAND, C. ROTH, M. WIEDERHOLT, AND J. WOHLFART (2026a): “Narratives about the Macroeconomy,” *Review of Economic Studies*.
- ANDRE, P., C. PIZZINELLI, C. ROTH, AND J. WOHLFART (2022): “Subjective Models of the Macroeconomy: Evidence From Experts and Representative Samples,” *The Review of Economic Studies*, 89, 2958–2991.
- ANDRE, P., P. SCHIRMER, AND J. WOHLFART (2026b): “Mental Models of the Stock Market,” *The Quarterly Journal of Economics*.
- ARGYLE, B. S., T. D. NADAULD, AND C. J. PALMER (2020): “Monthly payment targeting and the demand for maturity,” *Review of Financial Studies*, 33, 5416–5462.
- AUCLERT, A. (2019): “Monetary Policy and the Redistribution Channel,” *American Economic Review*, 109, 2333–2367.
- AUSSCHUSS FÜR FINANZSTABILITÄT (2026): “AFS nutzt neue Daten zur Überwachung der Kreditvergabestandards bei Wohnimmobilienfinanzierungen: erste Ergebnisse deuten auf mögliche Risiken hin,” Press release, 11 May 2026.
- BAJO, E. AND M. BARBI (2018): “Financial illiteracy and mortgage refinancing decisions,” *Journal of Banking & Finance*, 94, 279–296.
- BAKER, S. R., M. GELMAN, L. KUENG, AND S. H. LEE (2026): “Anticipatory Spending: Identifying Consumer Types with Pre-Payment MPCs,” Working paper.
- BARASINSKA, N., P. HAENLE, A. KOBAN, AND A. SCHMIDT (2023): “No reason to worry about German mortgages? An analysis of macroeconomic and individual drivers of credit risk,” *Journal of Financial Services Research*, 64, 369–399.

- BRACHA, A. AND J. TANG (2024): “Inflation Levels and (In)Attention,” *Review of Economic Studies*, 92, 1564–1594.
- BRACKE, P., M. EVERITT, M. FAZIO, AND A. VARADI (2024): “When refinancing meets monetary tightening: heterogeneous impacts on spending and debt via mortgage modifications,” Staff Working Paper 1105, Bank of England.
- BUCKS, B. AND K. PENCE (2008): “Do borrowers know their mortgage terms?” *Journal of Urban Economics*, 64, 218–233.
- BYRNE, S., K. DEVINE, M. KING, Y. MCCARTHY, AND C. PALMER (2023): “The Last Mile of Monetary Policy: Inattention, Reminders, and the Refinancing Channel,” Working Paper 31043, National Bureau of Economic Research.
- CAMPBELL, J. Y. (2006): “Household Finance,” *Journal of Finance*, 61, 1553–1604.
- CAMPBELL, J. Y. AND J. F. COCCO (2003): “Household Risk Management and Optimal Mortgage Choice,” *Quarterly Journal of Economics*, 118, 1449–1494.
- CHOPRA, F., C. ROTH, AND J. WOHLFART (2025): “Home Price Expectations and Spending: Evidence from a Field Experiment,” *American Economic Review*, 115, 2267–2305.
- CLOYNE, J., C. FERREIRA, AND P. SURICO (2020): “Monetary policy when households have debt: new evidence on the transmission mechanism,” *Review of Economic Studies*, 87, 102–129.
- COIBION, O., D. GEORGARAKOS, Y. GORODNICHENKO, G. KENNY, AND M. WEBER (2024): “The Effect of Macroeconomic Uncertainty on Household Spending,” *American Economic Review*, 114, 645–677.
- COIBION, O. AND Y. GORODNICHENKO (2026a): *Expectations Matter: The New Causal Macroeconomics of Surveys and Experiments*, Princeton University Press.
- (2026b): “Inflation expectations and monetary policy: what have we learned and to what end?” *The Econometrics Journal*.
- COIBION, O., Y. GORODNICHENKO, AND M. WEBER (2022): “Monetary Policy Communications and Their Effects on Household Inflation Expectations,” *Journal of Political Economy*, 130, 1537–1584.
- DEUTSCHE BUNDESBANK (2019): “Financial Stability Review 2019,” Tech. rep., Deutsche Bundesbank, Frankfurt am Main.
- DI MAGGIO, M., A. KERMANI, B. J. KEYS, T. PISKORSKI, R. RAMCHARAN, A. SERU, AND V. YAO (2017): “Interest rate pass-through: Mortgage rates, household consumption, and voluntary deleveraging,” *American Economic Review*, 107, 3550–3588.

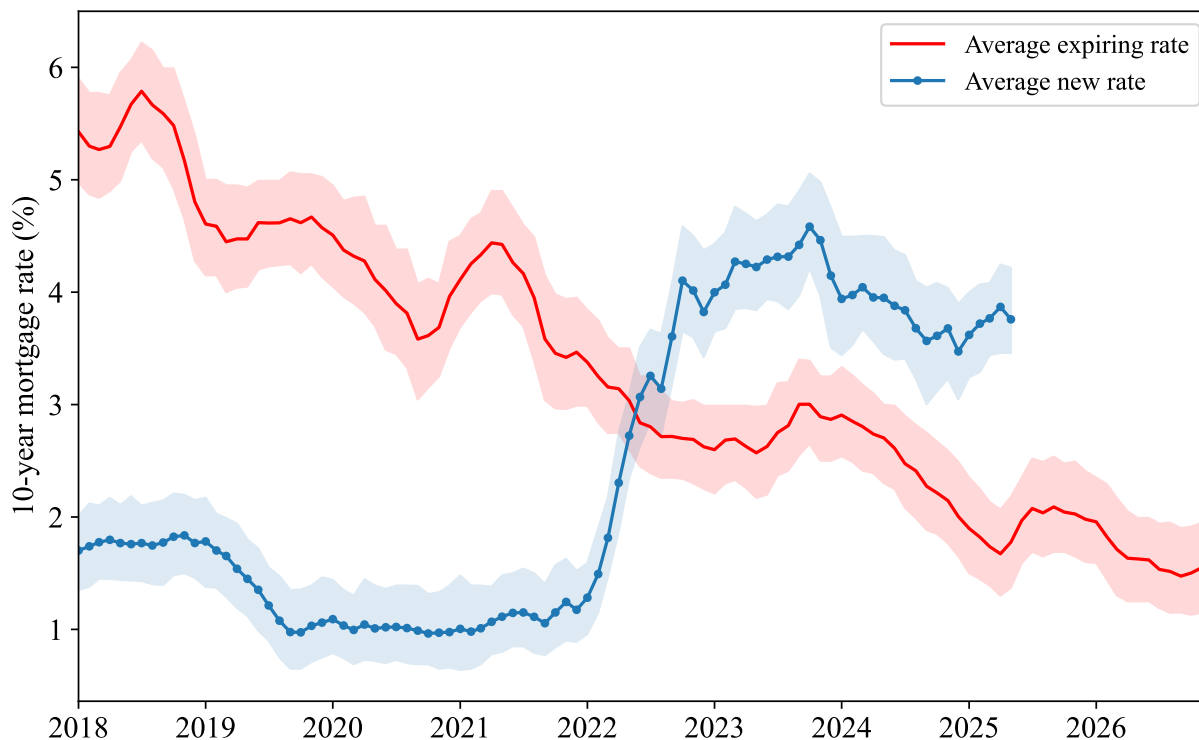
- DRUEDAHL, J., E. B. JENSEN, AND S. LETH-PETERSEN (2022): “The intertemporal marginal propensity to consume out of future persistent cash-flows: Evidence from transaction data,” Working paper, CEBI.
- DUTZ, D., I. HUITFELDT, S. LACOUTURE, M. MOGSTAD, A. TORGOVITSKY, AND W. VAN DIJK (2026): “Selection in Surveys: Using Randomized Incentives to Detect and Account for Nonresponse Bias,” *Review of Economic Studies*, forthcoming.
- ELIAS, M., C. GILLITZER, G. KAPLAN, G. L. CAVA, AND N. PRASAD (2025): “The Mortgage Debt Channel of Monetary Policy when Mortgages are Liquid,” Working Paper 34461, National Bureau of Economic Research.
- FISHER, J., A. GAVAZZA, L. LIU, T. RAMADORAI, AND J. TRIPATHY (2024): “Refinancing cross-subsidies in the mortgage market,” *Journal of Financial Economics*, 158, 103876.
- FLODÉN, M., M. KILSTRÖM, J. SIGURDSSON, AND R. VESTMAN (2021): “Household debt and monetary policy: revealing the cash-flow channel,” *Economic Journal*, 131, 1742–1771.
- FOULIS, A., J. HAZELL, A. MIAN, AND B. TRACEY (2026): “How Do Interest Rates Affect Consumption? Household Debt and the Role of Asset Prices,” Working paper.
- FUSTER, A., G. KAPLAN, AND B. ZAFAR (2021): “What Would You Do with \$500? Spending Responses to Gains, Losses, News, and Loans,” *Review of Economic Studies*, 88, 1760–1795.
- FUSTER, A. AND P. S. WILLEN (2017): “Payment Size, Negative Equity, and Mortgage Default,” *American Economic Journal: Economic Policy*, 9, 167–91.
- FUSTER, A. AND B. ZAFAR (2023): “Survey experiments on economic expectations,” in *Handbook of Economic Expectations*, ed. by R. Bachmann, G. Topa, and W. van der Klaauw, Academic Press, chap. 4, 107–130.
- GANONG, P. AND P. NOEL (2019): “Consumer Spending during Unemployment: Positive and Normative Implications,” *American Economic Review*, 109, 2383–2424.
- GEIGER, F., J. MUELLBAUER, AND M. RUPPRECHT (2016): “The housing market, household portfolios and the German consumer,” .
- GIANINAZZI, V. (2023): “Reference Points in Refinancing Decisions,” Working paper.
- HAALAND, I., C. ROTH, S. STANTCHEVA, AND J. WOHLFART (2025): “Understanding Economic Behavior Using Open-Ended Survey Data,” *Journal of Economic Literature*, 63, 1244–1280.
- HAALAND, I., C. ROTH, AND J. WOHLFART (2023): “Designing information provision experiments,” *Journal of Economic Literature*, 61, 3–40.

- HAPPEL, J., Y. KARABULUT, L. SCHÄFER, AND S. TÜZEL (2024): “Shattered housing,” *Journal of Financial Economics*, 156, 103835.
- JAPPELLI, T. AND A. SCOGNAMIGLIO (2018): “Interest rate changes, mortgages, and consumption: evidence from Italy,” *Economic Policy*, 33, 183–224.
- JOHNSON, E. J., S. MEIER, AND O. TOUBIA (2019): “What’s the catch? Suspicion of bank motives and sluggish refinancing,” *Review of Financial Studies*, 32, 467–495.
- JØRRING, A. (2024): “Financial Sophistication and Consumer Spending,” *Journal of Finance*, 79, 3773–3820.
- KARTASHOVA, K. AND X. ZHOU (2025): “How do mortgage rate resets affect consumer spending and debt repayment? Evidence from Canadian consumers,” *Journal of Money, Credit and Banking*, 57, 1167–1210.
- KEYS, B. J., D. G. POPE, AND J. C. POPE (2016): “Failure to refinance,” *Journal of Financial Economics*, 122, 482–499.
- KUENG, L. (2018): “Tax News Shocks and Consumption,” Working paper, Swiss Finance Institute.
- LINK, S., A. PEICHL, O. PFÄUTI, C. ROTH, AND J. WOHLFART (2025): “Attention to the Macroeconomy,” Working paper.
- LUKAS, M. AND M. NÖTH (2019): “Interest rate changes and borrower search behavior,” *Journal of Economic Behavior & Organization*, 163, 172–189.
- MALMENDIER, U. AND A. S. WELLSJO (2024): “Rent or Buy? Inflation Experiences and Homeownership within and across Countries,” *Journal of Finance*, 79, 1977–2023.
- MAĆKOWIAK, B., F. MATĚJKA, AND M. WIEDERHOLT (2023): “Rational Inattention: A Review,” *Journal of Economic Literature*, 61, 226–273.
- PFÄUTI, O. (2026): “The Inflation Attention Threshold and Inflation Surges,” Working paper.
- SCHNORPFEIL, P., M. WEBER, AND A. HACKETHAL (2023): “Households’ Response to the Wealth Effects of Inflation,” Working Paper 31672, National Bureau of Economic Research.
- (2025): “Inflation and Trading,” *Journal of Financial Economics*, 173, 104166.
- SCHOLNICK, B. (2013): “Consumption Smoothing after the Final Mortgage Payment: Testing the Magnitude Hypothesis,” *Review of Economics and Statistics*, 95, 1444–1449.
- STANTCHEVA, S. (2023): “How to Run Surveys: A Guide to Creating Your Own Identifying Variation and Revealing the Invisible,” *Annual Review of Economics*, 15.

- STEPHENS, MELVIN, J. (2008): "The Consumption Response to Predictable Changes in Discretionary Income: Evidence from the Repayment of Vehicle Loans," *Review of Economics and Statistics*, 90, 241–252.
- WEBER, M., B. CANDIA, H. AFROUZI, T. ROPELE, R. LLUBERAS, S. FRACHE, B. MEYER, S. KUMAR, Y. GORODNICHENKO, D. GEORGARAKOS, O. COIBION, G. KENNY, AND J. PONCE (2025): "Tell Me Something I Don't Already Know: Learning in Low- and High-Inflation Settings," *Econometrica*, 93, 229–264.

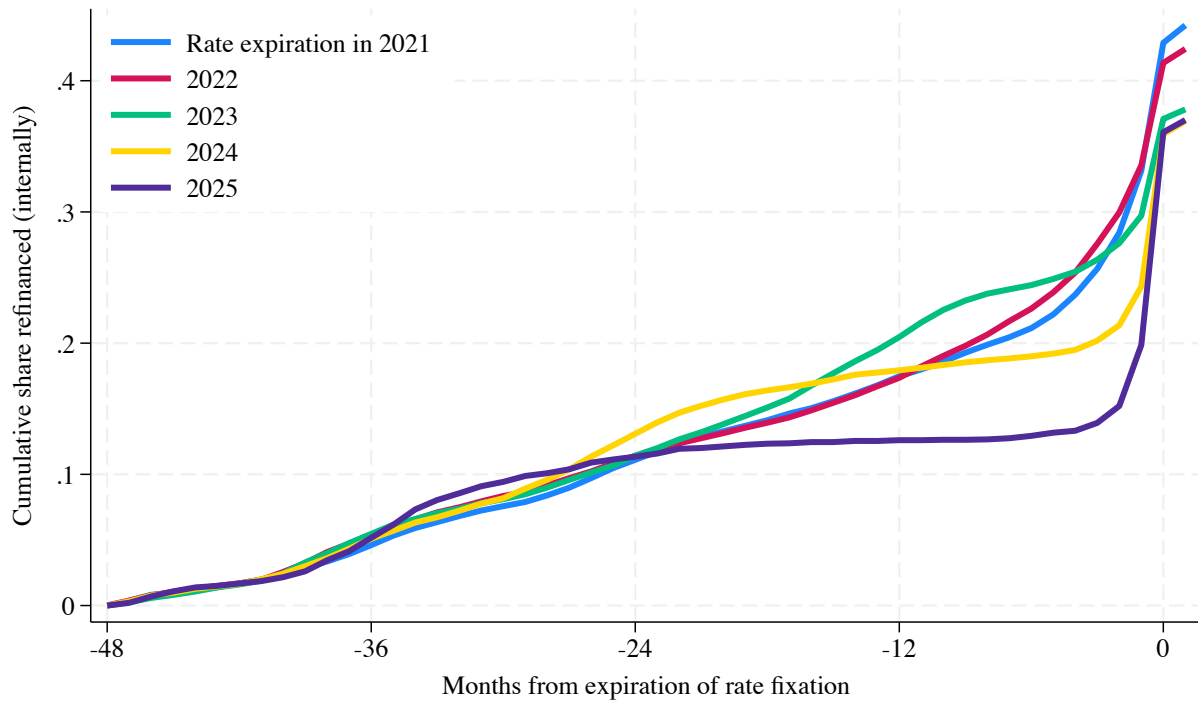
Figures and Tables

Figure 1: Rates on new and expiring mortgages at our partner bank



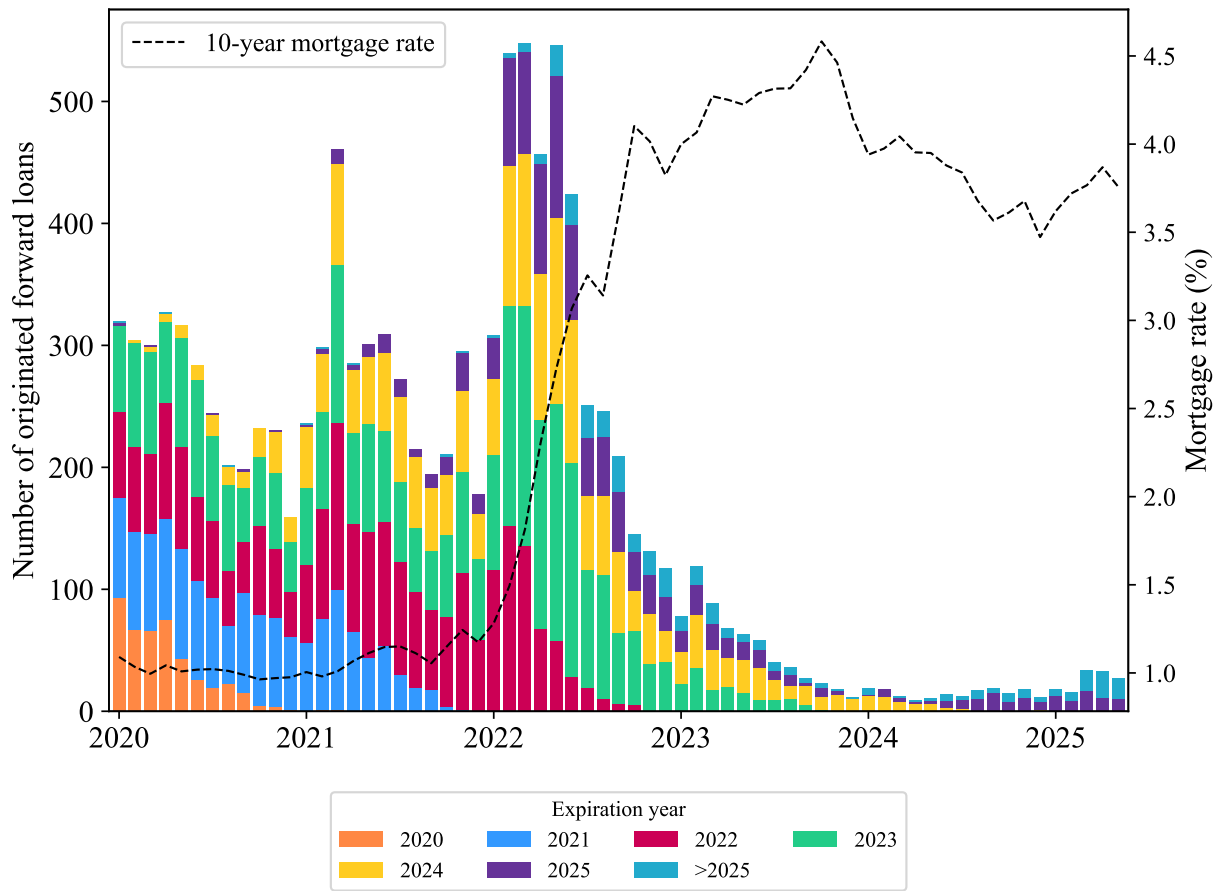
Notes: The figure plots average interest rates on 10-year fixed-rate mortgages at the partner bank. The blue dotted line shows the average interest rate on mortgages originated in a given month, while the red solid line shows the average interest rate on mortgages whose fixed-rate period expires in that month. The shaded area denotes the 10th and 90th percentiles. The horizontal axis reports calendar months, and the vertical axis reports interest rates in percent.

Figure 2: Cumulative hazard of internal refinancing across rate expiration vintages



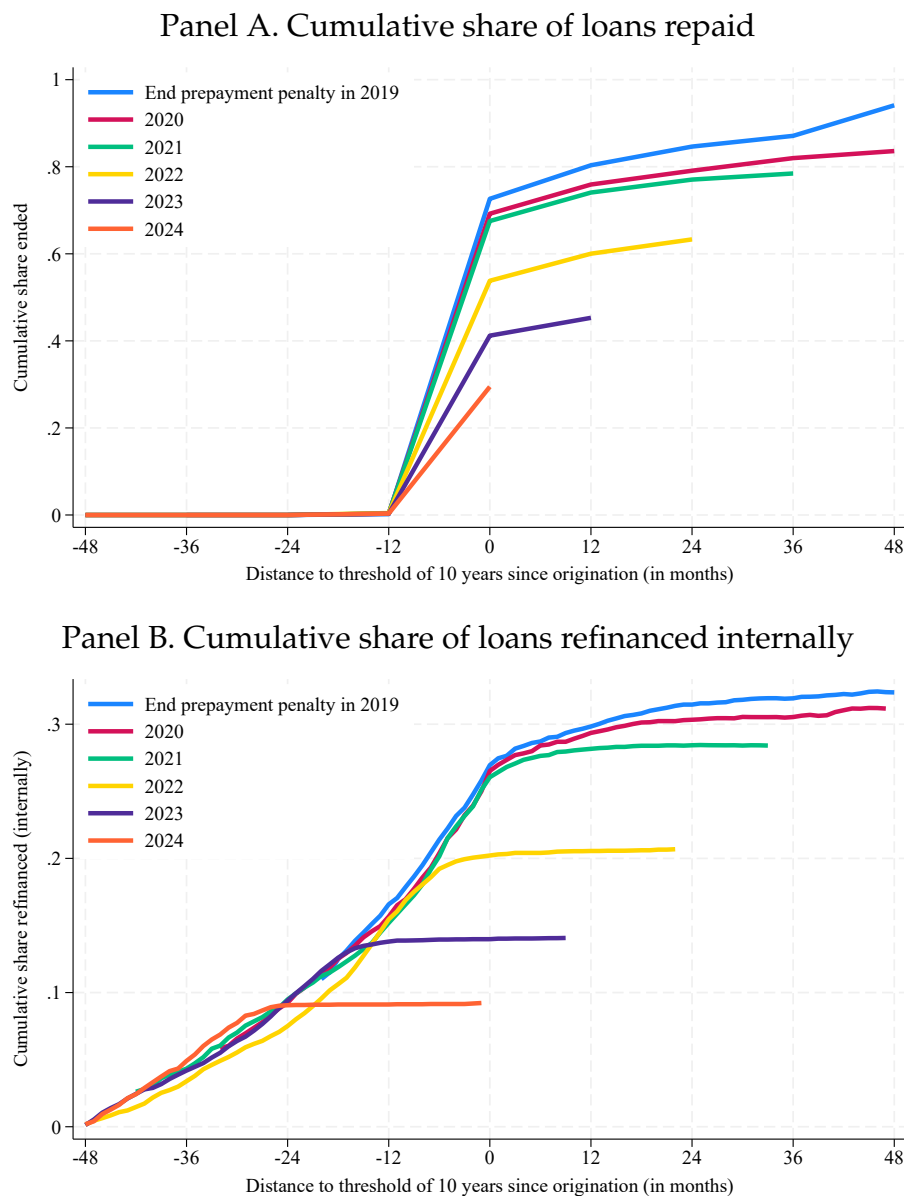
Notes: The figure plots the cumulative share of mortgages that refinance internally at our partner bank (via a forward mortgage or a regular prolongation) as a function of distance from expiration of the rate fixation, separately for different rate expiration vintages. The sample is restricted to loans with fixation period of up to 10 years, an expected residual balance of at least € 10,000 at expiration, and expiration between January 2018 and March 2025.

Figure 3: Origination of forward mortgages over time



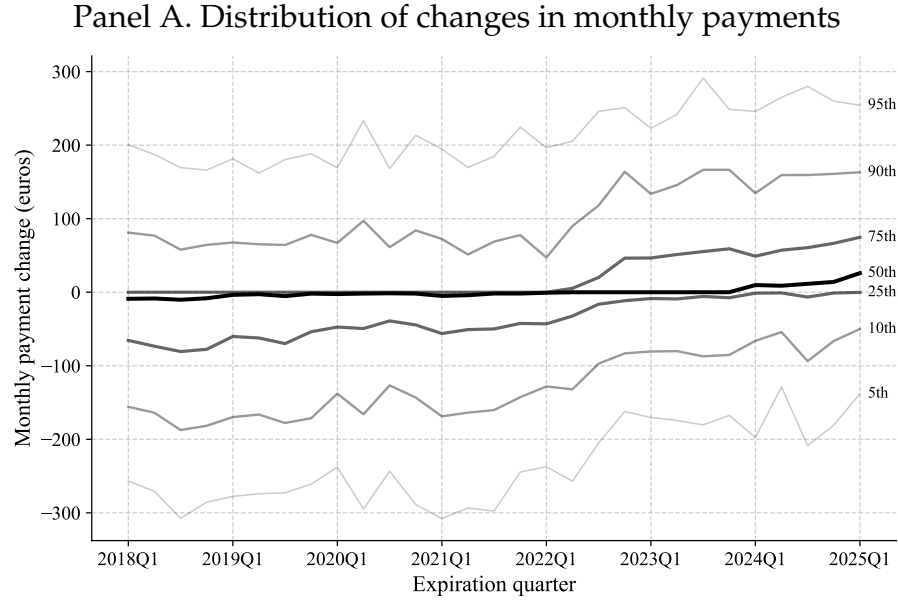
Notes: The figure shows the time series of forward loan originations at monthly frequency. Bars represent the total number of forward loans originated by the partner bank for internal refinancing, with colors indicating the expiration year of the old loan. The sample is restricted to forward loans used to refinance mortgages with a fixed-rate period of 10 years or less. The black dashed line shows the evolution of the mortgage rate, calculated as the average rate on newly originated 10-year fixed-rate mortgages at the partner bank.

Figure 4: Loan repayment and internal refinancing around the 10-year threshold

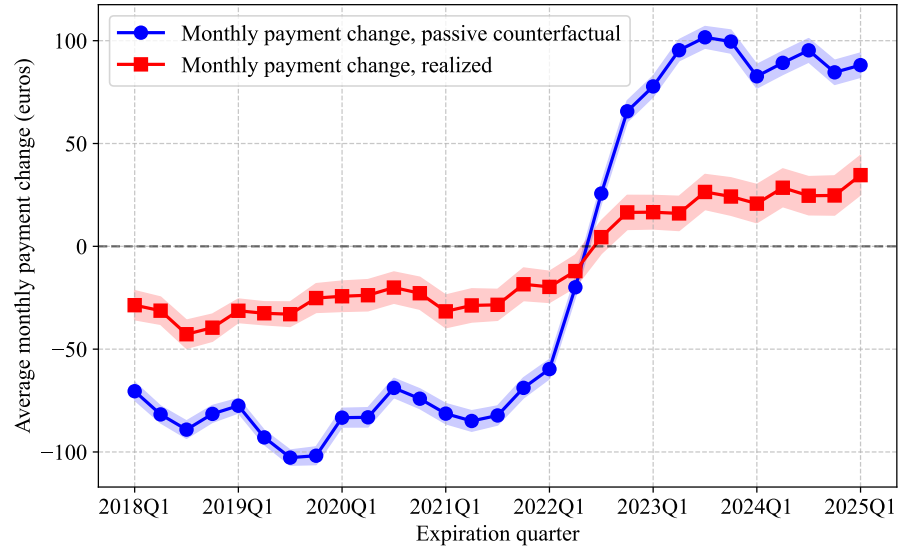


Notes: The figure shows the cumulative share of loan repayment and internal refinancing of borrowers with loans that have fixation periods of 15 years or longer around the threshold when refinancing without penalty becomes available (10 years plus a six-month notice period after the loan is fully paid out to the borrower). We study the share by year in which the prepayment penalty ends. We restrict the sample to loans with projected residual balance at fixation end of at least €10,000. It contains 29,631 such loans. Panel A plots the share of loans that are fully repaid, either through refinancing (internal or external), using own funds, or property sale. Panel B plots the share of mortgages that refinance internally, by months since the 10-year mark. Loans can be refinanced up to four years in advance via forward contracts. While we observe the date at which a loan is refinanced internally, loan repayment is observed only annually.

Figure 5: Changes in monthly payments following internal refinancing

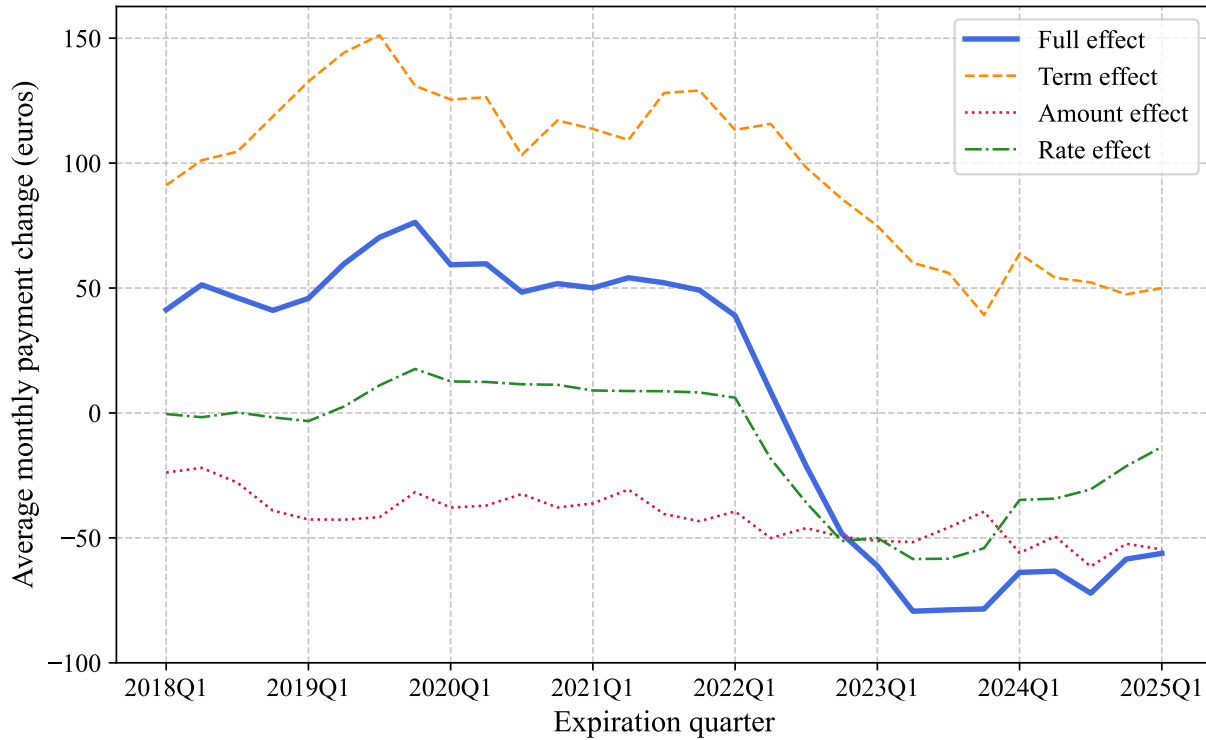


Panel B. Average actual vs. counterfactual monthly payment changes



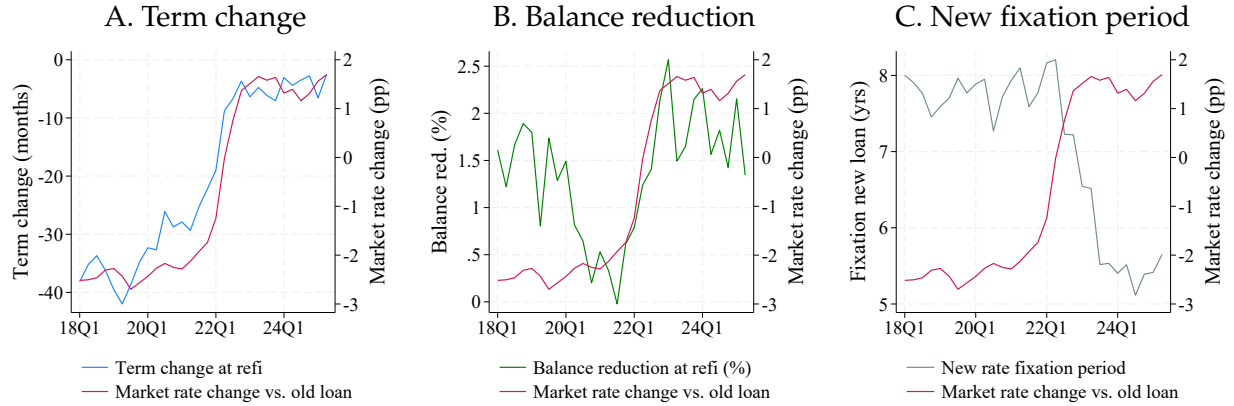
Notes: The figure shows the change in monthly payments upon internal refinancing. The sample is restricted to loans refinanced internally with fixation periods of up to 10 years and an expected residual balance of at least €10,000 at expiration. Panel A plots the distribution of the realized change in monthly payments, the difference between the new loan and the old loan, by quarter of fixation expiration. Panel B plots the average realized change in payments (red) together with the average estimated change in payments under a passive counterfactual scenario in which borrowers simply roll over the remaining debt at expiration, assuming no extra payments and no modifications to loan terms (blue). The shaded area represents the 95% confidence interval.

Figure 6: Effect of refinancing choices on monthly payments



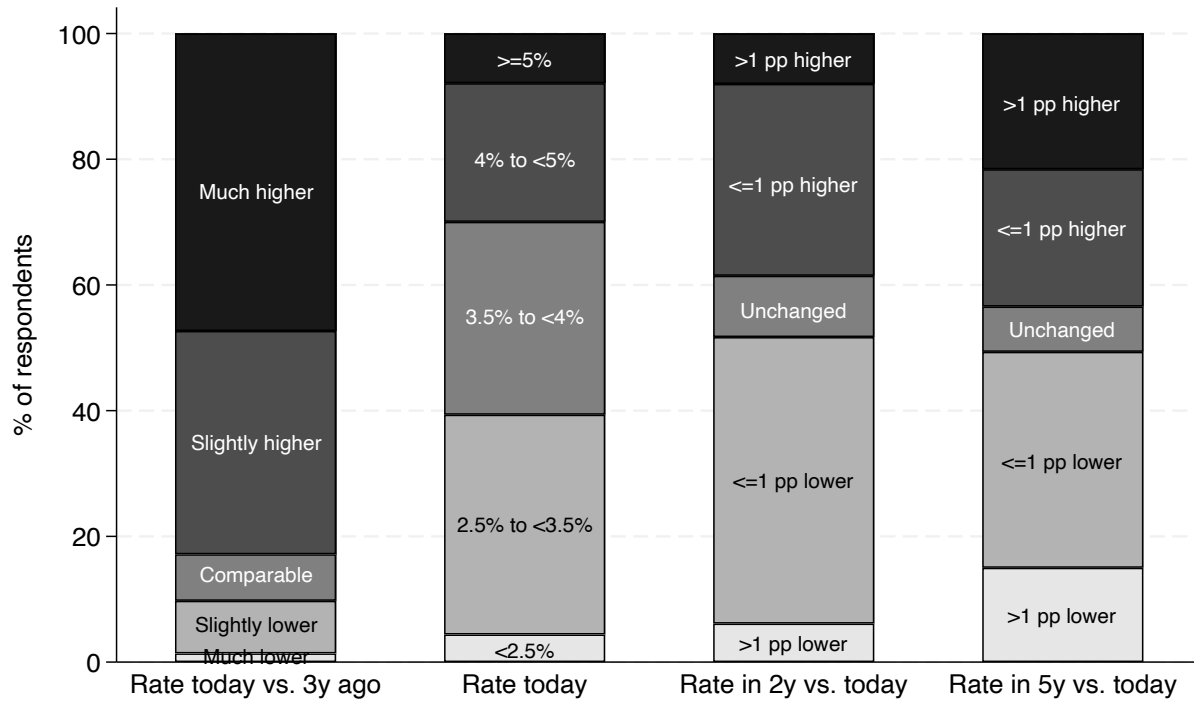
Notes: This figure decomposes the difference between actual and passive-counterfactual monthly payments into the contributions of changes in the loan term, loan amount, and interest rate, as defined in Equation 2. Positive values indicate that borrowers' choices increased monthly payments relative to the passive counterfactual. The thick blue line (full effect) shows the average deviation, i.e., the gap between the realized (red) and passive-counterfactual (blue) payment changes in Panel B of Figure 5. The orange line (term effect) shows how payments would have deviated from the counterfactual had borrowers modified only the loan term, with the loan amount and interest rate held at their counterfactual values. The red line (amount effect) and green line (rate effect) are defined analogously. The sample is restricted to loans refinanced internally with fixation periods of up to 10 years and an expected residual balance of at least €10,000 at expiration.

Figure 7: Refinancing choices by quarter of refinancing



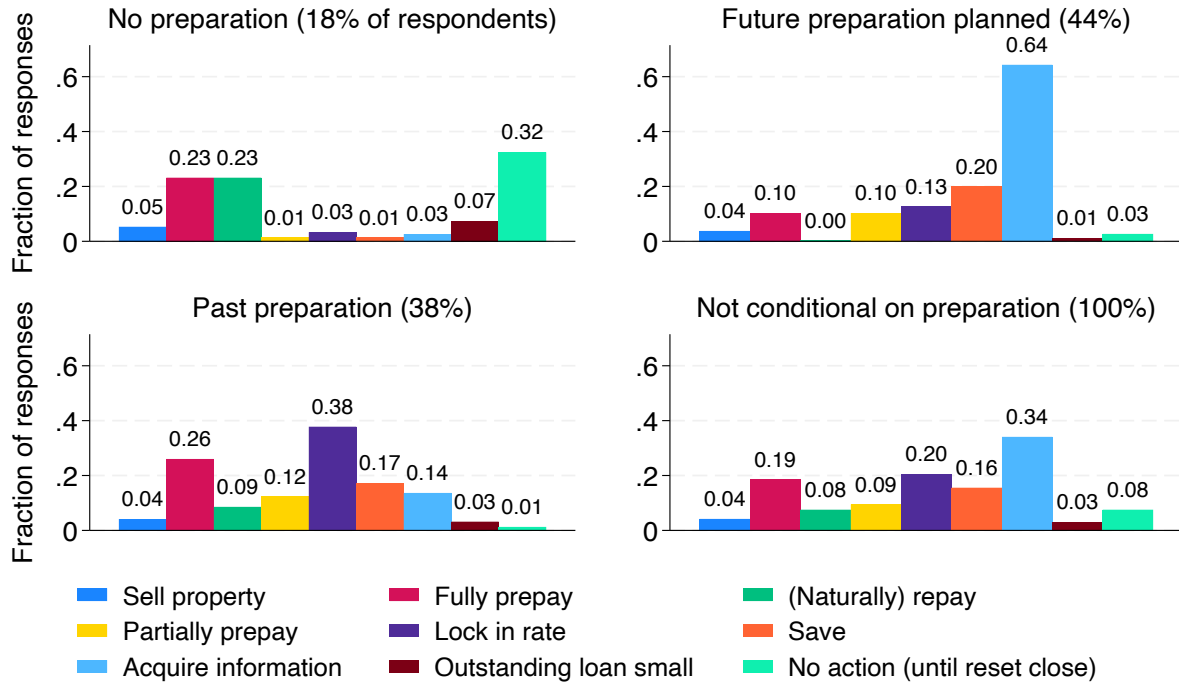
Notes: The figure plots modifications to loan characteristics upon refinancing. We restrict the sample to loans refinanced internally with fixation periods of up to 10 years and an expected residual balance of at least € 10,000 at expiration. The x-axis indicates the refinancing quarter, i.e., when the refinancing loan is originated, which may differ from the end of the fixation period. In each panel, the red line is the difference in the 10-year market rate between refinancing and origination quarter, averaged across loans refinancing in a quarter. The market rate is the average rate on loans originated by the partner bank. Panel A plots the average difference between the term on the new loan and that implied by passive refinancing absent loan modifications. Panel B plots the share of internally refinanced loans that involve a large balance reduction upon refinancing, defined as a reduction by more than 5% of the original loan amount. Panel C plots the average length of the new fixation period in years.

Figure 8: Perceptions and expectations of mortgage rates



Notes: The figure shows the distribution of beliefs about mortgage rates. The first stacked bar chart from the left plots responses to “How do you estimate the current mortgage rate in Germany compared to three years ago (i.e., in 2021)?” The remaining three charts show beliefs elicited based on the question: “How do you estimate the current rate on a new mortgage with a 10-year fixation period? And which interest rates do you expect in the future?” We ask respondents for point estimates, which we group into five buckets each for illustrative reasons. For the last two charts, we subtract the perceived current rate from the expected future rate, respectively.

Figure 9: Frequency of preparation types for prolongation, from open-text responses



Notes: The figure shows the frequency of categorized open-text explanations of whether and how mortgagors prepare for the interest-rate reset of their mortgage. We restrict the sample to respondents with at least € 10,000 expected mortgage debt at reset and whose responses we can classify, reducing the sample size to 887. We split the sample into three groups: categorized responses of those who state they have not prepared and will not do so are in the top-left figure, those who plan to prepare in the future are in the top-right, and those who have already prepared for a future reset are in the bottom-left. We pool across types of preparation in the bottom-right figure. Classified responses fall into the following categories: *sell property*; *fully prepay* in case of active payoff rather than through natural amortization alone; *(naturally) repay* if the borrower mentions paying off the mortgage without alluding to active repayment; *partially repay* for active repayment that reduces but not eliminates the loan balance; *lock in rate* through home-savings (Bausparen) or forward loan; *save* if the borrower builds up funds for the reset, whether in savings or investment products; *acquire information*, e.g., by comparing bank offers or seeking advice; *outstanding loan small* as a reason for not preparing; and *no action (until reset close)* to indicate no preparation, at least until being close to the rate reset. Each response can fall into more than one category.

Table 1: Summary statistics

	N	Mean	SD	P25	P50	P75
Panel A. Mortgage debt						
Original amount (€ k)	240,422	168.63	233.74	60.00	108.90	198.00
Outstanding amount (€ k)	240,422	132.42	205.27	35.74	79.08	156.00
Monthly payments (€)	240,422	895.00	1,195.42	400.00	641.55	1,022.05
Interest rate (%)	240,422	2.84	1.21	1.90	2.65	3.62
Fixation period (years)	240,422	11.64	4.41	10.00	10.08	15.00
Term (years)	240,422	19.99	9.03	12.65	19.25	26.94
Residual amount at expiration (€ k)	240,422	105.77	171.25	22.83	62.03	129.97
Residual term at expiration (years)	240,422	10.74	8.44	3.69	9.81	17.25
Ever on forward loan (0/1)	240,422	0.32	0.46	0.00	0.00	1.00
Allows partial prepayments (0/1)	240,422	0.56	0.48	0.00	1.00	1.00
Partly or fully rented out (0/1)	234,160	0.30	0.45	0.00	0.00	1.00
Panel B. Borrower characteristics						
Borrower age	239,991	51.14	11.55	42.50	51.00	58.50
Years at bank	199,486	12.72	9.55	5.00	10.50	18.00
Male (vs. female or couple) (0/1)	240,180	0.25	0.44	0.00	0.00	1.00
Couple (vs. female or male) (0/1)	240,180	0.62	0.49	0.00	1.00	1.00
Neighborhood quality (1–9)	220,572	5.91	2.44	4.00	6.00	8.00
Probability of default (%)	240,180	1.06	3.85	0.15	0.35	0.80
Deposits (€ k)	98,719	27.35	121.35	1.86	6.54	21.14
Has securities account (0/1)	240,422	0.07	0.25	0.00	0.00	0.00
Advisor meetings (yearly)	240,422	0.36	0.66	0.00	0.20	0.45
Simultaneously active loans	240,422	1.29	0.61	1.00	1.00	1.29

Notes: This table presents summary statistics for the main bank-data variables. We calculate variables as borrower-level averages across all years in which the borrower has outstanding regular mortgage debt. In case of multiple active loans in a given year for the same borrower, variables expressed in euros correspond to the total across loans; for all other variables, values are constructed as loan-amount-weighted averages. In Panel A, *residual amount at expiration* and *residual term at expiration* are the projected residual balance and the term at the end of the fixed rate period, respectively, assuming no extra payments. For loans with fixation period longer than 10 years, residual amount and residual term are those expected 10 years after origination. *Allows partial prepayments* is equal to one for borrowers with a partial prepayment right on at least one loan, typically allowing prepayments of 5% of the original loan amount. *Partly or fully rented out* is equal to one for borrowers with at least one loan on a rented-out property. In Panel B, *borrower age* is average age when the borrower is a couple. Both *neighborhood quality* (nine-point scale) and *probability of default* are from proprietary bank calculations. Probability of default is as of December 2017, or October 2024 when missing (18% of observations). *Deposits* is the total balance in checking and savings accounts at the bank, for borrowers who hold such accounts. *Has securities account* is equal to one if the borrower has a securities account with the bank. *Advisor meetings* is the average number of times a borrower meets with a bank advisor in a year.

Table 2: Refinancing outcomes over time

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Expiration year	N	Internal refi	Forward (FW)	Prolong	Prolong post-offer	Prolong bank offer	Prolong post-expiry	Ratio FW/prolong
2018	20,402	0.47	0.21	0.26	0.17	0.08	0.04	0.82
2019	25,634	0.45	0.23	0.22	0.14	0.06	0.03	1.05
2020	15,644	0.47	0.22	0.25	0.17	0.08	0.03	0.87
2021	14,943	0.45	0.22	0.23	0.16	0.09	0.03	0.93
2022	15,200	0.43	0.23	0.20	0.13	0.08	0.02	1.19
2023	14,898	0.39	0.24	0.15	0.11	0.06	0.02	1.56
2024	12,078	0.38	0.18	0.21	0.17	0.09	0.02	0.87
2025	2,778	0.39	0.12	0.27	0.22	0.14	0.02	0.46
Total	121,577	0.44	0.22	0.22	0.15	0.08	0.03	0.99

Notes: This table reports the distribution of refinancing outcomes by year of expiration. We restrict the sample to loans with a fixation period of up to 10 years and an expected residual balance of at least € 10,000 at expiration of the fixed rate period. To account for attrition due to loan extinction following property sales, we impose that loans be outstanding two years prior to expiration. Since our sample ends in May 2025, we only consider loans with expiration dates up to the end of March 2025. *Internal refi* is the fraction of loans refinanced internally; *Forward* is the fraction refinanced at least six months before expiration, via a forward loan (FW); *Prolong* is the fraction refinanced less than six months before expiration, via a regular prolongation; *Prolong post-offer* is the fraction refinanced internally six weeks or less before expiration (i.e., after receiving the prolongation offers) or after expiration; *Prolong bank offer* is the fraction of loans inferred to have accepted one of the bank's default prolongation offers, i.e., refinanced within six weeks of expiration with a loan amount within € 1,000 and a term within six months of the counterfactual loan; *Prolong post-expiry* is the fraction refinanced internally after expiration; and *Ratio FW/prolong* is the ratio of loans refinanced through forward loans to those refinanced through prolongation (i.e., column 4 divided by column 5).

Table 3: Proportional hazard models of internal refinancing

Panel A: Forward window

	(1)	(2)	(3)	(4)	(5)	(6)
$R_t^m - R_i$	-0.217*** (0.022)		-0.279*** (0.012)	-0.284*** (0.014)	-0.199*** (0.010)	-0.134*** (0.019)
ΔR_t^m		0.811*** (0.239)	2.233*** (0.229)	2.380*** (0.247)		
Controls	No	No	No	Yes	Yes	Yes
Year-month FE	No	No	No	No	Yes	Yes
Vintage-specific BH	No	No	No	No	No	Yes
Observations	2,986,228	2,986,228	2,986,228	2,985,009	2,985,009	2,985,009

Panel B: Prolongation window

	(1)	(2)	(3)	(4)	(5)	(6)
$R_t^m - R_i$	-0.077*** (0.008)		-0.079*** (0.008)	-0.091*** (0.010)	0.014 (0.012)	-0.087*** (0.015)
ΔR_t^m		-0.088 (0.077)	0.051 (0.070)	0.095 (0.072)		
Controls	No	No	No	Yes	Yes	Yes
Year-month FE	No	No	No	No	Yes	Yes
Vintage-specific BH	No	No	No	No	No	Yes
Observations	517,532	517,532	517,532	517,316	517,316	517,316

Notes: Reported estimates are hazard ratios minus one, representing proportional changes in the internal-refinancing hazard associated with a one-unit increase in the corresponding regressor, holding other covariates fixed. $(R_t^m - R_i)$ is the percentage-point gap between the market rate on a 10-year mortgage originated by the partnering bank at time t and the interest rate a borrower pays on their old loan. ΔR_t^m is the 3-month change in the market rate (from month $t - 3$ to month t). The sample is restricted to loans with fixation periods of up to 10 years and an expected residual balance of at least €10,000 at expiration. *Controls* include: expected residual loan balance at expiration (log), rate fixation period, occupancy type (owner-occ., fully rented out, partly rented out), borrower age, gender, couple status, proxy for neighborhood quality (9-point scale), indicator for missing neighborhood quality information, dummies for two-digit zip codes, probability of default (deciles), whether the borrower has other mortgage debt (from another regular mortgage loan or home savings contract) outstanding with the bank (0/1), and whether the borrower has non-mortgage products (checking, savings or securities account, or an insurance product) with the bank (0/1); the last three variables are measured four years before expiration or at the earliest available observation. *Year-month FE* are fixed effects capturing the calendar year and month of an observation. *Vintage-specific BH* means the baseline hazard is allowed to vary across origination vintages. Standard errors (two-way clustered at the loan and calendar-month levels) in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Determinants of preparation for end of rate fixation and awareness of rates

<i>Dependent variable:</i> <i>Specification:</i>	Preparation for rate-fixation end			Rate change	
		No prepay	<5y fix	Past 3y	Next 2y
	(1)	(2)	(3)	(4)	(5)
Perceived rate change past 3y (0–4)	0.042** (0.019)	0.039* (0.020)	0.081*** (0.027)		
Perceived rate today (%)	0.018 (0.014)	0.021 (0.015)	0.017 (0.016)		
Mortgage rate in 2y vs. today (%)	0.042** (0.019)	0.041** (0.020)	0.055** (0.026)		
Uncertainty about 2y f/cast (0–4)	0.000 (0.019)	–0.005 (0.021)	–0.013 (0.030)		
Risk tolerance (0–4)	0.000 (0.018)	–0.011 (0.019)	–0.036 (0.027)	0.023 (0.024)	–0.045** (0.022)
Debt aversion (0–4)	0.004 (0.016)	–0.017 (0.018)	–0.009 (0.024)	–0.022 (0.021)	0.026 (0.019)
Trust in bank (0–4)	–0.024 (0.015)	–0.035** (0.016)	–0.011 (0.025)	–0.050** (0.022)	0.034* (0.019)
Follow savings plan (0–4)	0.035** (0.015)	0.033** (0.016)	0.022 (0.022)	0.035* (0.020)	–0.011 (0.018)
Financial literacy (0–4)	0.003 (0.022)	–0.003 (0.023)	–0.012 (0.033)	0.110*** (0.030)	–0.064** (0.026)
Higher-rate effect on prices (0–4)	–0.033** (0.014)	–0.034** (0.015)	–0.045** (0.022)	–0.024 (0.020)	0.046** (0.018)
Higher-rate effect on RE (0–4)	–0.006 (0.016)	–0.005 (0.017)	0.033 (0.024)	–0.046** (0.022)	0.042** (0.020)
Own stocks (0/1)	0.036 (0.042)	0.021 (0.044)	0.156** (0.067)	0.003 (0.057)	–0.125** (0.052)
Investment property (0/1)	–0.013 (0.038)	0.011 (0.041)	0.054 (0.059)	0.027 (0.052)	–0.049 (0.047)
Income net of expenses (0–8)	0.020** (0.009)	0.015 (0.009)	0.018 (0.013)	0.000 (0.011)	–0.018 (0.011)
University completed (0/1)	–0.027 (0.039)	–0.012 (0.042)	0.033 (0.058)	0.085 (0.056)	–0.058 (0.049)
Borrower age (years)	–0.001 (0.002)	–0.004** (0.002)	–0.002 (0.003)	–0.005* (0.002)	0.001 (0.002)
Probability of default (deciles)	0.000 (0.007)	0.006 (0.007)	–0.012 (0.011)	0.011 (0.010)	–0.010 (0.009)
Checking/savings account (0/1)	0.030 (0.037)	0.032 (0.040)	–0.065 (0.058)	–0.061 (0.051)	–0.076* (0.045)
Mortgage rate (%)	–0.002 (0.022)	0.006 (0.023)	0.002 (0.032)	–0.076*** (0.027)	0.008 (0.023)

continued

Table 4 continued

<i>Dependent variable:</i> <i>Specification:</i>	Preparation for rate-fixation end			Rate change	
		No prepay	<5y fix	Past 3y	Next 2y
	(1)	(2)	(3)	(4)	(5)
Amount outstanding (log)	−0.045* (0.024)	−0.017 (0.026)	−0.077** (0.032)	0.088*** (0.028)	−0.073*** (0.023)
Mortgage origination (years)	−0.008 (0.006)	−0.014** (0.007)	−0.011 (0.008)	−0.011 (0.008)	0.000 (0.007)
Remaining maturity (years)	−0.002 (0.004)	−0.002 (0.005)	0.008 (0.015)	0.005 (0.006)	0.018*** (0.005)
Avg. Y	0.53	0.45	0.52	3.22	−0.05
Observations	836	722	363	1,595	1,595
R-squared	0.06	0.06	0.13	0.10	0.10

Notes: This table reports regression estimates of preparation for the end of the rate-fixation period and perceived and expected rate changes. In Columns 1–3, the dependent variable is an indicator equal to one if respondents explain in open-ended text they have prepared or plan to prepare for the end of the fixation period by considering selling their property, partially or fully prepaying their mortgage, locking in rates early, or saving. The sample includes respondents whose open-text explanations we can classify and with expected debt at reset of at least €10,000. Column 2 further excludes respondents from the sample who write they prepare by fully prepaying. Column 3 restricts the sample to respondents whose fixation end is in less than five years. Columns 4 and 5 restrict the sample to respondents with regular mortgages, reducing the sample size to 1,595. In Column 4, the dependent variable is the response, on a five-point scale from “much lower” (0) to “much higher” (4), to the question “How do you estimate the current mortgage rate in Germany compared to three years ago (i.e., in 2021)?” In Column 5, the dependent variable is the percentage-point difference between the expected mortgage rate in Germany in two years and the perceived rate today (1% tails winsorized). *Higher rate effect on prices* and *on RE* are responses spanning “strong decrease” (0) to “strong increase” (4) to a question on the effect of an unexpected increase in interest rates in the economy on the inflation rate and house prices (see Question 22 in [Appendix D](#)). *Probability of default* is from proprietary bank calculations. *Mortgage origination* captures the time passed since origination (in years) of the mortgage. We control for indicators of letter reception. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effects of changes in hypothetical refinancing situation

<i>Dependent variable (0/1):</i>	<i>Advisor</i>	<i>Search</i>	<i>Prepay</i>	<i>Forward</i>	<i>Spend ↓</i>	<i>Income ↑</i>	<i>Fix ≥10y</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
V2: 1.5% to 3.5% in 3y	−0.152*** (0.026)	−0.056*** (0.020)	−0.006 (0.024)	−0.037 (0.027)	−0.052* (0.028)	−0.070*** (0.027)	−0.001 (0.027)
V3: 3% to 3.5% in 1y	−0.045* (0.024)	−0.016 (0.019)	−0.039 (0.025)	0.096*** (0.027)	−0.003 (0.028)	−0.055** (0.027)	0.063** (0.027)
Controls	Y	Y	Y	Y	Y	Y	Y
Avg. Y V1	0.78	0.88	0.71	0.47	0.46	0.48	0.56
Observations	1,921	1,921	1,921	1,921	1,921	1,921	1,921
R-squared	0.07	0.08	0.13	0.10	0.08	0.14	0.10

Notes: The table reports estimates of regressions of actions in the context of a hypothetical refinancing situation. Other than in Column 7, the dependent variables are likelihoods of action measured on a four-point scale, covering “very unlikely,” “rather unlikely,” “rather likely,” and “very likely.” We construct an indicator for each action, equal to one if the action is rather or very likely, and zero if it is rather or very unlikely. *Advisor* refers to meeting an advisor from the current lender; *search* refers to comparing offers by different lenders; *prepay* is reducing the loan balance; *forward* means locking in rates using a forward loan; *spend ↓* is lowering spending; *income ↑* is increasing income; and *fix ≥10y* equals one if the borrower would choose a fixation period of 10 years or for as long as possible, and equals zero for one or five years. V2: 1.5% to 3.5% in 3y indicates vignette 2. V3: 3% to 3.5% in 1y indicates vignette 3. The effects are relative to vignette 1, which describes a rate increase from 1.5% to 3.5% in 1y. We list the *controls* in [Section 4.2](#). Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Letter effects observed in survey data

DV: Scale:	Letter read 0/1	Rate change Std.	Current rate 0/1	Pay change %	Prepay know Std.	Forward know Std.	Prepay hypo Std.	Forward hypo Std.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Indicator for each letter								
Letter 1	0.296*** (0.038)	-0.086 (0.092)	-0.012 (0.040)	0.006 (0.029)	0.149 (0.100)	0.129 (0.093)	0.191** (0.092)	-0.104 (0.093)
Letter 2	0.314*** (0.039)	-0.057 (0.096)	0.019 (0.041)	-0.016 (0.030)	0.111 (0.105)	0.102 (0.091)	0.154 (0.101)	-0.176* (0.100)
Letter 3	0.371*** (0.039)	-0.077 (0.092)	-0.051 (0.042)	0.012 (0.028)	0.210** (0.095)	0.107 (0.095)	0.144 (0.095)	0.002 (0.096)
Panel B. Indicator for any letter								
Any letter	0.328*** (0.028)	-0.074 (0.076)	-0.016 (0.034)	0.002 (0.024)	0.159* (0.086)	0.113 (0.079)	0.163** (0.079)	-0.089 (0.079)
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Avg. Y CG	0.08	0.00	0.77	0.05	0.00	0.00	0.00	0.00
N	881	881	881	659	881	881	881	881
R2	0.15	0.10	0.09	0.12	0.13	0.20	0.10	0.10

Notes: This table reports estimates from regressions of survey variables on letter dummies. We standardize dependent variables that are not binary or percentage changes. *Letter read* is an indicator that the respondent states they have read a letter from the bank on increases in mortgage rates in the past two months. *Rate change* is a qualitative estimate of the level of the mortgage rate today relative to three years ago, measured on a five-point scale ranging from “much lower” to “much higher.” *Current rate* is one if the perceived current rate on a new mortgage with a 10-year fixation period in Germany is 3–4%. *Pay change* captures the expected change in monthly payments relative to the current payment, estimated by respondents who report they need to refinance. Respondents first state whether they expect payments to go up, stay roughly the same, go down, or that they do not need to refinance. Those who expect payments to go up or go down then choose from a list the expected change in euros (in buckets of 50). *Prepay know* and *forward know* are self-reported measures of awareness of prepayment options / forward loans on a four-point scale, spanning “completely unknown” to “completely known.” *Prepay hypo* is reducing the loan balance and *forward hypo* means locking in rates using a forward loan, both measured as actions on a four-point scale from “very unlikely” to “very likely” as part of the hypothetical refinancing situation. Panel A shows the effect of each letter relative to the control group. Panel B shows the effect of an indicator that pools all letter groups. We explain the letter RCT in [Section 5](#). *Controls* include the bank controls described in [Section 4.2](#), measured in October 2024 prior to the intervention; household income and university education from the survey; and vignette-variant indicators for outcomes from the vignette section. *Avg. Y CG* shows the average of the dependent variable in the control group, except for the standardized variables for which the average is zero in the estimation sample. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Letter effects observed in bank data

<i>DV (0/100):</i>	Meet advisor	Refi	Pre-offer refi	Meet and refi	Extra paym	Sav product
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Aggregate effects of letter intervention						
Any letter	−0.059 (0.316)	0.159 (0.117)	0.085 (0.064)	0.235*** (0.090)	0.154 (0.406)	0.054 (0.125)
Panel B. Heterogeneous effects by distance to expiration						
Any letter	−0.185 (0.355)	−0.034 (0.039)	−0.012 (0.021)	0.003 (0.022)	0.181 (0.460)	0.127 (0.158)
Until May 2025	25.055*** (2.078)	22.505*** (1.815)	2.054*** (0.646)	11.461*** (1.390)	−3.958** (1.713)	0.167 (0.580)
June 2025 - May 2029	2.677*** (0.641)	0.436*** (0.156)	0.541*** (0.140)	0.402*** (0.128)	5.004*** (0.899)	−0.126 (0.245)
Any letter × Until May 2025	4.404* (2.385)	3.546* (2.113)	2.016** (0.812)	4.513*** (1.655)	2.610 (2.004)	−0.194 (0.658)
Any letter × June 2025 - May 2029	−0.320 (0.697)	0.092 (0.162)	0.003 (0.157)	0.071 (0.139)	−0.552 (0.996)	−0.201 (0.267)
Controls	Y	Y	Y	Y	Y	Y
Avg. Y CG	11.14	1.32	0.35	0.71	12.75	1.41
Observations	47,422	47,422	47,422	47,422	34,594	47,422

Notes: This table reports estimates from regressions of post-RCT outcomes in the bank data on indicators for being in the letter treatment group. All outcome variables measure actions between November 15, 2024 (the date of letter receipt) and May 31, 2025 (end of sample). *Meet advisor* indicates whether borrowers meet with a bank advisor at least once. *Refi* is an indicator for whether borrowers refinance internally. *Pre-offer refi* indicates whether borrowers refinance internally at least six weeks before loan expiration. *Meet and refi* indicates whether a borrower meets with an advisor and refinances within seven days from the meeting. *Extra paym* indicates whether borrowers make use of their right to make extra payments during the fixation period. The sample in Column (5) is restricted to borrowers who are allowed to make such payments. *Sav product* indicates whether borrowers open a new savings account, securities account or home savings account (*Bausparen*). Panel B interacts the letter dummy with indicators of when borrowers face the end of fixed rate period. The omitted group is borrowers whose fixed-rate period expires after May 2029. *Avg. Y CG* shows the average of the dependent variable in the control group of non-recipients. *Controls* include: current interest rate, expected residual loan balance at expiration (log), rate fixation period, borrower age, gender, couple status, proxy for neighborhood quality (9-point scale), indicator for missing neighborhood quality information, probability of default (deciles), whether the borrower has checking or savings accounts (0/1), a securities account (0/1), or multiple mortgage loans (including both regular mortgage loans and home savings contracts with immediate borrowing) outstanding with the bank (0/1); the last four variables are measured pre-treatment in October 2024. All specifications in Panel A further include expiration-year fixed effects. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Heterogeneity in propensity to read letter

DV (0/1):	Having read bank letter in past two months on topic of mortgage-rate increases						
Trait:	Loan size	Loan reset	Loan repay	Literacy	Income	Rate change	Current rate
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Letter	0.336*** (0.027)	0.341*** (0.028)	0.301*** (0.032)	0.239*** (0.037)	0.252*** (0.040)	0.191*** (0.066)	0.233*** (0.058)
Trait	-0.017 (0.017)	0.083 (0.053)	-0.015 (0.039)	-0.001 (0.037)	-0.042 (0.037)	0.005 (0.053)	-0.036 (0.048)
Letter x trait	0.070*** (0.026)	-0.019 (0.070)	0.105* (0.057)	0.182*** (0.052)	0.152*** (0.053)	0.168** (0.072)	0.133** (0.065)
Observations	881	881	877	884	884	884	884
R-squared	0.10	0.10	0.10	0.12	0.10	0.10	0.10

Notes: This table reports estimates from regressions of the propensity to read the bank letter. Specifically, the dependent variable equals one if the respondent states they have read a letter from the bank over the past two months on the topic of mortgage-rate increases, and zero otherwise (no letter received, letter received but not read, does not remember). *Letter* is one if the respondent has received any of the three letters, and zero if not. We explain the letter RCT in [Section 5.1](#). The sample includes the 884 survey respondents who are part of the RCT; columns with bank-data traits lose respondents we cannot match to the bank data. *Trait* is a loan (Columns 1–3) or borrower (Columns 4–7) characteristic we interact with *letter*: *Loan size* is standardized current total mortgage debt. *Loan reset* is one if the end to the interest-rate fixation period is within two years. *Loan repay* equals one if the borrower has reduced the loan balance in the past using an annual partial repayment option (Sondertilgung). *Literacy* captures above-median self-reported financial literacy. *Income* is one for above-median household income net of general expenses. *Rate change* is one if the respondent believes the current mortgage rate in Germany compared to three years ago (i.e., in 2021) is higher. *Current rate* is one if the perceived current rate on a new mortgage with a 10-year fixation period in Germany is 3–4%. We include no controls, given randomized letter assignment. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Online Appendix:

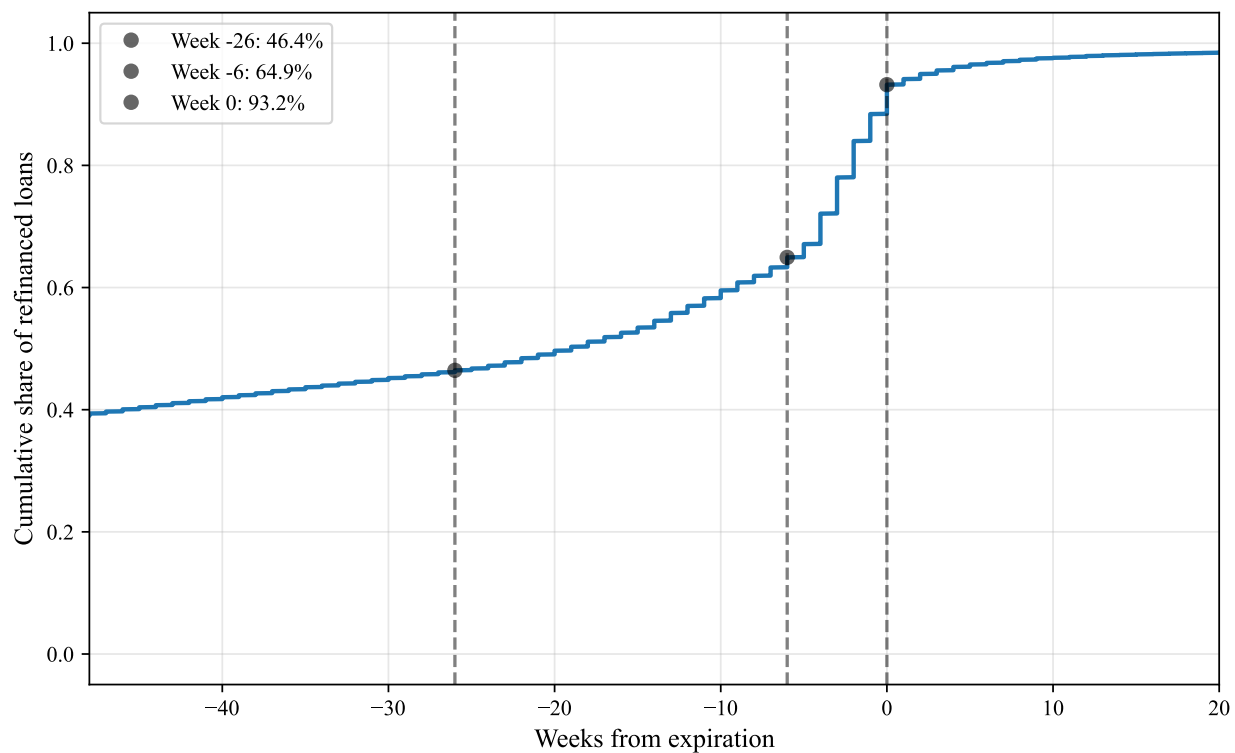
The Response of Debtors to Rate Changes

Andreas Fuster, Virginia Gianinazzi, Andreas Hackethal,
Philip Schnorpfeil, and Michael Weber

Not for Publication

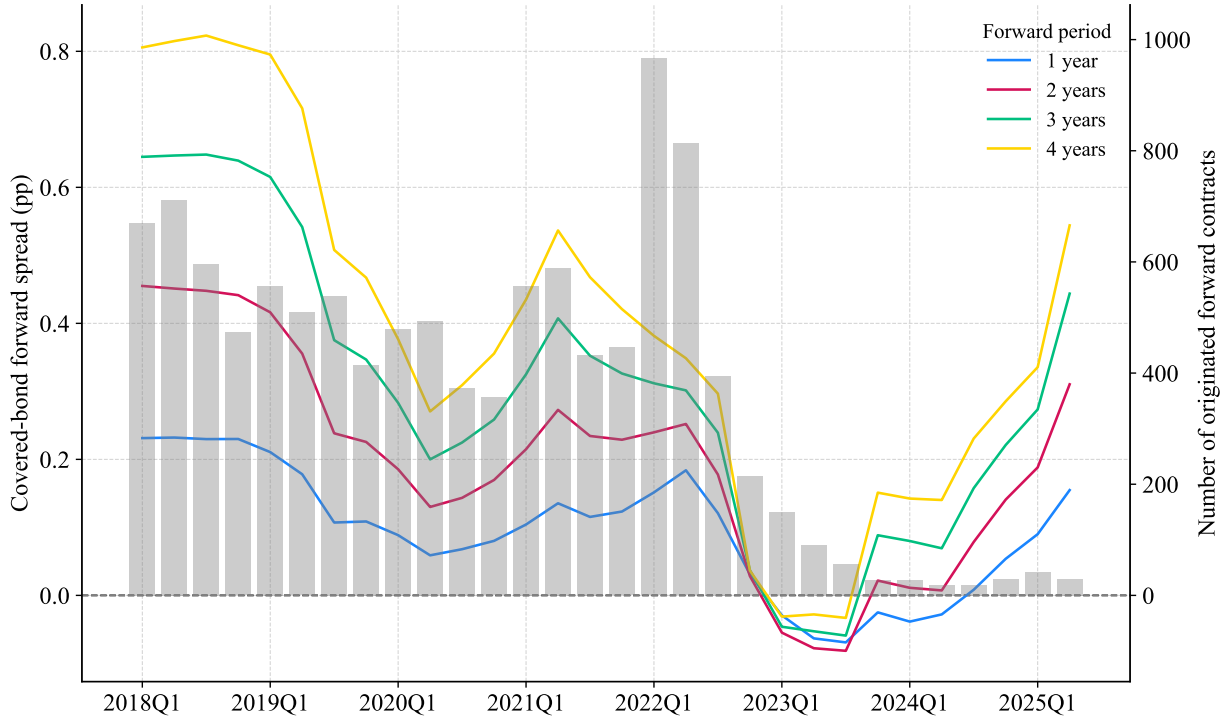
A Appendix figures and tables

Figure A.1: Timing of internal refinancing



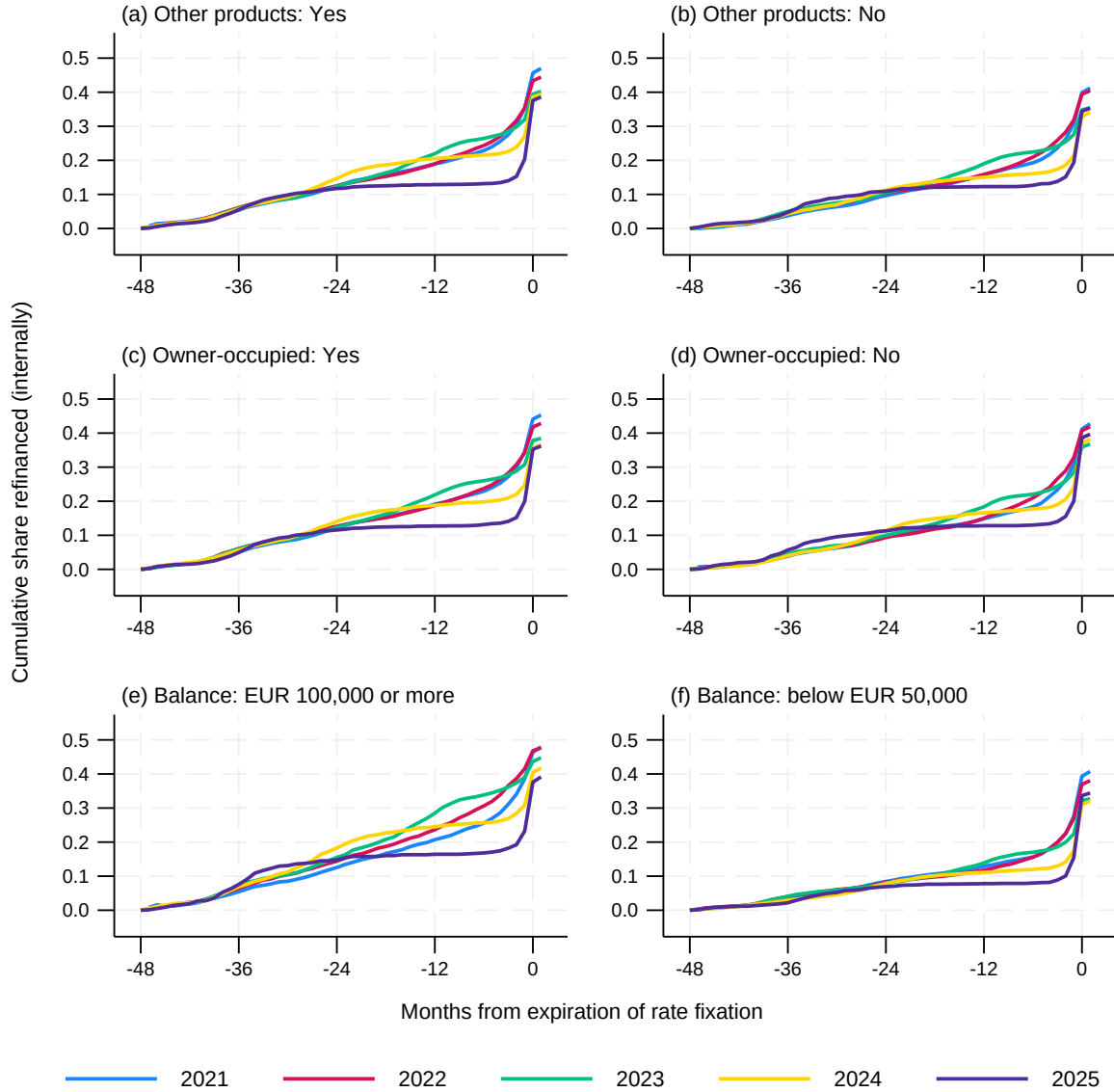
Notes: The figure plots the cumulative share of internally refinanced loans by distance to expiration of the fixation period (in weeks). The sample applies the same restrictions as in [Table 2](#) and is limited to loans refinanced internally. *Week -26* indicates when refinancing via regular prolongation becomes possible; *week -6* indicates when the bank sends prolongation offers; *week 0* corresponds to the end of the fixation period.

Figure A.2: Covered-bond forward spreads and forward loan originations over time



Notes: The figure plots covered-bond forward spreads, $f_{k,10} - R_{10}$, for forward periods of $k = 1, 2, 3$, and 4 years. $f_{k,10}$ is the 10-year rate starting in k years implied by the zero-coupon covered-bond yield curve, and R_{10} is the current 10-year zero-coupon yield (source: Deutsche Bundesbank). The gray bars (right axis) show the number of forward loans with a 10-year fixation originated by the partner bank in each quarter.

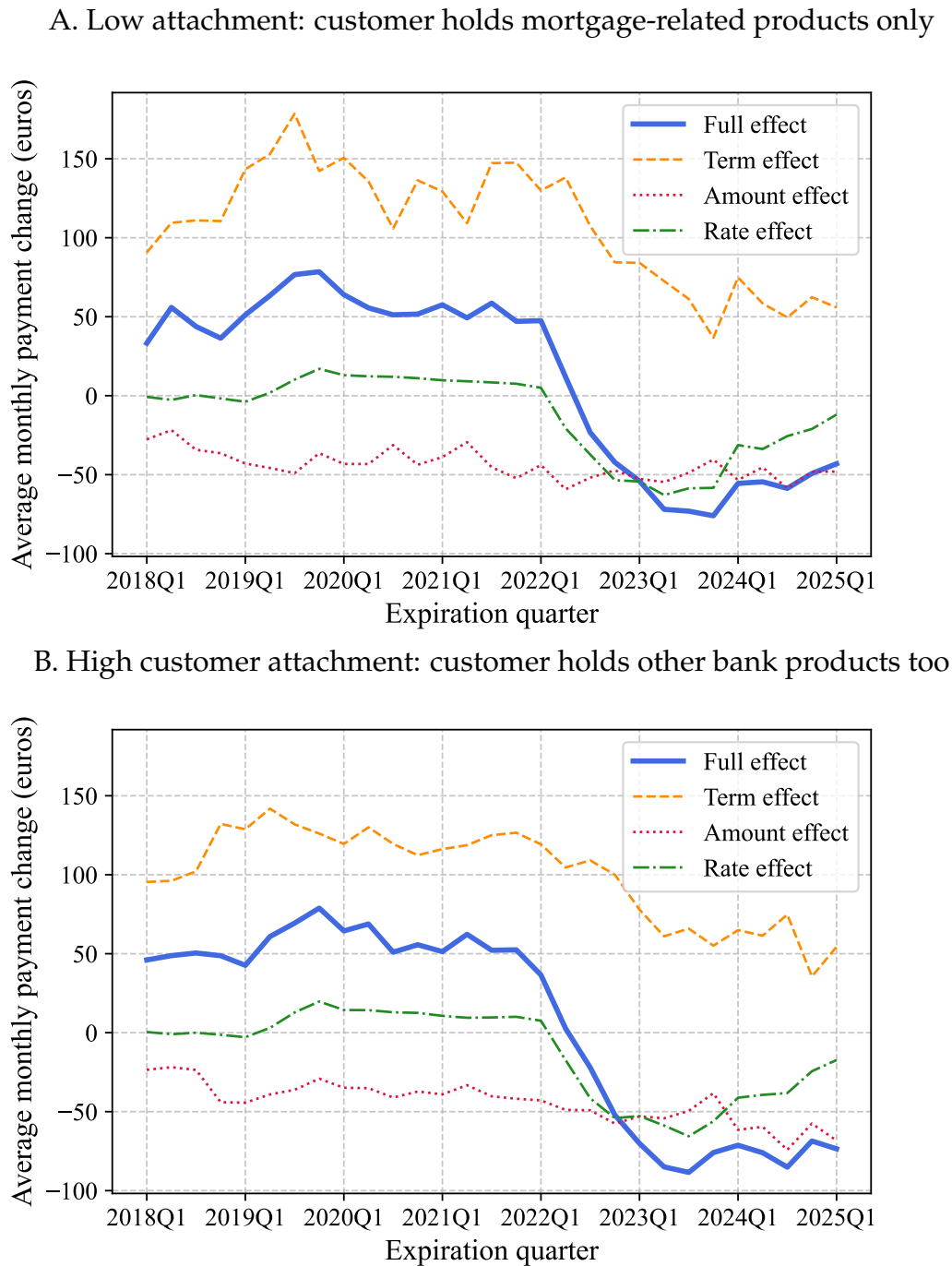
Figure A.3: Proportional hazard models of internal refinancing: Subsamples



Forward-window Cox models						
	Other products		Owner-occupied		Balance	
	Yes	No	Yes	No	≥ 100	< 50
$R_t^m - R_i$	-0.285*** (0.015)	-0.282*** (0.014)	-0.288*** (0.014)	-0.276*** (0.017)	-0.271*** (0.018)	-0.294*** (0.015)
ΔR_t^m	2.438*** (0.281)	2.314*** (0.227)	2.402*** (0.248)	2.358*** (0.291)	2.576*** (0.312)	2.175*** (0.248)
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Charts: Owner-occupied No = partly/fully rented out. Other products: checking/savings, brokerage, or insurance. Table: forward window (> 6 months); entries are $\exp(\beta) - 1$, not percentages. Standard errors (two-way clustered by calendar month and loan) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

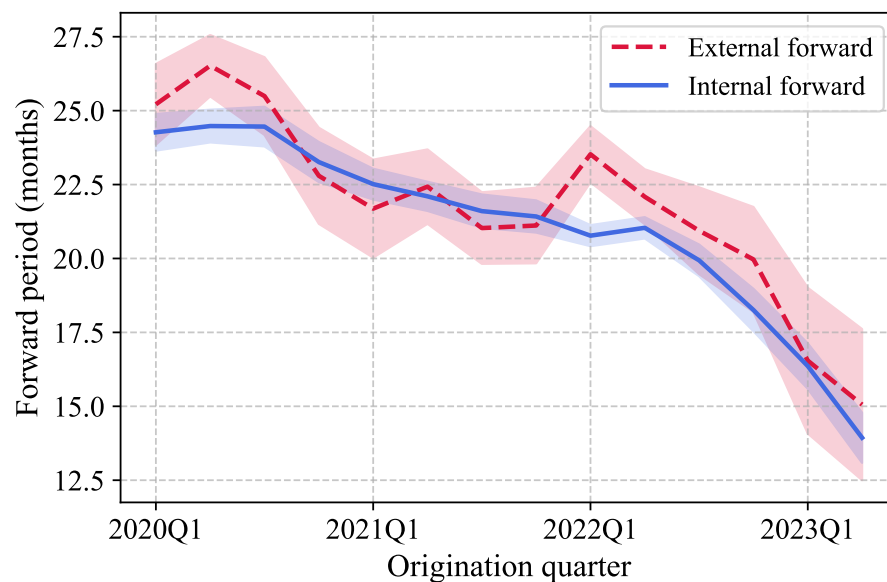
Figure A.4: Effect of refinancing choices on monthly payments, by bank attachment



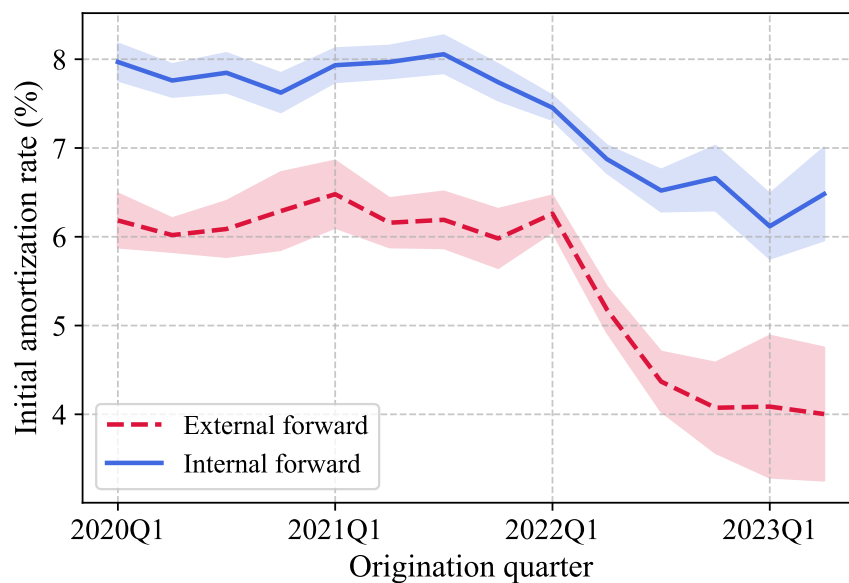
Notes: This figure reproduces [Figure 6](#) separately by borrowers' attachment to the bank. Borrowers with low bank attachment (Panel A) hold only mortgage-related products with the bank, i.e., besides their expiring regular mortgage, they have public-subsidy (KfW) loans, home-savings (Bausparen) loans, or no other product. Borrowers with high attachment (Panel B) additionally hold non-mortgage products with the bank, i.e., a checking or savings account, insurance, or a securities account. Attachment is measured four years before loan expiration, or at the earliest available observation. Panel A contains 19,910 observations and Panel B 22,252 observations.

Figure A.5: Forward loans to existing clients (internal) and switching borrowers (external)

A. Months between forward loan origination and start of the new fixation

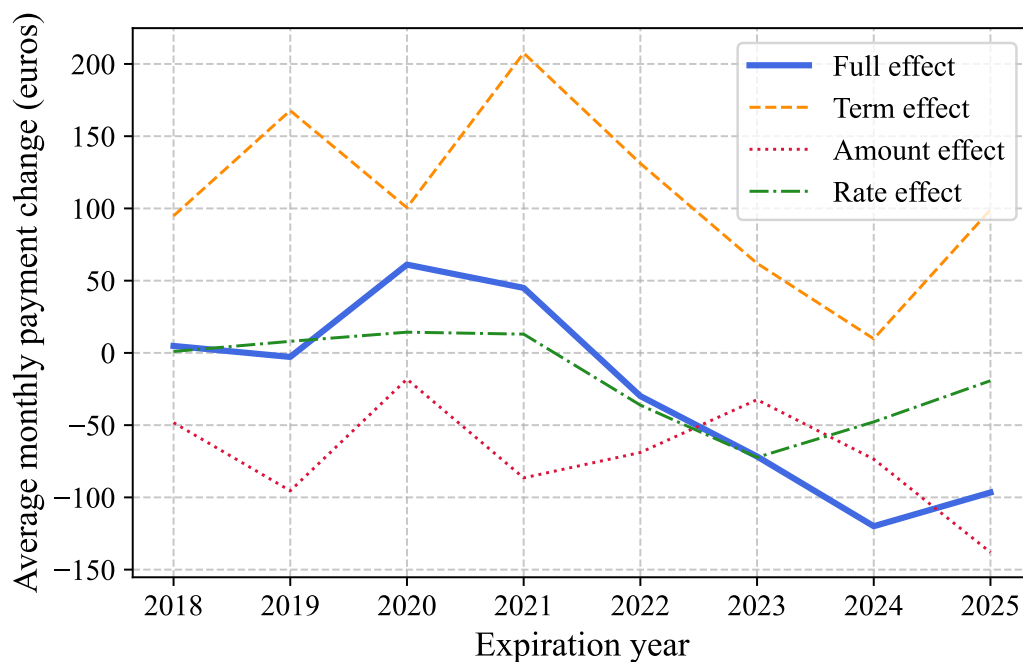


B. Annual principal repayment at origination, as a share of the loan amount (%)



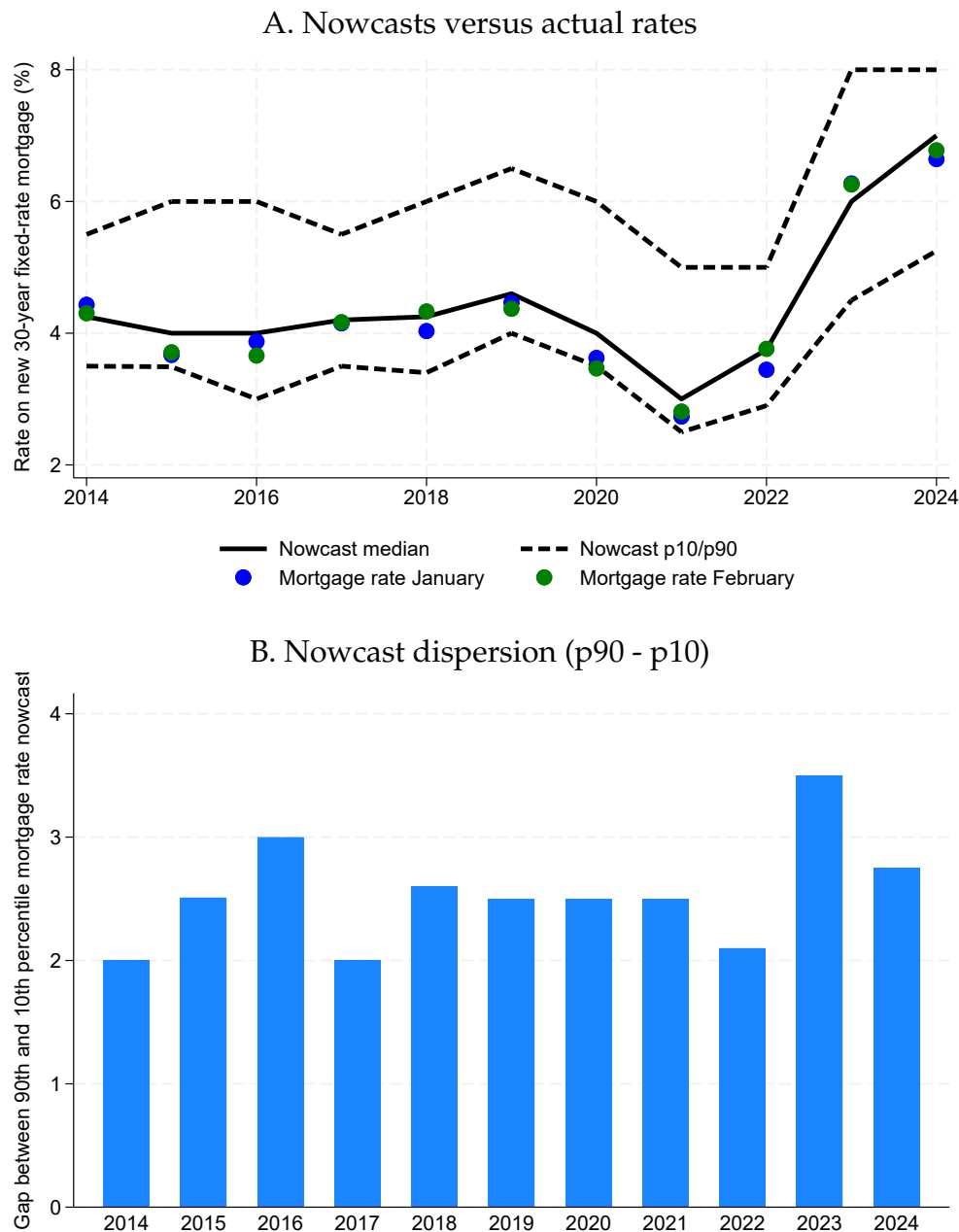
Notes: This figure compares average characteristics, by origination quarter, of forward loans originated by the partner bank to existing clients (internal forwards, blue solid line) and to borrowers switching from another lender (external forwards, red dashed line). Panel A shows the forward period, that is, the number of months between forward loan origination and the start of the new fixed-rate period. Panel B shows the initial amortization rate (*anfänglicher Tilgungssatz*), the annual principal repayment implied by the first monthly installment, as a share of the loan amount. Shaded areas show 95% confidence intervals.

Figure A.6: Effect of refinancing choices on monthly payments, survey respondents



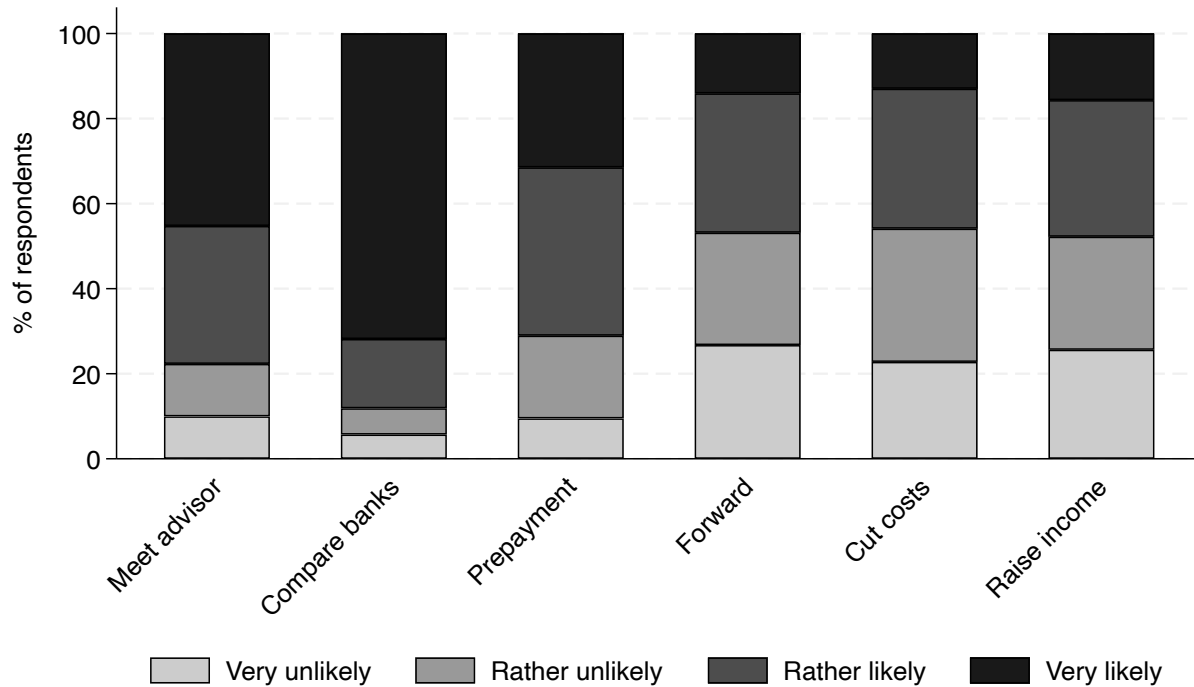
Notes: This figure reproduces [Figure 6](#) for borrowers who completed the survey. Specifically, the sample comprises survey respondents' loans with a fixation period of up to 10 years and an expected residual balance of at least €10,000 at expiration, which expired between 2018 and May 2025 and were refinanced internally. Because this subsample is small (494 loans), effects are averaged by year rather than quarter of expiration.

Figure A.7: Mortgage rate awareness in the NY Fed SCE



Notes: This figure shows the accuracy of mortgage rate nowcasts from the NY Fed Survey of Consumer Expectations (SCE) Housing Module, 2014–2024. The analysis is restricted to mortgagors (roughly 500–600 respondents per year) and uses the provided survey weights to compute the distribution of responses. Respondents are asked “What do you think is the average interest rate (for all borrowers) on a new 30-year fixed-rate mortgage as of today?” Panel A plots actual rates (the commonly used headline rate from the Freddie Mac PMMS, obtained via FRED) against the 10th, 50th, and 90th percentiles of respondent nowcasts. Since the survey is fielded in February, the figure shows both the mortgage rate in January and in February. Panel B shows the difference between the 90th and 10th percentiles of nowcasts for each survey year. For comparison, in our German survey from November/December 2024, this dispersion is 1.6 pp.

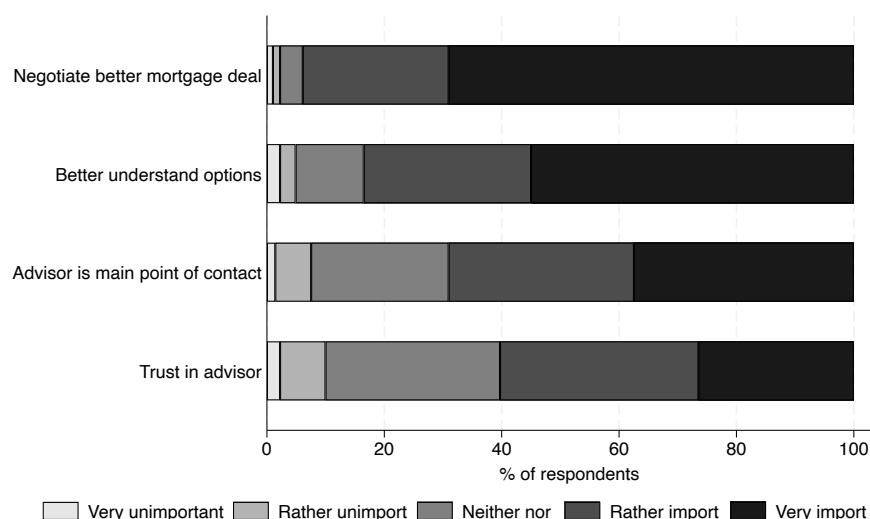
Figure A.8: Actions as part of hypothetical refinancing situation



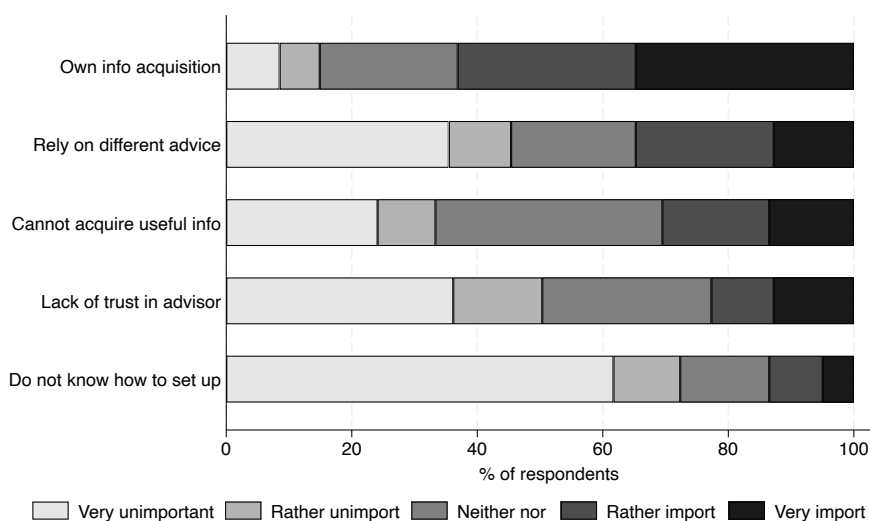
Notes: The figure shows the distribution of actions in the context of a hypothetical refinancing situation. We restrict the sample to 643 respondents who receive the baseline scenario of a hypothetical €100,000 mortgage with an interest rate of 1.5% that resets in one year to a rate of 3.5% (vignette 1). We study likelihoods of action measured on a four-point scale, covering “very unlikely,” “rather unlikely,” “rather likely,” and “very likely.” The actions are as follows. *Meet advisor* refers to meeting an advisor from the current lender; *compare banks* refers to comparing offers by different loan providers; *prepayment* is reducing the loan balance; *forward* means locking in rates using a forward loan; *cut costs* is lowering spending; and *raise income* is increasing income.

Figure A.9: Reasoning about advisor meeting as part of hypothetical refinancing situation

Panel A. Reasons of the 78% of borrowers who would likely meet advisor



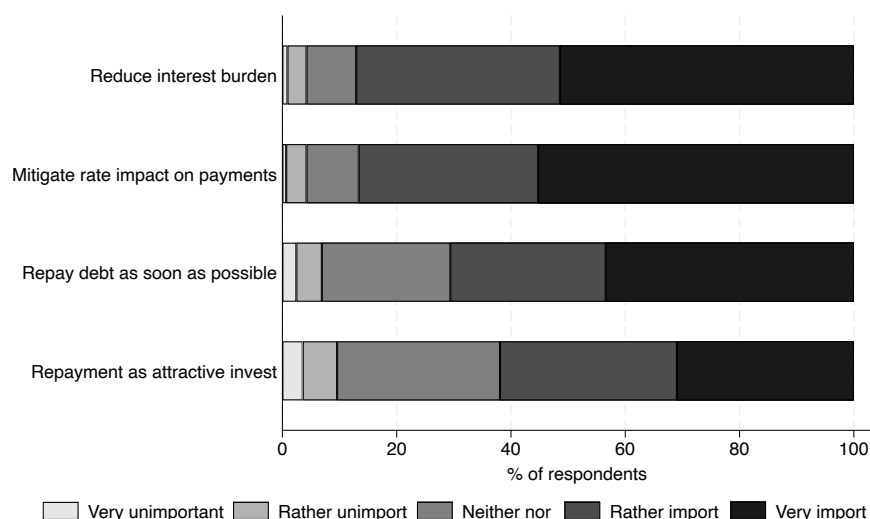
Panel B. Reasons of the 22% of borrowers who would be unlikely to meet advisor



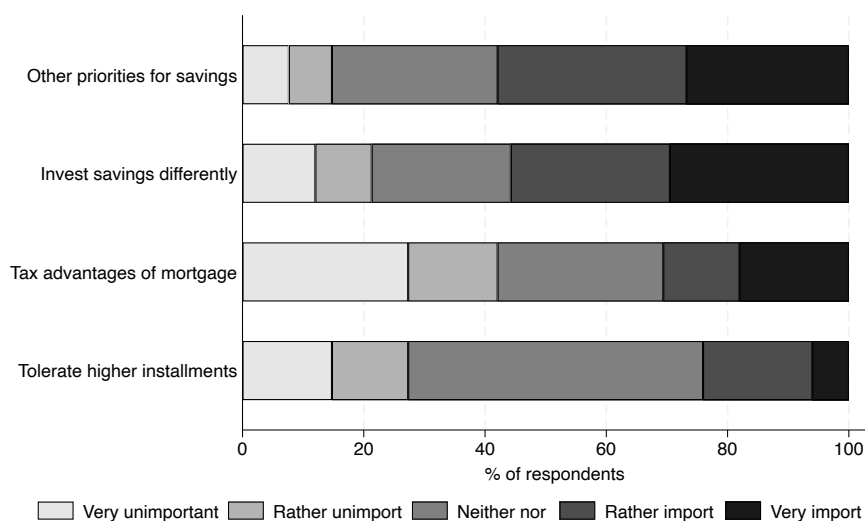
Notes: The figure shows how respondents reason about meeting an advisor from the current lender in the context of a hypothetical refinancing situation. We restrict the sample to 643 respondents who receive the baseline scenario of a hypothetical €100,000 mortgage with an interest rate of 1.5% that resets in one year to a rate of 3.5% (vignette 1). Among them, the 78% of borrowers who state they would be “rather” or “very” likely to meet an advisor subsequently explain why (Panel A). Accordingly, the remaining 22% of borrowers who would be “rather” or “very” unlikely to meet an advisor explain why not (Panel B). For each predefined reason, respondents choose from five answers ranging from “very unimportant” to “very important.”

Figure A.10: Reasoning about prepayment as part of hypothetical refinancing situation

Panel A. Reasons of the 71% of borrowers who would likely prepay



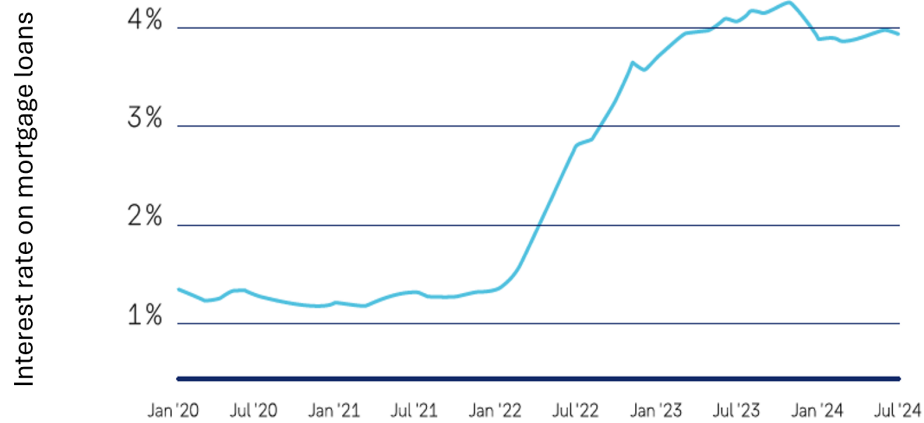
Panel B. Reasons of the 29% of borrowers who would be unlikely to prepay



Notes: The figure shows how respondents reason about prepaying a mortgage in the context of a hypothetical refinancing situation. Our definition of prepayments, which respondents read, is broad, encompassing any voluntary, extra reduction in the loan balance. We restrict the sample to 643 respondents who receive the baseline scenario of a hypothetical €100,000 mortgage with an interest rate of 1.5% that resets in one year to a rate of 3.5% (vignette 1). Among them, the 71% of borrowers who state they would be “rather” or “very” likely prepay subsequently explain why (Panel A). Accordingly, the remaining 29% of borrowers who would be “rather” or “very” unlikely to prepay explain why not (Panel B). For each predefined reason, respondents choose from five answers ranging from “very unimportant” to “very important.”

Figure A.11: Graphics shown in the letter

Panel A. Recent increase in mortgage rates in Germany



Source: Deutsche Bundesbank SUD 131

Panel B. Initial impact of higher mortgage rates on monthly payments

	Initial effect on interest payments	
	Annual	Monthly
Interest rate increase from 2% to 4% on a €100,000 loan amount	+ 2,000 euros	+ 167 euros

Notes: The figures show the graphics embedded in the letter. We describe the letter experiment in [Section 5.1](#) and show the letter content in [Appendix E](#). Panel A shows the development of the average effective fixed interest rate including costs on newly originated mortgages for households in Germany. The data come from Deutsche Bundesbank (SUD 131). Panel B illustrates how an increase in the interest rate by 2 pp of a €100,000 mortgage affects interest payments initially on an annualized and on a monthly basis.

Table A.1: Comparison of means across samples

<i>Data source:</i>	PHF	Partner bank		
<i>Sample:</i>		All mortgagors	Survey invited	Survey completed
Demographic characteristics				
Age (years)	50.22	50.51	48.71	48.08
University completed (0/1)	0.36			0.62
Household net income (€)	4,257.28			5,680.12
Male (vs. F or C) (0/1)		0.26	0.35	0.42
Couple (vs. F or M) (0/1)		0.60	0.47	0.41
Neighborhood quality (1–9)		5.88	5.88	6.09
Mortgage characteristics				
Original amount (€ k)	203.17	182.61	195.57	211.74
Current amount (€ k)	139.85	148.17	163.14	175.89
Monthly payment (€)	985.20	859.57	886.27	946.52
Fixation period (years)	11.46	11.92	12.25	12.37
Interest rate (%)	2.16	2.29	2.16	2.03
Value property >2015 (€ k)	311.09	389.35	383.15	372.01
>1 collateral property (%)	9.35	4.23	4.51	6.02
Relation with partner bank				
Individual advisor (0/1)		0.27	0.34	0.43
Advisor meetings (no.)		1.72	1.89	2.05
Credit card (0/1)		0.11	0.20	0.26
Checking account (0/1)		0.41	0.58	0.61
Observations	989	190,375	40,357	1,705

Notes: This table presents summary statistics for borrowers' demographic characteristics (survey and bank data), mortgage characteristics (bank data), and their relation with the partnering bank (bank data). We report the mean of each variable for four samples: a representative German sample from the Bundesbank's Panel on Household Finances (PHF), mortgagors we observe in the bank data (see [Section 3.1](#)), those who receive a survey invitation (see [Section 4.2](#)), and those who complete the survey. We restrict the PHF sample to households with a mortgage, within which we focus on the person most knowledgeable about financial matters. As the latest PHF data are from 2021, we require an active mortgage in 2021 in the bank samples, and report mortgage characteristics from that year (we only have current data on demographics and the bank relation). Mortgage amounts and payments are sums at the borrower level; fixation period and interest rate are averages across loans of the borrower. *Household net income* is a point estimate in the PHF and the mid point of the income bucket selected by a respondent of our survey (e.g., €4,000 to 4,999). *Male* equals one if the bank customer is male rather than female (F) or a couple (C). *Original amount* refers to loan origination or, if available, refinancing. *Value property >2015* is the self-reported purchase price of a property if bought after 2015 in the PHF, and the initial value of the collateral underlying a loan initiated after 2015 in the bank samples. *>1 collateral property* indicates whether the mortgagor uses more than one property as collateral.

Table A.2: Proportional hazard models of internal refinancing – robustness to using market rate at origination

	Forward window				Prolongation window			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$R_t^m - R_i^m$	-0.316*** (0.013)	-0.303*** (0.016)	-0.223*** (0.012)	-0.072** (0.030)	-0.082*** (0.009)	-0.092*** (0.010)	0.095*** (0.021)	0.066* (0.039)
ΔR_t^m	2.663*** (0.262)	2.639*** (0.277)			0.058 (0.071)	0.100 (0.073)		
Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Year-month FE	No	No	Yes	Yes	No	No	Yes	Yes
Vintage-specific BH	No	No	No	Yes	No	No	No	Yes
Observations	2,969,769	2,968,550	2,968,550	2,968,550	502,075	501,859	501,859	501,859

Notes: This table presents from proportional hazard models similar to [Table 3](#), except that R_i is replaced by R_i^m , the average interest rate on 10-year mortgages originated by the bank in the month that a given loan i was originated. Reported estimates are hazard ratios minus one. *Year-month FE* are fixed effects capturing the calendar year and month of an observation. *Vintage-specific BH* means the baseline hazard is allowed to vary across origination vintages. Standard errors (two-way clustered at the loan and calendar-month levels) in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Heterogeneity in refinancing choices

DV (0/1):	Internal refi	Forward	Prolong	Conditional on internal refinancing			
	(1)	(2)	(3)	Prolong	Post-offer	Bank offer	Post-expiry
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Mortgage rate (%)	0.014*** (0.004)	0.022*** (0.004)	−0.008 (0.005)	−0.036*** (0.009)	−0.016* (0.008)	−0.015** (0.006)	−0.004 (0.003)
Residual loan balance (log)	0.051*** (0.007)	0.083*** (0.007)	−0.032*** (0.005)	−0.134*** (0.008)	−0.097*** (0.006)	−0.113*** (0.006)	0.004 (0.004)
Initial rate fixation (years)	0.005*** (0.001)	0.016*** (0.002)	−0.011*** (0.003)	−0.031*** (0.005)	−0.031*** (0.005)	−0.019*** (0.003)	−0.008*** (0.002)
Allows partial prepayments (0/1)	−0.037*** (0.007)	0.008 (0.006)	−0.044*** (0.002)	−0.058*** (0.007)	−0.031*** (0.007)	−0.034*** (0.005)	−0.010*** (0.003)
Borrower age	−0.001** (0.000)	−0.001*** (0.000)	0.000 (0.000)	0.001*** (0.000)	0.000 (0.000)	0.002*** (0.000)	−0.001*** (0.000)
Male (0/1)	−0.009* (0.005)	−0.009** (0.004)	0.001 (0.004)	0.012* (0.007)	0.006 (0.008)	0.004 (0.008)	0.004 (0.004)
Couple (0/1)	−0.044*** (0.005)	−0.007 (0.005)	−0.037*** (0.004)	−0.028*** (0.009)	−0.031*** (0.006)	−0.009* (0.005)	−0.010*** (0.002)
Neighborhood quality (1–9)	−0.004*** (0.001)	−0.002* (0.001)	−0.002** (0.001)	−0.001 (0.001)	0.000 (0.001)	0.001 (0.001)	−0.001 (0.001)
Probability of default (deciles)	0.009*** (0.001)	0.001 (0.000)	0.008*** (0.001)	0.007*** (0.001)	0.009*** (0.001)	0.006*** (0.001)	0.004*** (0.001)
Checking/savings account (0/1)	0.057*** (0.006)	0.038*** (0.006)	0.019*** (0.004)	−0.016** (0.008)	−0.006 (0.006)	0.005 (0.003)	0.005* (0.002)
Securities account (0/1)	−0.026*** (0.007)	−0.010* (0.005)	−0.016*** (0.005)	−0.009 (0.009)	−0.028*** (0.007)	−0.021** (0.007)	−0.016** (0.006)
Other mortgage debt (0/1)	0.035*** (0.006)	0.051*** (0.004)	−0.016*** (0.004)	−0.068*** (0.006)	−0.058*** (0.006)	−0.060*** (0.005)	0.006* (0.003)
Partially rented out (0/1)	0.006 (0.011)	−0.015* (0.008)	0.020** (0.007)	0.036*** (0.009)	0.025*** (0.006)	0.013* (0.007)	0.009*** (0.002)
Fully rented out (0/1)	−0.030*** (0.007)	−0.050*** (0.007)	0.020*** (0.003)	0.079*** (0.008)	0.063*** (0.006)	0.043*** (0.005)	0.018*** (0.005)
Expiration quarter × location FE	Y	Y	Y	Y	Y	Y	Y
Avg. Y	0.44	0.22	0.22	0.50	0.35	0.18	0.07
Observations	121,478	121,478	121,478	53,111	53,111	53,111	53,111
R-squared	0.05	0.09	0.06	0.20	0.16	0.18	0.07

Notes: This table reports estimates of regressions of actual refinancing choices on loan and borrower characteristics. We restrict the sample to loans with a fixation period of up to 10 years, an expected residual balance of at least €10,000 at expiration of the fixed rate period, and expiration between January 2018 and March 2025. Columns 4–7 restrict the sample to loans that refinance internally. *Post-offer* is equal to one for loans that refinance internally six weeks or less before expiration of the fixation period (i.e., after receiving the prolongation offers) or after expiration. *Bank offer* is equal to one for loans inferred to have accepted one of the bank's default prolongation offers, i.e., loans refinanced within six weeks of expiration with a new loan amount within €1,000 and a new term within six months of the counterfactual loan. *Post-expiry* is equal to one for loans refinanced internally after expiration. *Other mortgage debt* equals one if the borrower has other mortgage loans outstanding (regular mortgages, home-savings contracts, and/or public subsidy loans) four years before expiration or, if unavailable, at the earliest observed date. *Partially rented out* and *fully rented out* indicate properties that are partly or entirely rented out; the omitted category is owner-occupied properties. All regressions also include indicators for missing neighborhood quality (6.6% of observations) and missing occupancy status (3.6%), which are not reported. All columns include expiration quarter × location (2-digit zipcode) fixed effects. Robust standard errors are two-way clustered at the origination year and borrower level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Loan adjustments at refinancing

<i>Dependent variable:</i>	Term adjustment (months)			Balance reduction (%)			New fixation (years)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Market rate change (pp)	8.538*** (0.395)	8.000*** (0.516)	9.302*** (0.936)	0.196*** (0.040)	0.135** (0.064)	-0.096 (0.182)	-0.507*** (0.051)	-0.174*** (0.060)	0.683*** (0.082)
Residual loan balance (log)		-9.009*** (1.022)	-9.675*** (0.975)		2.127*** (0.119)	2.247*** (0.112)		1.173*** (0.052)	1.131*** (0.038)
Initial rate fixation (years)		-0.017 (0.229)	0.341 (0.293)		-0.028 (0.050)	-0.114 (0.078)		0.477*** (0.020)	0.706*** (0.037)
Borrower age		0.568*** (0.053)	0.550*** (0.051)		0.095*** (0.010)	0.104*** (0.009)		-0.051*** (0.003)	-0.050*** (0.003)
Male (0/1)		3.196*** (0.794)	3.839*** (0.747)		-0.514* (0.268)	-0.638** (0.266)		-0.248*** (0.090)	-0.316*** (0.077)
Couple (0/1)		-0.345 (0.772)	0.021 (0.814)		-0.503** (0.228)	-0.613*** (0.209)		-0.035 (0.071)	-0.141** (0.058)
Neighborhood quality (1-9)		0.703*** (0.097)	0.684*** (0.108)		-0.124*** (0.038)	-0.131*** (0.036)		-0.062*** (0.010)	-0.035*** (0.010)
Probability of default (deciles)		1.238*** (0.161)	1.356*** (0.160)		-0.129*** (0.030)	-0.146*** (0.031)		0.047*** (0.009)	0.053*** (0.009)
Checking/savings account (0/1)		4.610*** (0.649)	4.730*** (0.646)		-0.288 (0.186)	-0.300 (0.189)		-0.036 (0.044)	-0.085** (0.041)
Securities account (0/1)		0.704 (1.102)	0.140 (1.034)		0.115 (0.288)	0.182 (0.281)		-0.136 (0.093)	-0.177** (0.079)
Other mortgage debt (0/1)		-5.373*** (0.772)	-6.404*** (0.792)		-1.187*** (0.113)	-1.092*** (0.130)		0.710*** (0.045)	0.644*** (0.038)
Allows partial prepayments (0/1)		-11.129*** (0.816)	-12.046*** (0.749)		-0.438*** (0.136)	-0.316** (0.136)		-0.368*** (0.041)	-0.551*** (0.040)
Partially rented out (0/1)		6.386*** (1.063)	7.202*** (1.234)		-0.600** (0.271)	-0.668** (0.300)		-0.117* (0.061)	-0.085 (0.066)
Fully rented out (0/1)		8.583*** (0.826)	8.914*** (0.894)		-1.213*** (0.193)	-1.219*** (0.208)		-0.523*** (0.069)	-0.395*** (0.059)
Refinancing quarter x location FE	No	No	Yes	No	No	Yes	No	No	Yes
Avg. Y	-26.36	-26.36	-26.43	1.23	1.23	1.23	7.52	7.52	7.53
Observations	39,305	39,304	39,149	39,305	39,304	39,149	39,305	39,304	39,149
R-squared	0.08	0.15	0.22	0.00	0.04	0.10	0.05	0.25	0.34

Notes: This table reports estimates of regressions of the form $Y_i = \beta_0 + \beta_1(R_t^m - R_i^m) + \Gamma'X_i + \mu_t + \varepsilon_i$, where the dependent variable Y_i represents a particular loan choice at the time of internal refinancing. The sample is restricted to loans with fixation periods of up to 10 years, an expected residual balance of at least €10,000 at expiration, expiration between January 2018 and March 2025, and which are refinanced internally. Our main explanatory variable of interest is $(R_t^m - R_i^m)$, the change in the market rate on a 10-year mortgage in the month of refinancing versus the month in which the borrower had taken out their original loan. The dependent variable in Columns 1–3, *term adjustment*, is the difference between the term of the new loan and the term of the passive counterfactual loan. In Columns 4–6, *balance reduction* is the reduction in outstanding balance at refinancing as a percentage of the original loan amount. In Columns 7–9, *new fixation* is the rate fixation period after refinancing. Controls (X_i): *Residual loan balance* is the natural logarithm of the expected residual loan balance at expiration assuming no extra payments. *Initial rate fixation* is the number of years the expiring interest rate was fixed for. *Borrower age* is borrower age at expiration. *Male* is equal to one if the borrower is a single male (versus female or couple). *Couple* is equal to one if the borrower is a couple (versus male or female). Controls for both *neighborhood quality* (9-point scale) and *probability of default* (deciles) are from proprietary bank calculations. *Checking/savings account* is equal to one for borrowers with a current or savings account at the bank. *Securities account* is equal to one for borrowers with a securities account at the bank. *Other mortgage debt* equals one if the borrower has other mortgage loans outstanding (both regular mortgage and *Bausparen*). The last three variables are measured four years before expiration or at the earliest available observation. *Refi-month FE* captures the year-month of loan refinancing (μ_t). Constant included but not reported. Robust standard errors are two-way clustered at the refinancing-month and borrower level and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Letter effects on propensity to participate in survey

<i>Dependent variable:</i>	Survey started (0/1)		Survey completed (0/1)	
	(1)	(2)	(3)	(4)
Panel A. Indicator for each letter				
Letter 1: baseline, all sections included	0.007 (0.005)	0.008* (0.005)	0.003 (0.004)	0.004 (0.004)
Letter 2: section on rate uncertainty omitted	0.003 (0.005)	0.003 (0.005)	0.000 (0.004)	0.000 (0.004)
Letter 3: section on payment impact omitted	0.004 (0.005)	0.004 (0.005)	0.004 (0.004)	0.004 (0.004)
Panel B. Indicator for any letter				
Any of the three letters received	0.004 (0.004)	0.005 (0.004)	0.002 (0.003)	0.003 (0.003)
Controls from bank data	N	Y	N	Y
Avg. Y control group	0.053	0.053	0.045	0.045
Observations	19,477	19,477	19,477	19,477
R-squared	0.00	0.01	0.00	0.01

Notes: This table reports estimates from regressions of indicators for starting or completing the survey on indicators for receiving the letter. The estimation sample includes mortgagors from the partner bank who are part of the letter experiment (including the control group) and receive an email invitation to participate in the survey. *Survey started* equals one if a mortgagor clicks on the survey link embedded in the email invitation. *Survey completed* is one if the mortgagor makes it to the end of the survey. Panel A shows the effect of each letter relative to the control group. Panel B is on the effect of an indicator that pools all letter groups. We explain the letter RCT in [Section 5.1](#). A description of the *controls from the bank data* used in the survey analysis is in [Section 4.2](#). Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Association between actual choices and hypothetical refinancing

<i>Dependent variable (0/1):</i>	Advisor	Search	Prepay	Forward	Fix $\geq 10y$
	(1)	(2)	(3)	(4)	(5)
Advisor meeting past 12m (0/1)	0.041* (0.022)	-0.044** (0.018)	0.058*** (0.021)	0.001 (0.024)	-0.041* (0.024)
Any past prepayment (0/1)	0.034 (0.023)	-0.001 (0.018)	0.212*** (0.021)	-0.044* (0.026)	-0.014 (0.025)
Any forward loan (0/1)	0.049* (0.025)	-0.056*** (0.021)	-0.023 (0.025)	0.119*** (0.028)	0.133*** (0.027)
Any home-savings loan (0/1)	0.053** (0.024)	-0.036* (0.019)	-0.041* (0.024)	0.079*** (0.026)	0.062** (0.026)
Rate fixation length (0-2)	0.015 (0.016)	0.014 (0.013)	-0.021 (0.016)	-0.002 (0.017)	0.080*** (0.017)
Avg. Y V1	0.78	0.88	0.71	0.47	0.56
Observations	1,921	1,921	1,921	1,921	1,921
R-squared	0.03	0.02	0.05	0.03	0.03

Notes: The table reports estimates of regressions of actions in the context of a hypothetical refinancing situation. Other than in Column 5, the dependent variables are likelihoods of action originally measured on a four-point scale, covering “very unlikely,” “rather unlikely,” “rather likely,” and “very likely.” We construct an indicator equal to one if the action is rather or very likely, and zero if it is rather or very unlikely. *Advisor* refers to meeting an advisor from the current lender; *search* refers to comparing offers by different loan providers; *prepay* is reducing the loan balance through voluntary, extra payments; *forward* means locking in rates using a forward loan; and *fix $\geq 10y$* equals one if the borrower would choose a fixation period of at least 10 years, and equals zero for one or five years. Explanatory variables of interest are from the bank data. *Advisor meeting past 12m* captures whether the borrower met an advisor over the past 12 months. *Any past prepayment* indicates whether the mortgagor has made any voluntary, extra repayments in the past. *Any forward loan* equals one if the borrower has had a forward mortgage at some point. *Any home-savings loan* is one if the borrower currently has a home-savings (Bausparen) product. *Rate fixation length* is zero if the original rate fixation period is less than 10 years, one if it is 10 to less than 15 years, and two for at least 15 years. We control for mortgage-status and vignette-variant fixed effects. *Avg. Y V1* is the average of the dependent variable for vignette-1 respondents. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Letter sample

	Control	Treat: group 1	Treat: group 2	Treat: group 3	p-value (1)=(2)	p-value (1)=(3)	p-value (1)=(4)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Months to expiration	80.03 (55.50)	80.32 (55.82)	79.67 (55.01)	80.22 (55.33)	0.69	0.62	0.79
Residual amount at expiration (€ k)	111.59 (136.38)	108.33 (133.16)	111.26 (137.57)	110.12 (137.41)	0.06*	0.85	0.41
Interest rate (%)	1.94 (0.86)	1.95 (0.86)	1.95 (0.86)	1.94 (0.86)	0.23	0.15	0.77
Monthly payments (€)	936.35 (875.94)	918.74 (868.23)	925.60 (876.03)	915.16 (867.77)	0.12	0.34	0.06*
Fixation period (years)	12.76 (4.24)	12.80 (4.29)	12.75 (4.20)	12.75 (4.23)	0.42	0.92	0.90
Allowed prepayments (0/1)	0.73 (0.45)	0.73 (0.44)	0.73 (0.45)	0.73 (0.44)	0.37	0.89	0.36
Borrower age	49.31 (11.20)	49.42 (11.27)	49.40 (11.09)	49.23 (11.11)	0.45	0.52	0.57
Male (0/1)	0.28 (0.45)	0.27 (0.45)	0.27 (0.44)	0.27 (0.44)	0.64	0.33	0.11
Couple (0/1)	0.60 (0.49)	0.60 (0.49)	0.60 (0.49)	0.61 (0.49)	0.88	0.35	0.05**
Has bank account	0.50 (0.50)	0.50 (0.50)	0.51 (0.50)	0.49 (0.50)	0.54	0.67	0.16
Met advisor (0/1)	0.19 (0.39)	0.18 (0.38)	0.19 (0.39)	0.18 (0.39)	0.07*	0.74	0.34
Observations	11,854	11,853	11,857	11,858			

Notes: This table reports means and standard deviations (in parentheses) for loan characteristics and borrower demographics in the control group (Column 1) and the three letter treatment groups (Columns 2–4). Columns 5–7 report the p-value of the t-test of equal means across treated and control groups. *Months to expiration* the remaining time (in months) until the end of the current fixation period as of the end of October 2024. *Residual balance* is the expected outstanding loan amount at expiration. *Interest rate*, *monthly payments*, *fixation period*, and *allowed prepayments* refer to the terms of the current loan. To determine exposure to the end of the fixed rate period we only consider loans with an expected remaining balance of at least €10,000 at expiration, and for which the post-expiration interest rate has not been locked in through a forward contract. For borrowers with multiple loans, classification is based on the earliest expiration date. *Has bank account* is equal to one for borrowers with a deposit account at the bank. *Meets advisor* is equal to one for borrowers who met with a bank advisor at least once in the year prior to the experiment. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Predictors of reading the bank letter on mortgage-rate increases

<i>Dependent variable (0/1):</i>	Having read bank letter in past two months			
	(1)	(2)	(3)	(4)
Total outstanding mortgage debt (std.)	0.043** (0.020)	0.047** (0.021)	0.050* (0.026)	0.056** (0.027)
Rate reset within next 2y (0/1)	0.072 (0.047)	0.081* (0.048)	0.052 (0.055)	0.062 (0.055)
Partial repayment in past (0/1)	0.070* (0.042)	0.075* (0.043)	0.055 (0.047)	0.065 (0.048)
Financial literacy (0/1)	0.133*** (0.039)	0.149*** (0.043)	0.145*** (0.040)	0.153*** (0.044)
Household income net of expenses (0/1)	0.051 (0.039)	0.062 (0.040)	0.038 (0.041)	0.044 (0.042)
Perceived rate increases (0/1)	0.140*** (0.049)	0.158*** (0.051)	0.135*** (0.051)	0.151*** (0.053)
Accurate perceived current rate (0/1)	0.057 (0.043)	0.064 (0.043)	0.061 (0.044)	0.068 (0.045)
Survey controls	N	Y	N	Y
Bank controls	N	N	Y	Y
Avg. Y	0.41	0.41	0.41	0.41
Observations	669	669	668	668
R-squared	0.07	0.09	0.12	0.14

Notes: This table reports estimates from regressions of the propensity to read the bank letter. Specifically, the dependent variable equals one if the respondent states they have read a letter from the bank over the past two months on the topic of mortgage-rate increases, and zero otherwise. The sample only includes survey respondents who receive one of the three letters. We explain the letter RCT in [Section 5.1](#). The coefficients of interest are the traits from [Table 8](#), included jointly rather than one at a time. *Total outstanding mortgage debt (std.)* is the standardized current total mortgage debt. *Rate reset within next 2y (0/1)* is one if the end to the interest-rate fixation period is within two years. *Partial repayment in past (0/1)* equals one if the borrower has reduced the loan balance in the past using an annual partial repayment option. *Financial literacy (0/1)* is one for self-reported financial literacy that is above the median. *Household income net of expenses (0/1)* is one for above-median household income net of general expenses. *Perceived rate increases (0/1)* is one if the respondent believes the current mortgage rate in Germany compared to three years ago (i.e., in 2021) is higher. *Accurate perceived current rate (0/1)* is one if the perceived current rate on a new mortgage with a 10-year fixation period in Germany is 3–4%. *Survey* and *bank controls* are those described in [Section 4.2](#), excluding the variables that enter as traits. All columns include letter-variant fixed effects. *Avg. Y* is the mean of the dependent variable in the estimation sample. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

B Institutional features of the German mortgage market

This appendix provides additional discussion of various institutional features of the German mortgage market: (i) market structure and loan funding; (ii) loan structure and underwriting; (iii) loan pricing; and (iv) public subsidy loans (KfW).

Market structure and loan funding Germany's residential mortgage market is dominated by banks and building and loan associations (*Bausparkassen*).² Life insurers are the principal non-bank mortgage lenders, but accounted for only around 3% of residential mortgage financing in 2021.³ Although online brokers and fintech platforms play an increasingly visible role in mortgage distribution, the loans they arrange are generally originated and held by banks, building and loan associations, or insurers.

Mortgage securitization exists, but estimates indicate that at most 5–6% of outstanding loans are securitized in residential mortgage-backed securities (RMBS).⁴ Instead, mortgages are commonly used as collateral for covered bonds (*Pfandbriefe*), which provide banks with long-duration funding. Unlike an MBS, a covered bond is not a pass-through claim on a fixed pool of mortgages that is moved off-balance-sheet: the underlying loans remain on the issuing bank's balance sheet, while investors receive the bond's contractual coupon and principal according to a predetermined schedule. Mortgage prepayments therefore do not accelerate payments to covered-bond investors or directly shorten the bond's duration. Instead, the issuing bank bears the associated prepayment, reinvestment, and borrower-default risks and must maintain sufficiently large eligible collateral throughout the life of the bond, replacing prepaid or otherwise ineligible mortgages when necessary. Covered bonds are generally issued as conventional fixed-rate bullet bonds, most commonly at maturities of roughly 5, 7, or 10 years.

Loan structure and underwriting As described in [Section 2](#), the typical residential mortgage in Germany is an annuity loan with a fixed interest rate for a predetermined fixation period, typically 5, 10, or 15 years. The fixation period is generally shorter than the time required to fully amortize the loan, implying that borrowers typically need to refinance at least once.

²German *Bausparkassen* are specialized housing-finance banks, broadly analogous to US savings and loan associations (thrifts), but organized around home-savings schemes (*Bausparen*).

³Source: <https://www.gdv.de/gdv/medien/medieninformationen/versicherer-mit-solidem-baukredit-geschaeft-nachfrageknick-zeichnet-sich-jedoch-ab--131486>.

⁴Outstanding volume of RMBS is from AFME (<https://www.afme.eu/>); data on the volume of outstanding residential mortgages is from the Deutsche Bundesbank banking statistics.

German loans cannot generally be prepaid in full during the fixation period. Early repayment beyond any contractual prepayment option requires the lender's consent and is subject to a prepayment penalty (*Vorfälligkeitsentschädigung*), which compensates the lender for the resulting loss of interest income. In particular, a borrower who simply wishes to refinance because market interest rates have fallen has no legal right to prepay the loan, even if willing to compensate the lender for the associated loss of interest income. The principal statutory exception is §490(2) of the German Civil Code (*Bürgerliches Gesetzbuch*, BGB), which grants borrowers a right to early termination if they have a legitimate interest in ending the loan, most commonly because they are selling the mortgaged property. In such cases, the lender must release the borrower in exchange for a prepayment penalty. Lenders often waive the prepayment penalty in case of a sale.

Another important exception is the statutory termination right under §489(1) of the BGB. Borrowers may terminate a fixed-rate mortgage without penalty 10 years after the loan has been fully disbursed, provided they give six months' notice. For mortgages with interest rate fixation periods longer than 10 years (e.g., 15 years), borrowers can therefore refinance or repay the loan after year 10 without incurring a prepayment penalty, despite the remaining fixation period.

Turning to underwriting standards, there is no single authoritative source that measures loan-to-value ratios in Germany consistently over time.⁵ However, several sources converge on a typical LTV at origination of around 80% in recent years. Supervisory survey data on lending practices report an average of 82% in 2016, rising to 84% in 2018 ([Deutsche Bundesbank, 2019](#)). Figures from the market platform Europace show 75–80% up to 2016, rising to 83% by 2020 ([Barasinska et al., 2023](#)), and the new WIFSta reporting puts the average LTV on newly granted residential mortgages at 83% in the fourth quarter of 2025 ([Ausschuss für Finanzstabilität, 2026](#)). Earlier estimates are somewhat lower, at 70–75% ([Geiger et al., 2016](#); [Lukas and Nöth, 2019](#)). Our bank data do not contain information on LTV at origination.

German mortgages are full-recourse debt: if the property is worth less than the outstanding mortgage balance at foreclosure, the borrower remains personally liable for the shortfall, which may be enforced against other assets and against income through wage garnishment, subject to statutory exemption thresholds (§850c ZPO). This removes the

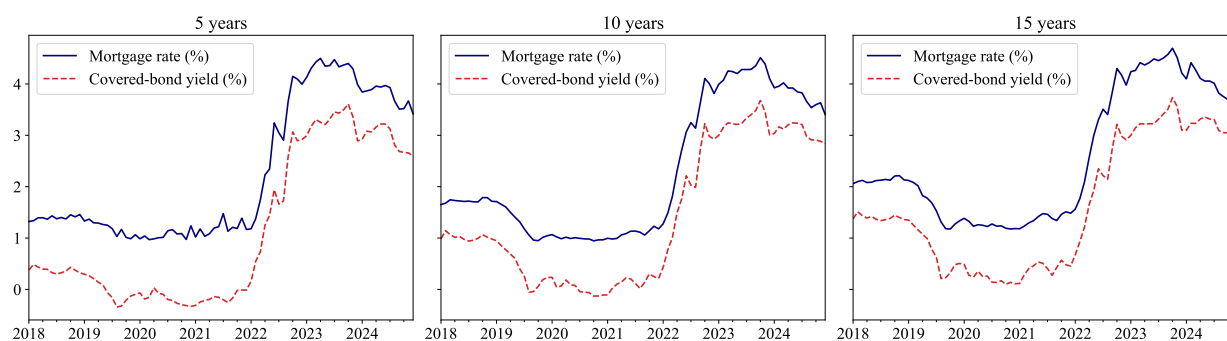
⁵Germany imposes no binding quantitative limits on mortgage lending standards. A 2017 law created the legal basis for an LTV cap and an amortization requirement, but neither has been activated ([Deutsche Bundesbank, 2019](#)).

option to walk away, so that negative equity is unlikely to trigger strategic default. Consistent with this, [Barasinska et al. \(2023\)](#) find that in Germany defaults are driven mostly by income and liquidity rather than LTV ratios, which instead are the key determinant of loss-given-default. Foreclosure rates in Germany are historically very low, averaging 0.8% since 1991, and in the PHF survey only 0.2% of mortgagor households report arrears ([Barasinska et al., 2023](#)).

German law does not require lenders to reassess creditworthiness on internal refinancing unless the loan amount is significantly increased (*deutlich erhöht*; see §505a(3) BGB).

Loan pricing German mortgages are fairly standardized. Mortgage rates are typically determined as a mark-up to a bank's cost of funding the mortgages through covered bonds. We confirm in our bank data that average mortgage rates at each fixation length closely track the covered-bond yield at the corresponding maturity (see [Figure A.12](#)).⁶

Figure A.12: Mortgage rates and covered-bond yields



Notes: The figure plots the average mortgage rate on new loan originations at the partner bank by month of origination and fixation length (blue solid line), alongside the German covered-bond yield of matching maturity (red dashed line, source: Bundesbank). The sample comprises 3,644, 30,122 and 23,896 newly originated loans with 5-, 10- and 15-year fixation, corresponding to 43, 359, and 284 originations per month on average, respectively.

In the cross-section (for a given fixation length at a point in time), anecdotally the rate offered on a specific loan is primarily determined by a small set of observable loan and property characteristics: the length of the interest-rate fixation, the LTV ratio, the amortization rate, the loan amount, the location of the property, and the type of property. [Table A.9](#) documents the determinants of mortgage rates on new loan originations in the bank data. Column 1 shows that origination-month and location fixed effects ex-

⁶The figure indicates that the spread for five-year fixation loans was higher than for longer fixations during the low-rate period. This could reflect that the bank imposed some lower bound on the benchmark yield or the actual loan rates.

plain 86% of the variation in mortgage rates. Column 2 adds interactions of origination month with fixation length and loan term. These account for 44% of the remaining variation, reducing the residual standard deviation from 38 to 29 basis points. Column 3 controls for loan amount, deciles of the bank’s internal default-probability score, and partial prepayment rights, increasing the within R-squared from 44% to 51%.⁷ A higher loan amount significantly decreases the loan interest rate, though at a decreasing rate. Default probability also has significant effects, though the resulting spread is relatively small—going from lowest to highest decile increases the predicted rate by 15 basis points. The partial prepayment option similarly has a rather small effect (6 basis points). Finally, column 4 additionally controls for the initial amortization rate, borrower age and its square, gender, couple status, neighborhood quality (dummies), property type and occupancy (owner-occupied vs. rented out). Together they add little explanatory power.

As explained in [Section 2](#) of the main text, borrowers in Germany can secure the terms of their refinancing in advance through a forward loan, available up to 3–5 years before the current fixation expires, depending on the lender. Forward loans carry a premium (*Aufschlag*) over otherwise comparable loans originated at the same time. The forward premium depends on the forward period (the number of months remaining on the borrower’s current fixation when the forward contract is signed) and the term structure of covered-bond yields. Lenders generally publish forward costs as a per-month premium that may vary with the length of the forward period. The total premium is the per-month premium multiplied by the number of months in the forward period. For illustration, if a lender charges 0.01 pp per month for forward periods of 13–24 months, a forward agreed two years in advance carries a total premium of 0.24 pp; if the same lender charges 0.03 pp per month for forward periods of 25–36 months, a three-year forward carries a total premium of 1.08 pp. It is fairly common for lenders to waive the premium entirely for forward periods below one year, or to deduct a fixed number of months, usually 6 or 12, from the forward period, so that a 24-month forward is charged for only 18.

[Figure A.13](#) documents forward pricing in the bank data. For this purpose, we restrict the sample to loans originated since 2015 with a 10-year rate fixation (the modal contract), which account for 48% of forward loans and 39% of new originations, and exclude prolongations. We end the sample in December 2022 because only few forward loans are originated afterwards. Panel A plots the estimated average interest-rate premium on for-

⁷The default probability is the value reported in December 2017, if available, otherwise the one reported in October 2024.

Table A.9: Pricing of new loan originations

<i>Dependent variable:</i>	Interest rate (%)			
	(1)	(2)	(3)	(4)
Loan amount (log)			−0.854*** (0.035)	−0.941*** (0.035)
Loan amount (log), squared			0.031*** (0.001)	0.035*** (0.001)
Probability of default (deciles)			0.015*** (0.000)	0.014*** (0.000)
Allows partial prepayments (0/1)			0.056*** (0.003)	0.058*** (0.003)
Origination-month FE	Y	Y	Y	Y
Location FE	Y	Y	Y	Y
Origination-month $\times$ fixation FE	N	Y	Y	Y
Origination-month $\times$ term FE	N	Y	Y	Y
Other controls	N	N	N	Y
Avg. Y	1.92	1.92	1.92	1.92
Observations	89,262	89,262	89,262	89,262
R-squared	0.86	0.92	0.93	0.93
Within R-squared	0.00	0.44	0.51	0.52
Root MSE	0.38	0.29	0.27	0.27

Notes: This table presents results of regressions in which the dependent variable is the mortgage interest rate. We restrict the sample to loans originated between 2017 and May 2025 and exclude forwards and prolongations. Column 1 includes origination-month and location (two-digit zip code) fixed effects; column 2 adds origination-month $\times$ fixation-length fixed effects and origination-month $\times$ loan-term fixed effects; column 3 adds log loan amount and its square, the decile of the bank's default-probability score (entered linearly), and a dummy for whether the loan allows partial prepayments; column 4 additionally controls for initial amortization rate, borrower age and its square, gender, couple status, neighborhood quality (dummies), property type and occupancy (owner-occupied vs. rented out). Standard errors (clustered at borrower level) are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

ward mortgages relative to regular new originations in the same month, by length of the forward period, controlling for loan and borrower characteristics. Controls are the same as in the last column of [Table A.9](#). Forwards agreed 7–12 months in advance carry essentially no premium, consistent with lenders waving the surcharge for forward periods below 12 months. Beyond that, each additional year of forward period adds roughly 20

basis points.⁸ Panel B plots the estimated monthly premia against the corresponding forward spreads implied by the covered-bond term structure. Specifically, we first calculate the k -year forward 10-year rate from the zero-coupon covered-bond curve as

$$f_{k,10} = \left[(1 + R_{k+10})^{k+10} / (1 + R_k)^k \right]^{1/10} - 1,$$

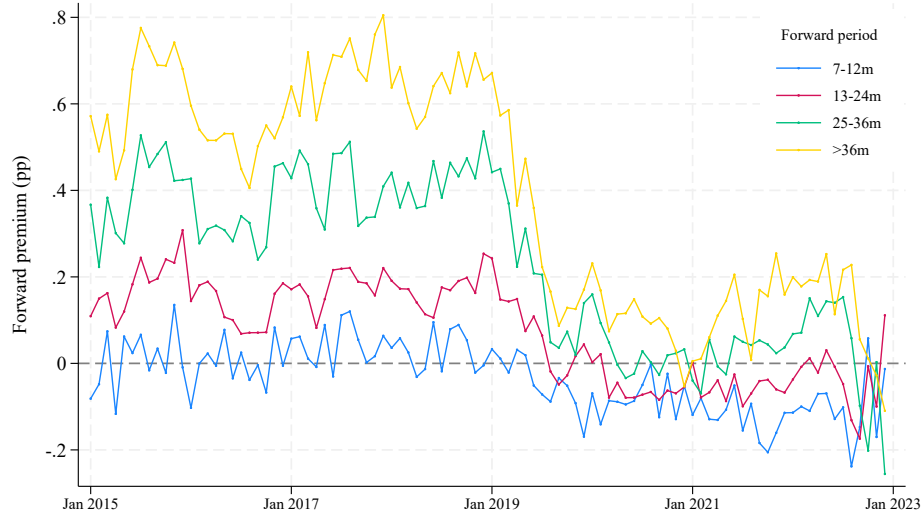
where R_m is the zero-coupon yield at maturity m . The covered-bond forward spread is then $f_{k,10} - R_{10}$, the difference between this forward rate and the current 10-year spot yield. The figure shows that estimated premia rise nearly one-for-one with the covered-bond forward spread across all forward periods. Forward premia, however, systematically lie below the 45-degree line, which might be explained by the bank waiving the surcharge for the first months.

Public subsidy loans (KfW) In addition to regular mortgages, homebuyers in Germany can in some cases finance part of a residential purchase through loans originated by Germany's state-owned development bank (KfW). These loans primarily support energy-efficient new construction and deep energy-efficiency renovations. KfW loans are not subject to income eligibility requirements and are originated through commercial banks, typically as one tranche within a broader mortgage financing package rather than as a substitute for a regular mortgage. Like regular mortgages, KfW loans are exposed to interest rate risk once the initial fixed-rate period expires. Their refinancing process, however, differs from that of regular mortgages. First, at expiration borrowers receive an unsubsidized follow-on offer from KfW through the originating bank, although they have the option to refinance using the bank's standard mortgage products. Second, contractual flexibility is more limited: partial prepayment is only available for certain KfW programs and refinancing through forward mortgages at our partner bank is only possible for the most recent KfW loan tranches.

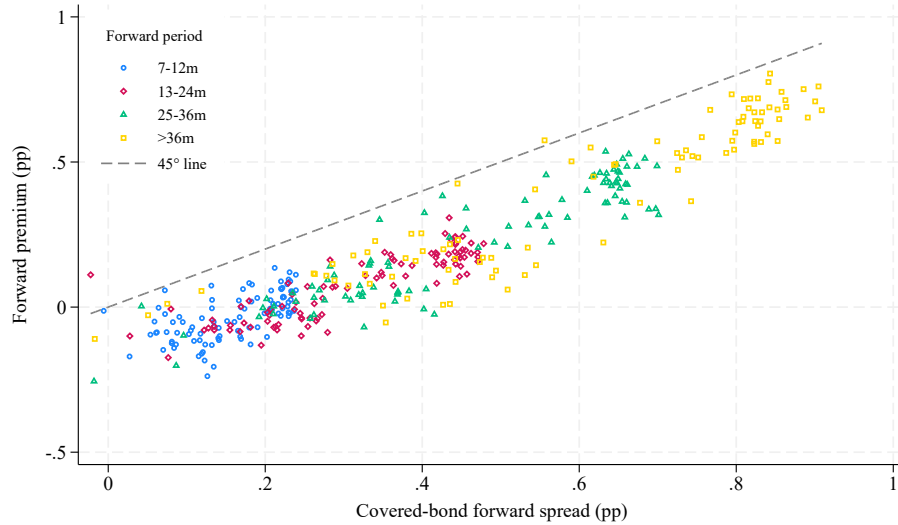
⁸The figure shows negative premia for some months; these could reflect compositional differences that are not absorbed by the loan characteristics we are controlling for. For instance, they could reflect differences in LTV that are unobservable to us.

Figure A.13: Pricing of forward mortgages

Panel A. Cost of forward loans relative to new originations



Panel B. Forward mortgage premia and covered-bond forward spreads



Notes: The figure shows the pricing of forward mortgages. The sample comprises loans with a 10-year fixation originated between January 2015 and December 2022. Panel A plots the estimated average interest-rate premium on forward mortgages relative to regular new originations in the same month, excluding prolongations, by length of the forward period, using the same controls and fixed effects as column 4 of [Table A.9](#). Panel B plots the estimated monthly premia against the corresponding covered-bond forward spreads, $f_{k,10} - R_{10}$, using the 1-, 2-, 3-, and 4-year forward spreads for forward periods of 7–12, 13–24, 25–36, and more than 36 months, respectively. The dashed line is the 45-degree line.

C Passive counterfactual estimation

This section explains in more detail the methodology to estimate the offered interest rate on the counterfactual passive prolongation loan and to calculate the corresponding counterfactual monthly payment.

We begin by training a random forest model to predict the interest rate on prolongation offers. The model uses as inputs (i) the month of loan origination, (ii) loan amount, (iii) loan term, (iv) length of the fixation period, and (v) the time gap between the origination of the prolongation and the expiration of the existing loan (recall that prolongations can be agreed upon as early as six months before the end of the fixation period). The training sample consists of about 39,000 observed prolongation loans.

We then use the trained model to predict counterfactual prolongation rates for internally refinanced loans, under the assumption that the borrower passively rolls over the residual debt at fixed-rate expiration. In this counterfactual, the loan characteristics are defined as follows: (i) month of origination is set to the month of expiration of the existing loan; (ii) residual loan amount is the balance at expiration assuming no extra payments; (iii) the term is either the residual term at expiration or such that the initial amortization rate of the prolongation loan matches the final amortization rate (*Tilgungssatz*) of the original loan at the end of the fixed-interest period;⁹ (iv) the fixation period is the minimum of the fixation on the expiring loan and the residual term; and (v) the time gap between the origination of the prolongation and expiration is set to zero by definition, since the prolongation is assumed to be taken out at expiration. Finally, the counterfactual monthly payment is computed for each loan based on these characteristics and the predicted interest rate. [Table A.10](#) provides an illustrative example.

Table A.10: Example of counterfactual and actual loan choice

Loan type	Amount (€)	Term (months)	Fixation (months)	Rate (%)	Payment (€)
Old loan	155,000.00	230	120	2.45	850.00
Passive counterfactual	83,892.39	101	101	4.52	994.72
New loan	67,816.00	96	96	4.45	848.27

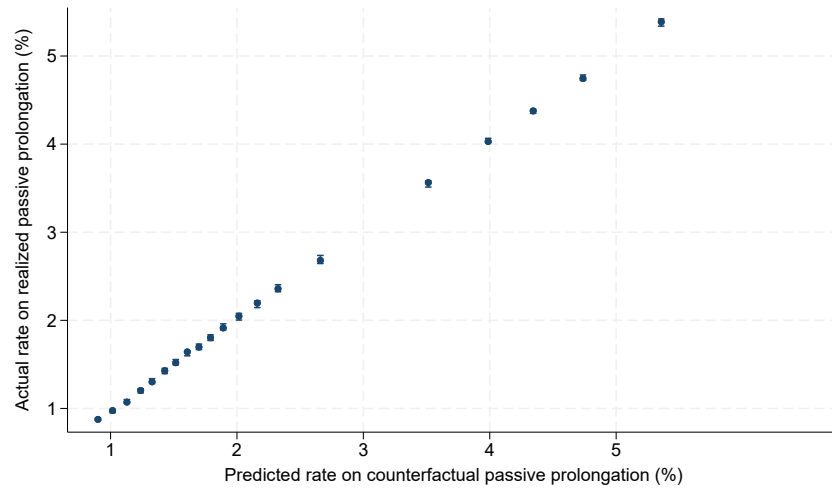
To verify the accuracy of the counterfactual, we focus on the subset of loans that appear to have passively rolled-over their debt, as for these the counterfactual should match

⁹This approach mirrors the bank's actual refinancing offers. The goal of the latter is to prevent a reduction in amortization speed upon prolongation.

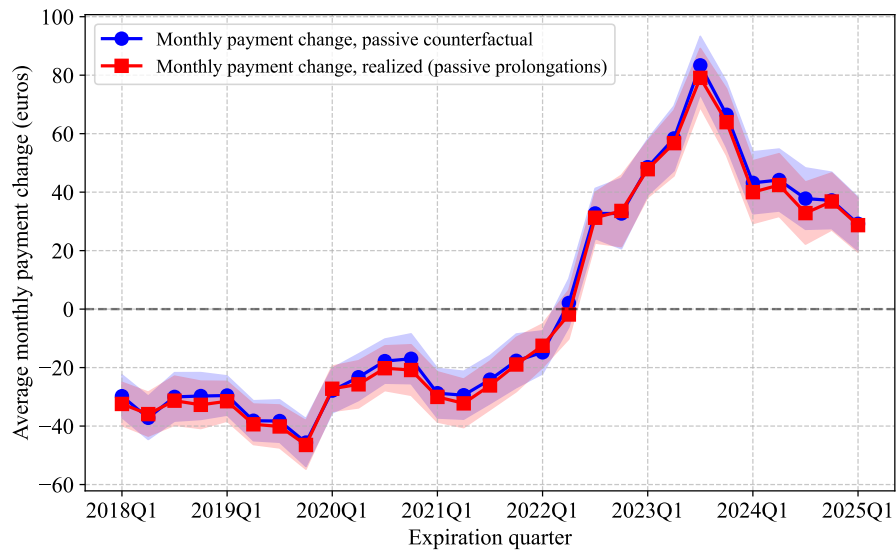
the observed prolongation. Since the bank data do not contain information on whether borrowers selected an offer sent by the bank, we infer this based on the features of the refinanced loan. Specifically, we classify a loan as a passive rollover if the new amount is within €1,000 of the expected outstanding balance, and both the term and interest-rate fixation differ by no more than three months from their respective counterfactuals. This identifies about 5,000 such prolongations. Panel A of [Figure A.14](#) shows a binned scatter plot of the realized rates plotted against bins of the predicted rate. The predicted rates align closely with the observed rates. Panel B then replicates Panel B of [Figure 5](#) on this subset, confirming that the counterfactual accurately reproduces roll-over offers.

Figure A.14: Prediction accuracy for passively rolled-over loans

Panel A. Binned scatter of predicted and actual rates



Panel B. Average actual vs. counterfactual monthly payment changes



Notes: The figure shows the accuracy of the estimated passive prolongation rates. Accuracy is evaluated using the sample of internally refinanced loans that, based on timing and loan characteristics, are likely to have been passively rolled over. Panel A presents a binned scatter plot of realized and predicted prolongation rates; the vertical bars denote 95% confidence intervals. Panel B plots the average realized change in monthly payments upon refinancing (red) together with the average estimated change under the passive counterfactual (blue); the shaded area represents the 95% confidence interval.

D Survey text

This appendix provides the survey instructions translated from German to English. We use **green text in brackets** to highlight aspects of the survey design. We show non-numerical response options using a), b), c), and so forth.

D.1 Welcome screen

Welcome to this survey by Goethe University Frankfurt!

The survey will take **around 10 minutes**. Your responses will be aggregated and only used for scientific research. If you feel unfamiliar with some of the survey topics, that is fine. We ask you to provide your best estimate. Since we are interested in your unfiltered opinion, please refrain from relying on external sources (e.g., using a Google search).

As a thank-you for completing the survey, you will receive an **Amazon voucher worth €10**. To receive the voucher, you can enter your e-mail address at the end of the survey. Your e-mail address will only be used to send you the voucher.

Please do not use the “Back” button in your browser, as this may require restarting the survey. Do you have any questions? Please contact us at **umfrage@finance.uni-frankfurt.de**.

D.2 Pre-vignette section

Q1: How do you assess the current interest rate for mortgage loans in Germany compared to three years ago (i.e., in 2021)?

- a) The current interest rate is much lower
- b) Somewhat lower
- c) Similar
- d) Somewhat higher
- e) Much higher

[Ask Q2 only if Q1 = d) or e).]

Q2: Do you think the increase in mortgage interest rates has an impact on your **household’s financial situation**?

Impact on your **current** financial situation

[Likert scale from 1 (“Very negative”) to 3 (“No impact”) to 5 (“Very positive”)]

Impact on your **expected future** financial situation

[Likert scale from 1 (“Very negative”) to 3 (“No impact”) to 5 (“Very positive”)]

[Allow non-negative values only; at most one decimal place. If a negative entry is attempted, display: “Entries cannot be negative.” Use a comma as the decimal separator.]

Q3: What do you think is the current **interest rate** of a newly originated mortgage with a 10-year fixed-rate period? And what interest rates do you expect for the future?

Please enter a positive number with at most one decimal place (use a comma as the decimal separator).

Current annual interest rate: _____ %

Annual interest rate in two years: _____ %

Annual interest rate in five years: _____ %

Q4: What do you think is the probability that **the actual mortgage interest rate in two years will be higher than (your answer in Q3 for “in two years” + 2 percentage points)**?

[Likert scale from 1 (“Very unlikely”) to 5 (“Very likely”)]

Q5: If your mortgage is not fully repaid at the end of the fixed-rate period, you will need follow-up financing. Have you already prepared for the **future follow-up financing of your loan**, or do you plan to do so?

- a) No, I have not prepared and will not do so
- b) Yes, I plan to prepare
- c) Yes, I have already prepared

[If Q5 = a), ask Q5a.]

Q5a: Please describe why you are not preparing for the future follow-up financing of your loan.

Your response is very important for our research project. Please take the time to answer carefully.

[Open-ended response field.]

[If Q5 = b), ask Q5b]

Q5b: Please describe how you plan to prepare for the future follow-up financing of your loan and what the reasons are.

Your response is very important for our research project. Please take the time to answer carefully.

[Open-ended response field]

When would you like to begin preparing?

- a) Less than six months before follow-up financing
- b) Six to 12 months before
- c) 1–2 years before
- d) 2–4 years before
- e) More than four years before
- f) No response

[If Q5 = c), ask Q5c]

Q5c: Please describe how you prepared for the future follow-up financing of your loan and what the reasons were.

Your response is very important for our research project. Please take the time to answer carefully.

[Open-ended response field]

When did you begin preparing?

- a) Less than six months before follow-up financing

- b) Six to 12 months before
- c) 1–2 years before
- d) 2–4 years before
- e) More than four years before
- f) No response

Q6: What do you expect will happen to your **monthly payment (interest + principal)** with the future follow-up financing of your loan?

Please do not take into account planned actions (e.g., Sondertilgungen, that is, voluntary extra repayments) that could affect the future payment amount.

- a) The payment will decrease
- b) The payment will remain roughly unchanged
- c) The payment will increase
- d) I will not need follow-up financing—the loan will be fully repaid

[If Q6 = a), ask Q6a]

Q6a: By how much do you expect your **monthly payment (interest + principal)** to decrease with the future follow-up financing of your loan?

- a) At least € 250 less
- b) € 200–€ 250 less
- c) € 150–€ 200 less
- d) € 100–€ 150 less
- e) € 50–€ 100 less
- f) Up to € 50 less

[If Q6 = c), ask Q6b]

Q6b: By how much do you expect your **monthly payment (interest + principal)** to increase with the future follow-up financing of your loan?

- a) Up to € 50 more
- b) € 50–€ 100 more
- c) € 100–€ 150 more
- d) € 150–€ 200 more
- e) € 200–€ 250 more
- f) At least € 250 more

Q7: In the past two months, have you received and read a **letter from your bank about increases in mortgage interest rates**?

- a) No, I have not received a letter
- b) I have received a letter but have not read it
- c) Yes, I have received a letter and read it
- d) I do not remember

D.3 Vignette section

Q8: In the next questions, we would like to learn how you would behave in the following role-play. There are no right or wrong answers; we are interested in your opinion. **Your responses are very important for our research project.** Please take your time when answering.

[Randomly assign respondents to one of the following three variants]

[Variant 1]

Role-play: Suppose the remaining balance of your mortgage is €100,000 and the **fixed-rate period ends in one year**. Your current interest rate is 1.5%. Due to changing market conditions, you expect your interest rate to **increase by two percentage points to 3.5%**, with uncertainty about the exact path.

[Variant 2]

Role-play: Suppose the remaining balance of your mortgage is €100,000 and the **fixed-rate period ends in three years**. Your current interest rate is 1.5%. Due to changing market conditions, you expect your interest rate to **increase by two percentage points to 3.5%**, with uncertainty about the exact path.

[Variant 3]

Role-play: Suppose the remaining balance of your mortgage is €100,000 and the **fixed-rate period ends in one year**. Your current interest rate is 3%. Due to changing market conditions, you expect your interest rate to **increase by 0.5 percentage points to 3.5%**, with uncertainty about the exact path.

[Place immediately below the framed text above. For all variants]

We will now ask you about several options you might consider in this situation. We will ask you in more detail about a) and c); the other options only briefly:

- a) Appointment with your bank advisor
- b) Comparing loan offers from different lenders
- c) Reducing the outstanding loan balance (prepayment or partial repayment at the end of the fixed-rate period)
- d) Fixing future loan terms today (forward loan)
- e) Building up savings
- f) Length of the fixed-rate period

[Show the respective framed text summarizing the role-play on top of question]

Q9: How likely is it that you would **schedule an appointment with an advisor at your bank** to prepare for the expected interest-rate increase?

[In Variant 2, append: “within the next 12 months”]

[Response options: “Very unlikely,” “Rather unlikely,” “Rather likely,” “Very likely”]

[If Variant 1 or 3 and Q9 = 1 or 2, ask Q9a]

Q9a: You answered that it is unlikely that you would schedule an **appointment with your bank**. How important are the following reasons for your answer?

[Randomize order of response options; for each of them, rate from 1 “Not important at all” to 5 “Very important”]

- a) I have concerns about whether I can trust the advisor’s recommendations
- b) I do not expect to obtain useful information in a meeting
- c) I am already well informed or prefer to think things through on my own
- d) I use other advisory services
- e) I do not know with whom to schedule or how to schedule an appointment

[If Variant 2 and Q9 = 1 or 2, ask Q9b]

Q9b: You answered that it is unlikely that you would schedule an **appointment with your bank** within the next 12 months. How important are the following reasons for your answer?

[Randomize order of response options; for each of them, rate from 1 “Not important at all” to 5 “Very important”]

- a) I have concerns about whether I can trust the advisor’s recommendations
- b) I do not expect to obtain useful information in a meeting
- c) I am already well informed or prefer to think things through on my own
- d) I use other advisory services
- e) I do not know with whom to schedule or how to schedule an appointment
- f) I will only deal with the follow-up financing later

[If Variant 1 or 3 and Q9 = 3 or 4, ask Q9c; if Variant 2 and Q9 = 3 or 4, ask Q9d]

Q9c/Q9d: You answered that it is likely that you would schedule an **appointment with your bank** [in Variant 2, add: “within the next 12 months”]. How important are the following reasons for your answer?

[Randomize order of response options; for each of them, rate from 1 “Not important at all” to 5 “Very important”]

- a) I trust the advice of the bank advisor
- b) I would like to better understand my situation and options
- c) I try to negotiate more attractive loan terms
- d) I use my bank advisor as my main point of contact for loan questions

Q10: Would you use **other information sources** to prepare for the expected interest-rate increase?
Multiple answers possible.

[Randomize order of response options; keep “No, I would not use other sources” at the bottom]

- a) Independent advisors/brokers
- b) Other banks
- c) Finance apps or online tools
- d) Social media (e.g., Instagram, Facebook, X/Twitter, online forums)
- e) Printed guides or traditional media (print, TV, radio)
- f) Family, friends, or coworkers

g) No, I would not use other sources [Exclusive response]

[Show the respective framed text summarizing the role-play on top of question]

Q11: How likely is it that, to prepare for the expected increase in interest rates, you would **reduce the remaining loan balance**? Such a reduction can occur through annual prepayments during the fixed-rate period (i.e., voluntary extra repayments) or partial repayments at the end of the fixed-rate period.

[Response options: 1 ("Very unlikely"), 2, 3, 4 ("Very likely")]

[If Q11 = 1 or 2, ask Q11a]

Q11a: You answered that it is unlikely that you would **reduce the remaining loan balance**. How important are the following reasons for your answer?

[Randomize order of response options; rate each item from 1 "Not important at all" to 5 "Very important"]

- a) I have other priorities for my savings (e.g., larger purchases, emergency fund)
- b) I prefer to invest my savings elsewhere
- c) I benefit from tax advantages due to the outstanding loan
- d) I am willing to tolerate higher monthly payments due to the rate increase

[If Q11 = 3 or 4, ask Q11b]

Q11b: You answered that it is likely that you would **reduce the remaining loan balance**. How important are the following reasons for your answer?

[Randomize order of response options; rate each item from 1 "Not important at all" to 5 "Very important"]

- a) I want to reduce interest costs
- b) I want to get rid of outstanding debt quickly
- c) I view paying down the loan as an attractive "investment" of my savings
- d) I want to avoid a (large) increase in the monthly payment

[If Variant 1 or 3 and Q11 = 3 or 4, ask Q12a; if Variant 2 and Q11 = 3 or 4, ask Q12b]

Q12a/Q12b: How would you reduce the remaining loan balance?

Multiple answers possible.

- a) Prepayment(s) before the end of the fixed-rate period [In Variant 2 instead allow: "Prepayment(s) within the next 12 months" and "Prepayment(s) at later dates"]
- b) Partial repayments at the end of the fixed-rate period
- c) Don't know [Exclusive response]

[Show the respective framed text summarizing the role-play on top of question]

Q13: We would now like to ask about **other possible actions**. There will be no follow-up questions on these actions. [In V2, add: "Please focus on actions planned within the next 12 months."]

How likely is it that you would already fix future loan terms such as the interest rate—so that this uncertainty is eliminated? This is possible via a **forward loan**.

[Response options: 1 ("Very unlikely") to 4 ("Very likely")]

How likely is it that you would **spend less to save** for the future interest-rate increase?

[Response options: 1 ("Very unlikely") to 4 ("Very likely")]

How likely is it that you would try to **increase your income** to save for the future interest-rate increase?

[Response options: 1 ("Very unlikely") to 4 ("Very likely")]

How likely is it that you would **compare loan offers from different providers**?

[Response options: 1 ("Very unlikely") to 4 ("Very likely")]

For how long would you want to fix the **3.5% interest rate** after the fixed-rate period ends? Assume the remaining loan term exceeds 15 years.

- a) One year
- b) Five years
- c) Ten years
- d) As long as possible

Q14: How familiar are you with prepayments during the fixed-rate period or partial repayments at the end of the fixed-rate period as ways to reduce the remaining balance of a mortgage loan?

[Likert scale from 1 ("Completely unfamiliar") to 4 ("Very familiar")]

Q15: How familiar are you with forward loans as a way to lock in future loan terms already today?

[Likert scale from 1 ("Completely unfamiliar") to 4 ("Very familiar")]

D.4 Post-vignette section

Q16: You have now reached the final part of the survey. We would like to ask a few questions about your attitudes and your situation.

To what extent do you agree with the statements below?

[Likert scale from 1 ("Strongly disagree") to 5 ("Strongly agree")]

- a) I am willing to take risks when making financial decisions
- b) I try to get rid of outstanding debt as quickly as possible
- c) A bank's reputation plays an important role in my choice of lender
- d) My bank will only present me with a product offer if it is in my best interest
- e) Because loan interest rates have increased, I have started to save more money in recent months than before
- f) For my saving and spending decisions, I create and follow a plan or rules
- g) I have good financial knowledge

Q17: After paying your essential monthly expenses such as groceries, utilities, and debt payments, how much money typically remains for other spending or for saving?

- a) € 0 (no money left)
- b) Less than € 100
- c) € 100 to € 249
- d) € 250 to € 499
- e) € 500 to € 749
- f) € 750 to € 999

- g) € 1,000 to € 1,249
- h) € 1,250 to € 1,499
- i) € 1,500 or more

Q18: Do you own real estate?

Multiple answers possible.

- a) Property I live in
- b) Rental property
- c) Vacation home
- d) Other real estate
- e) I do not own any real estate [Exclusive response]

Q19: Do you own the following financial products?

Multiple answers possible.

- a) Savings accounts (e.g., instant-access or time deposits)
- b) Life insurance policies
- c) Bonds
- d) Stocks
- e) Gold or other precious metals
- f) Cryptocurrencies
- g) Consumer loans
- h) Mortgage loans
- i) I do not own any of these financial products [Exclusive response]

Q20: What is your **household's total monthly net income**?

Note: This refers to the sum of wages, salaries, income from self-employment, pensions, or retirement income, each after deduction of taxes and social security contributions. Please also include income from public benefits, rental income, housing benefits, child benefits, and any other income.

If your household's net income was irregular over the past 12 months, please provide an average over the last 12 months.

- a) Under € 1,000
- b) € 1,000 to € 1,999
- c) € 2,000 to € 2,999
- d) € 3,000 to € 3,999
- e) € 4,000 to € 4,999
- f) € 5,000 to € 5,999
- g) € 6,000 to € 7,999
- h) € 8,000 or more
- i) Prefer not to say

Q21: What is your highest educational attainment?

- a) No degree
- b) Lower secondary school certificate (Hauptschulabschluss)
- c) Intermediate secondary school certificate (Realschulabschluss)

- d) Upper secondary school leaving certificate ((Fach-)Abitur)
- e) Vocational training/apprenticeship
- f) Master craftsman/technician qualification
- g) University of applied sciences degree
- h) University degree
- i) Doctorate

Q22: What do you think is the effect of an **unexpected increase in interest rates** in the economy (by the central bank) on the following macroeconomic variables?

[Response options: "Strong decrease", "Slight decrease", "No effect", "Slight increase", "Strong increase"]

- a) Inflation rate
- b) Economic growth
- c) Stock prices
- d) House prices

Q23: How interesting did you find this survey?

[Likert scale from 1 ("Not interesting at all") to 5 ("Very interesting")]

Q24: Do you have any suggestions or feedback about our survey? Please share them here (**optional**).
[Open-ended response field]

Q25: Thank you very much for participating in our survey!

As a thank-you for your participation, you will receive an Amazon voucher worth €10. To receive the voucher, please confirm that you agree to be contacted by us for the purpose of sending the voucher, and in the next step provide your e-mail address.

- a) Yes, I would like to receive the voucher
- b) No, I do not want to receive the voucher

[If Q25 = a), show Q25a on the same screen]

Q25a: Please enter your e-mail address for sending the voucher:

[Allow continuation only if the two e-mail entries match]

Enter e-mail address:

Confirm e-mail address:

Q26: Once again, thank you very much for your participation! Your responses have been saved.
You can now close this window in your browser.

E Letter text

This section reports the text of the letter, translated from German to English. The gray text indicates which letter variants include which letter sections and the position of two figures that are also part of the letter. Our partner bank asks us not to display the original letter, so we show letter figures separately in [Figure A.11](#). The original letter also includes the bank letterhead and contact details of the respective mortgage advisor.

How do you deal with higher mortgage rates?

Dear XXXX,

—— Section 1: included in variants 1,2, and 3 ——

Since the beginning of 2022, market interest rates have risen significantly. Together with Goethe University Frankfurt, we investigate how borrowers deal with the increase in interest rates. As part of this study, we would like to provide you with relevant information some time before the fixed interest rate on your mortgage comes to an end.

—— Figure 1: recent increase in mortgage rates ——

According to the Bundesbank, the average fixed interest rate on newly originated mortgages in Germany is currently 4% per year. The interest rate at which we can continue your contract may be higher or lower than this value. This is because the final value depends primarily on market developments[.]

—— Section 2: included in variants 1 and 3 ——

[...] that cannot be predicted. For example, a current study by the Bundesbank (Survey on Consumer Expectations, date: August 2024) shows that 40% of Germans expect an increase in loan rates. 20% of respondents believe that lending rates will fall.

[page 1]

—— Section 3: included in variants 1 and 2 ——

An increase in interest rates can imply significantly higher monthly payments after the end of your fixed interest period. This example illustrates the possible impact:

—— Figure 2: initial impact of higher rates on interest payments ——

—— Section 4: included in variants 1, 2, and 3 ——

In principle, there are different options to deal with increased mortgage rates. These include:

- You reduce the loan amount by making extra payments during the fixation period and/or a (partial) prepayment. Extra payments are possible once a year, provided that the contract allows them; prepayments can be made after 10 years from origination for fixation periods of more than 10 years and at expiration of the fixation period.
- You take out a forward loan or a home savings plan. This gives you planning certainty, as future fixed interest rates and payments are predetermined.
- You build up savings to be able to make higher payments in the future.

Which option is most appropriate for your situation and best suited to you depends on various factors. We are always available for a consultation.

[Signature from a manager of the bank responsible for retail customers]

As part of the joint study with Goethe University, you may receive an email in the coming weeks inviting you to participate in an online survey run by Goethe University. We appreciate your participation and thank you in advance.

[page 2]